

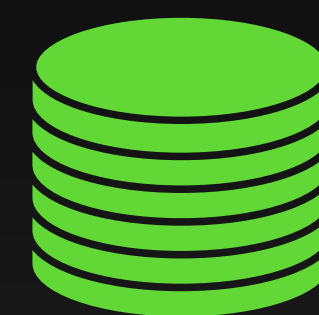
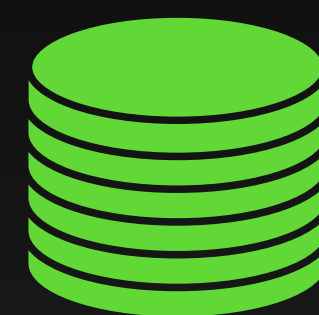
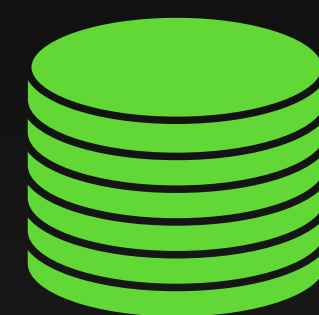
DAG-Based Consensus

Introduction and practical Aspects

Alberto Sonnino

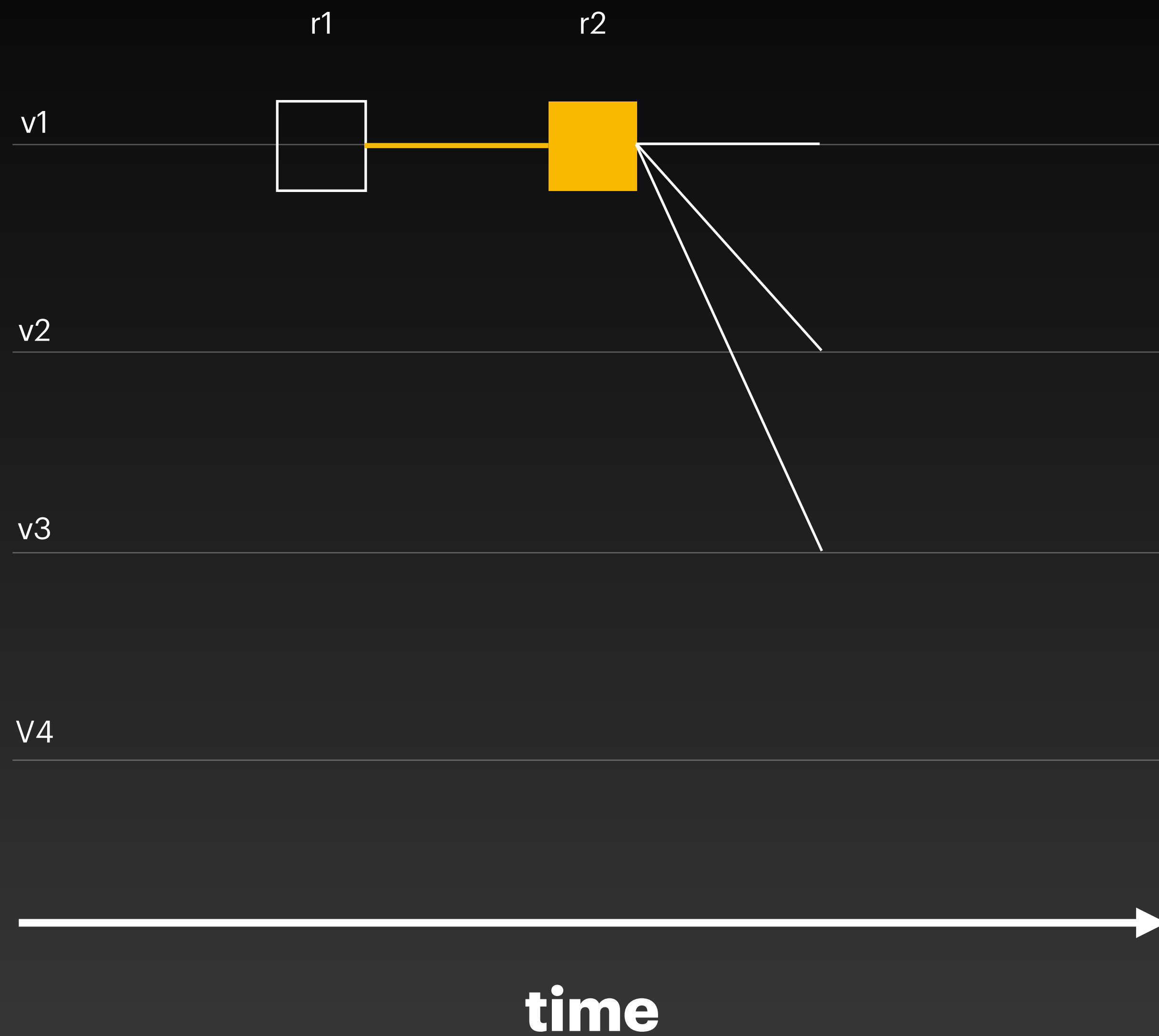


> 2/3

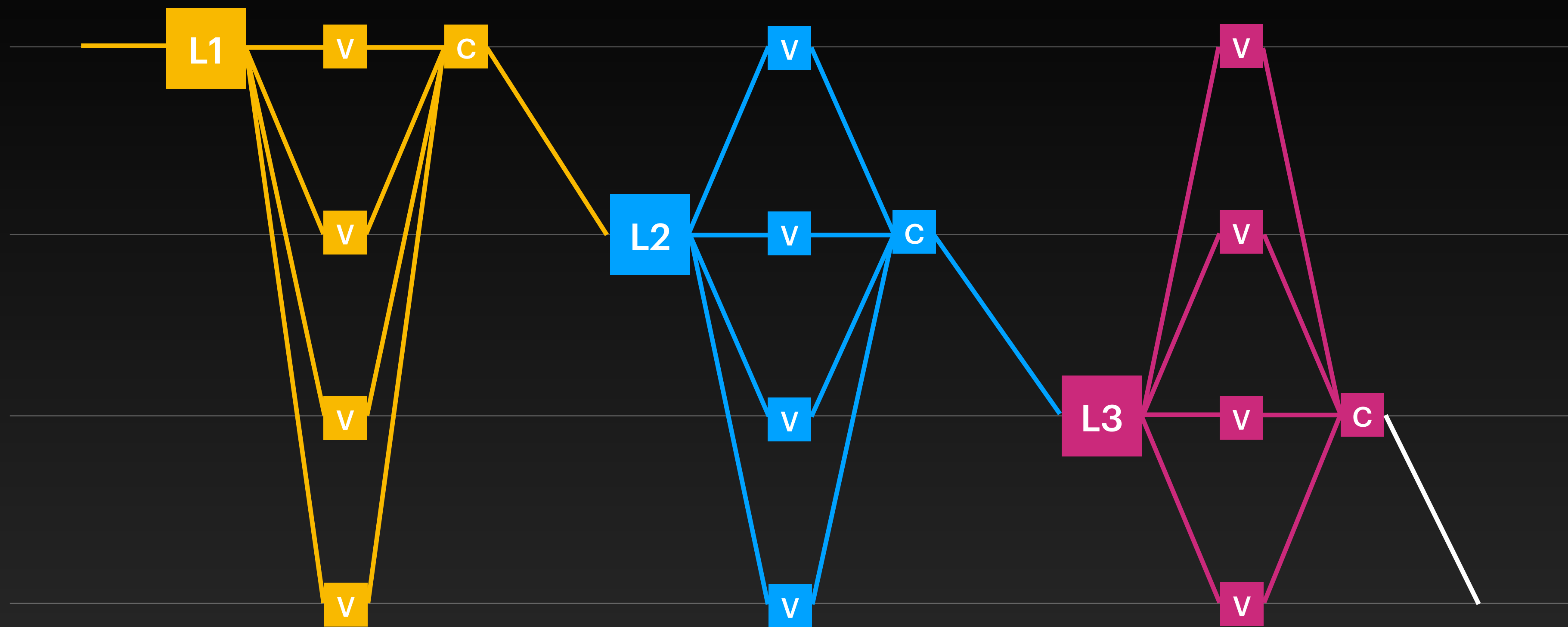


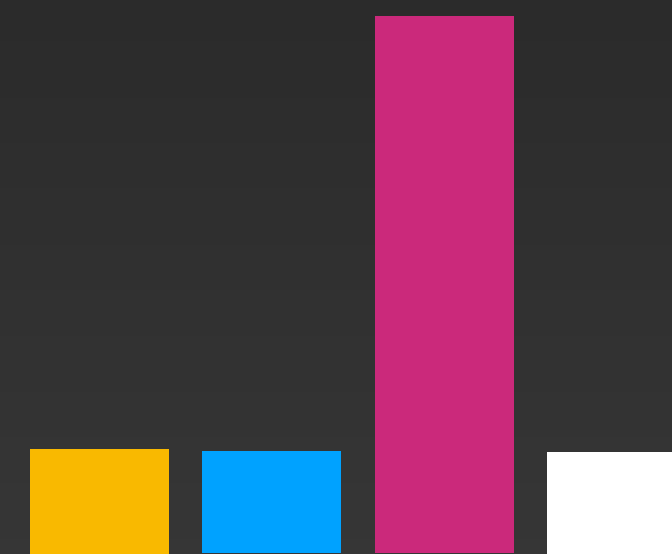
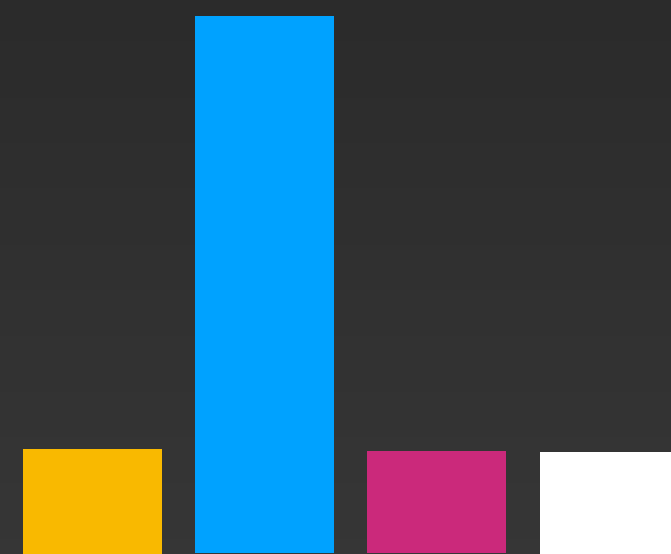
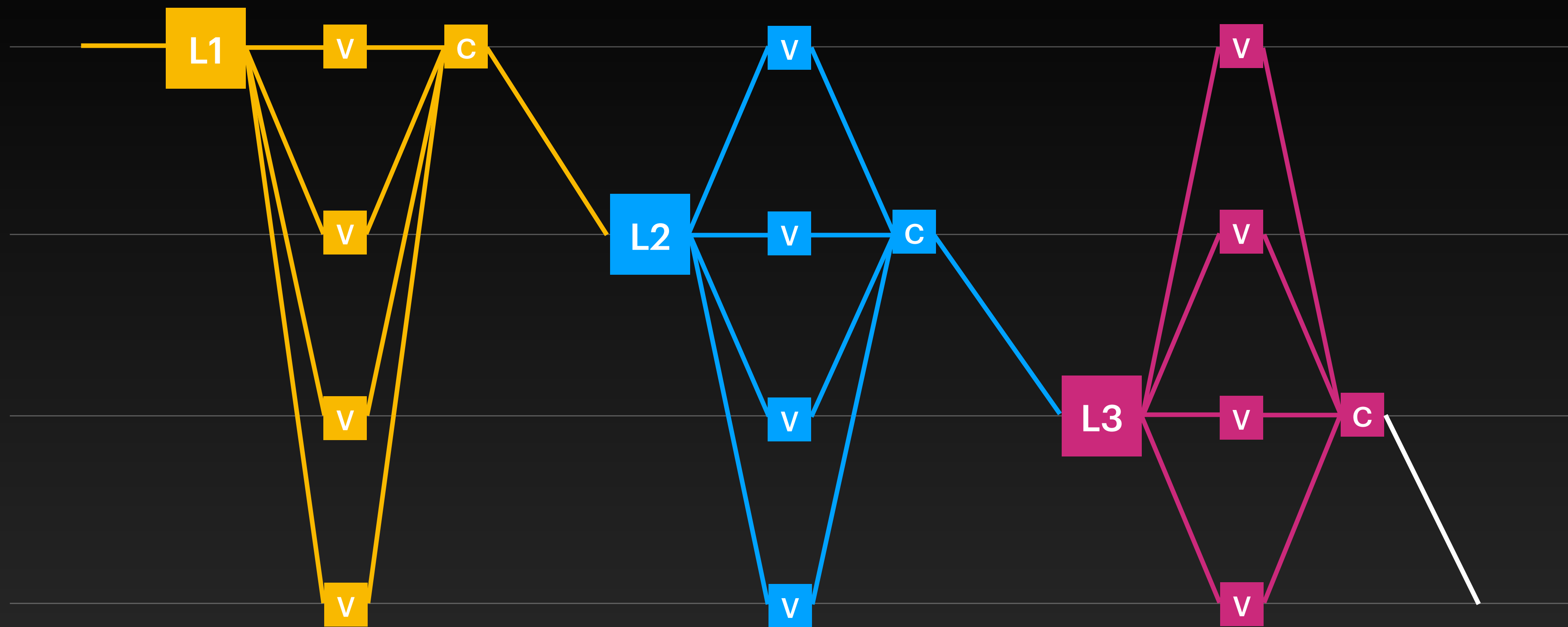
Partial Synchrony

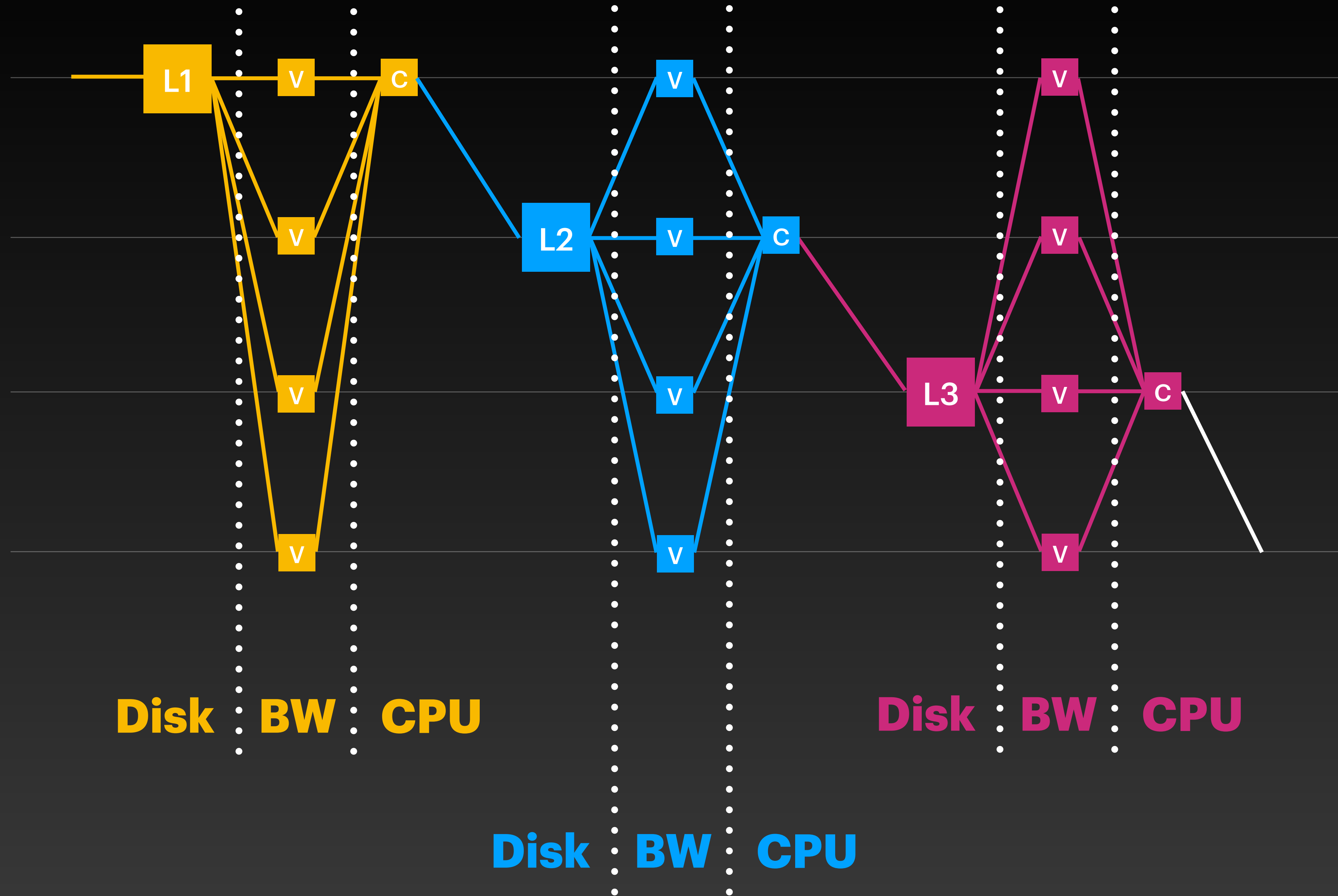


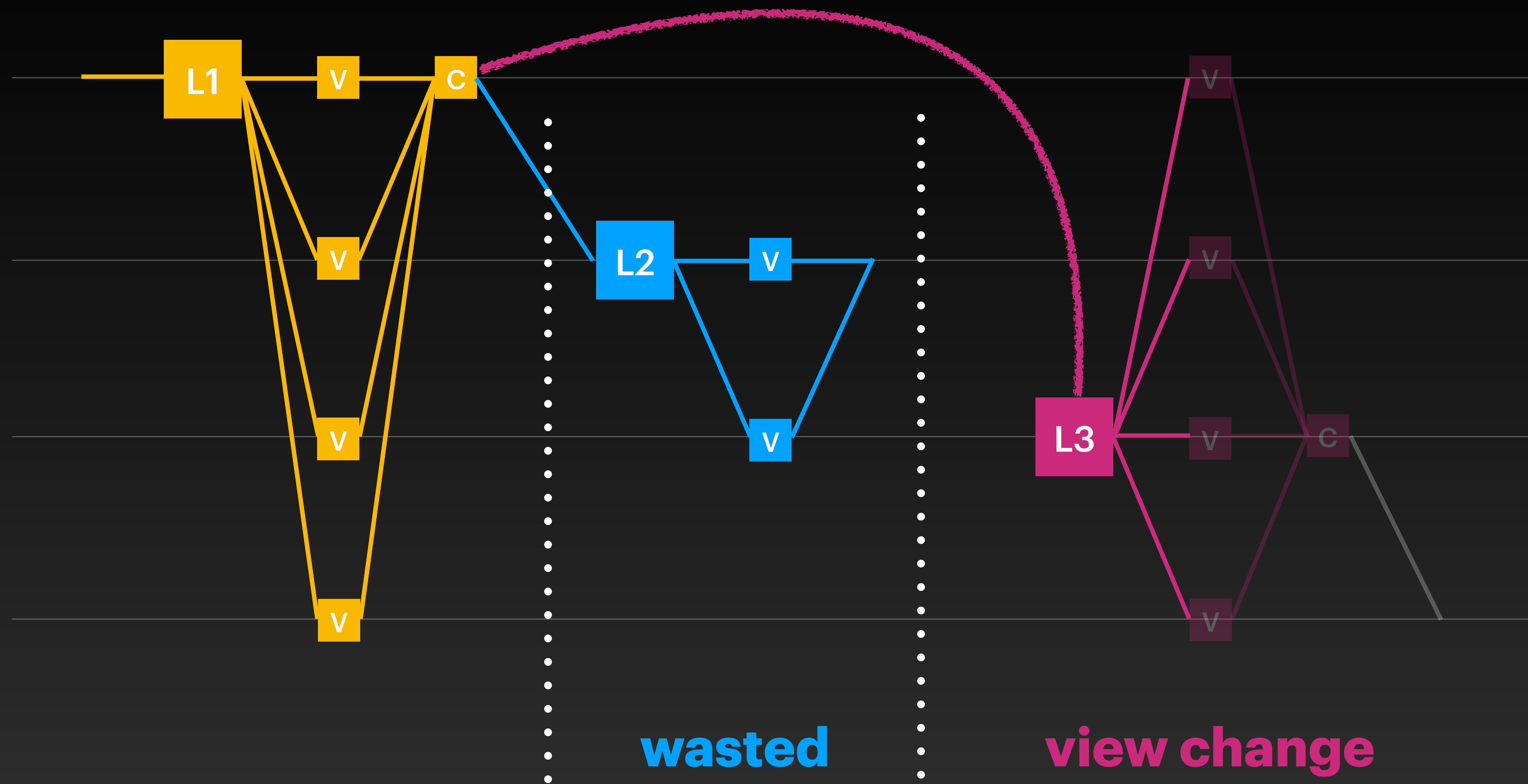


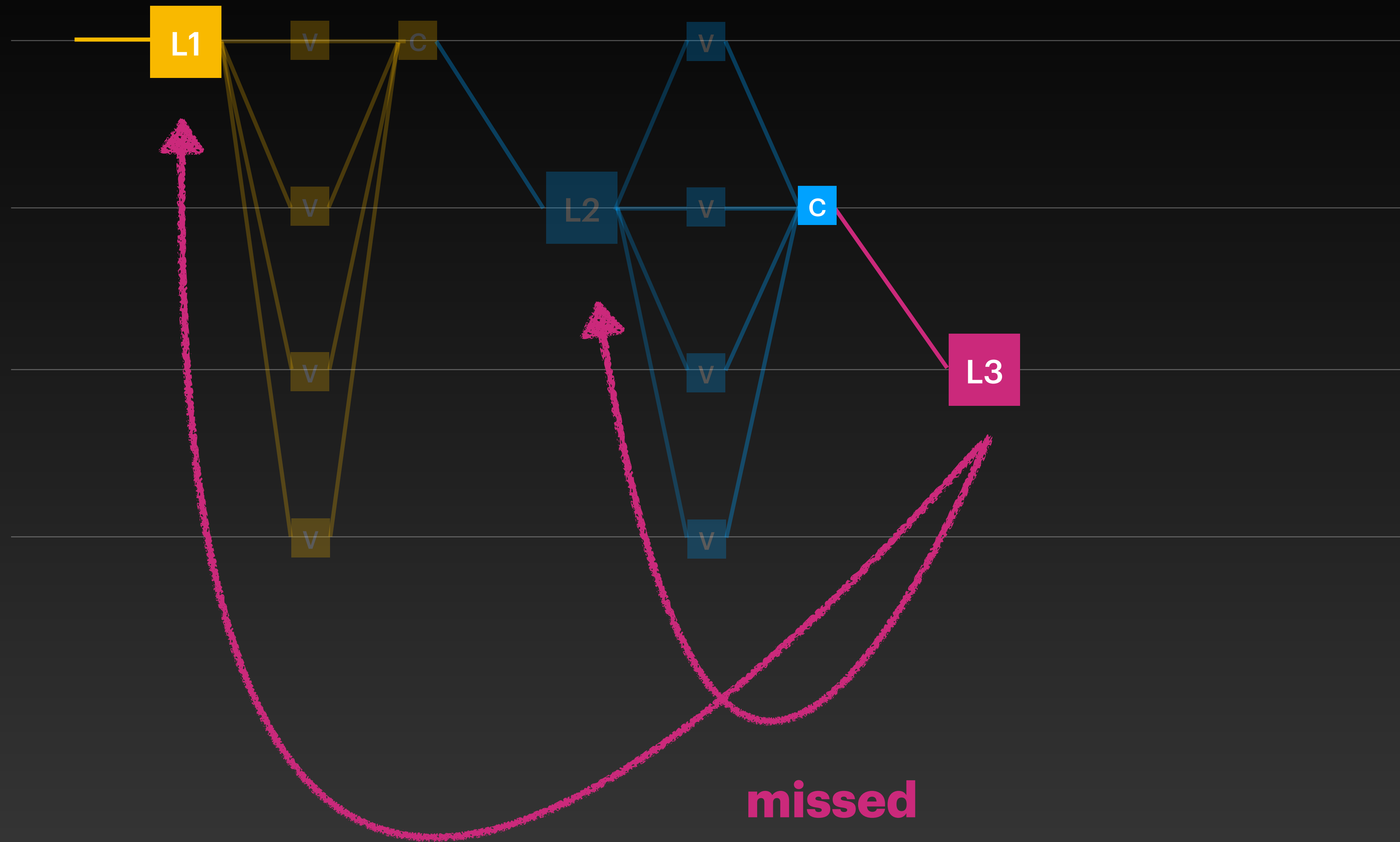
- Round number
- Author
- Payload (txs)
- Parent
- Signature

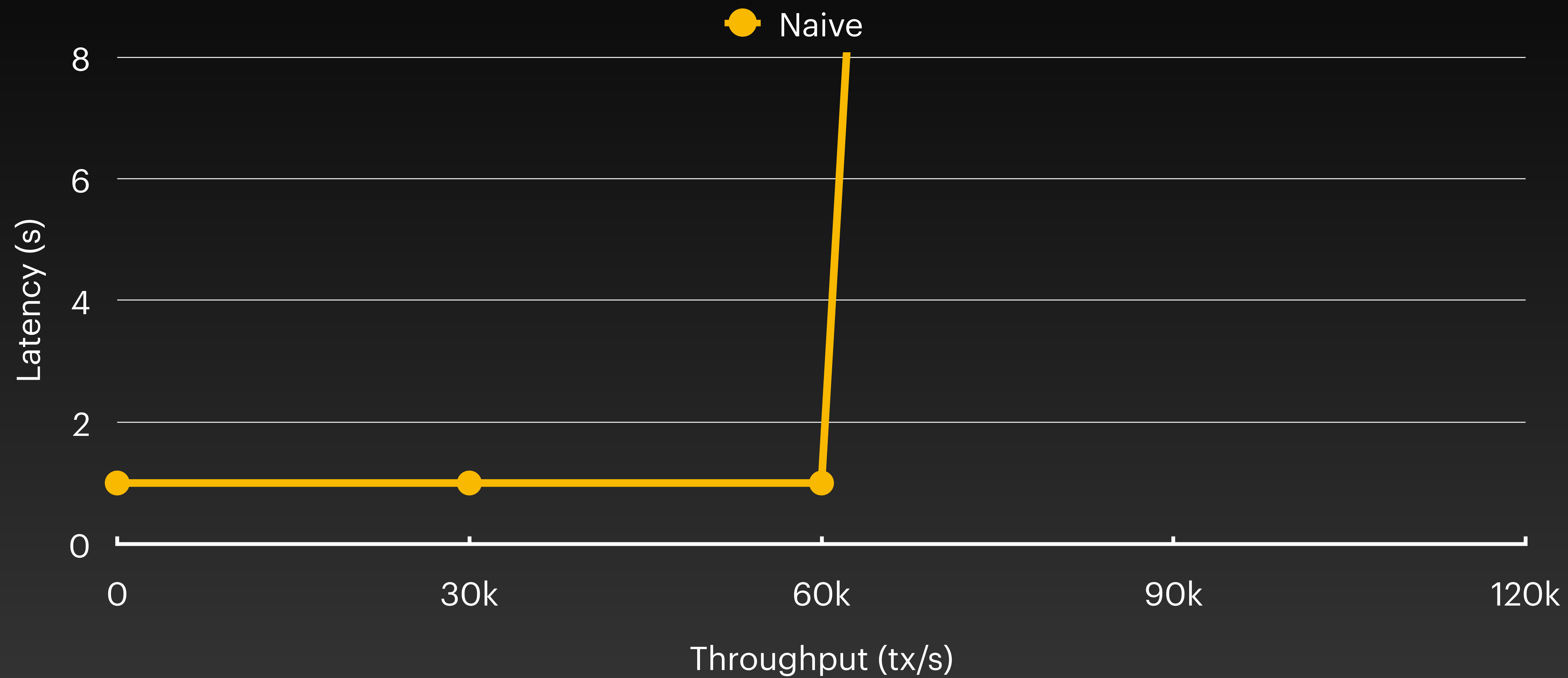


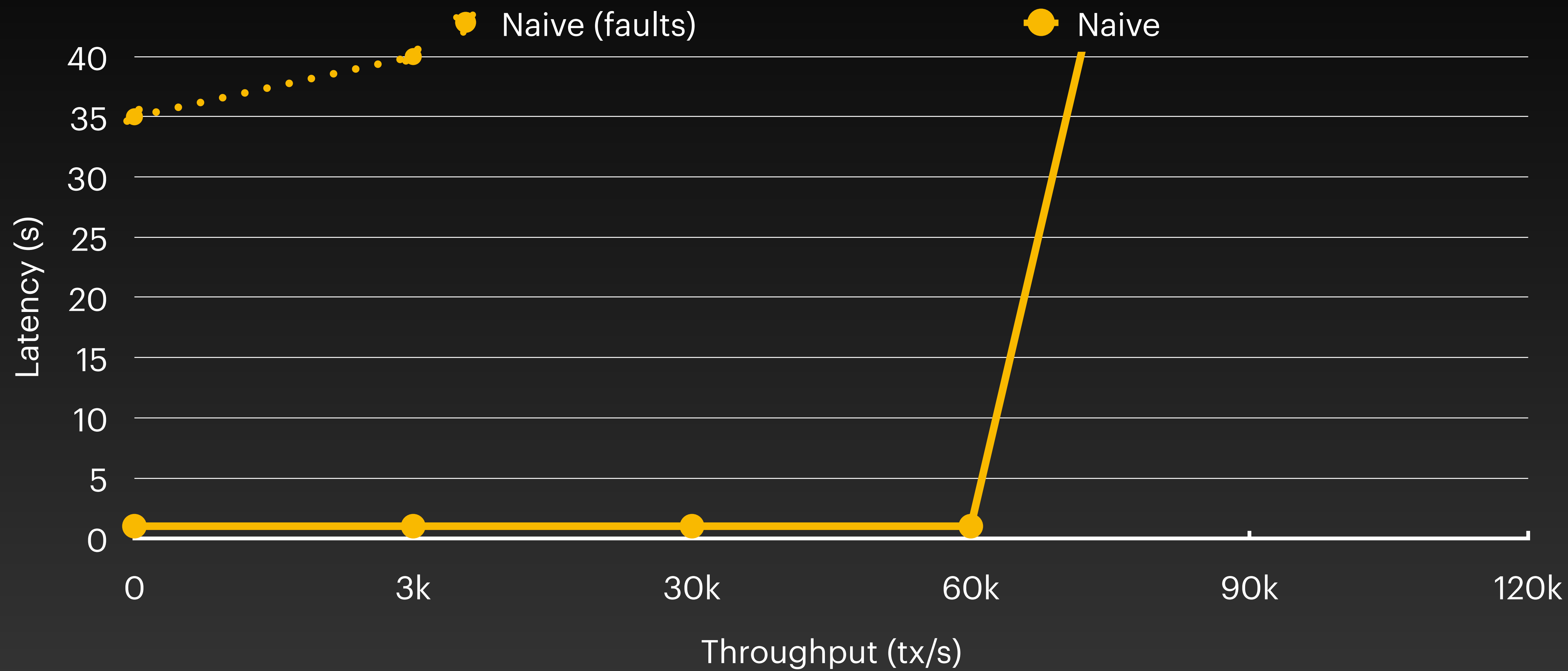


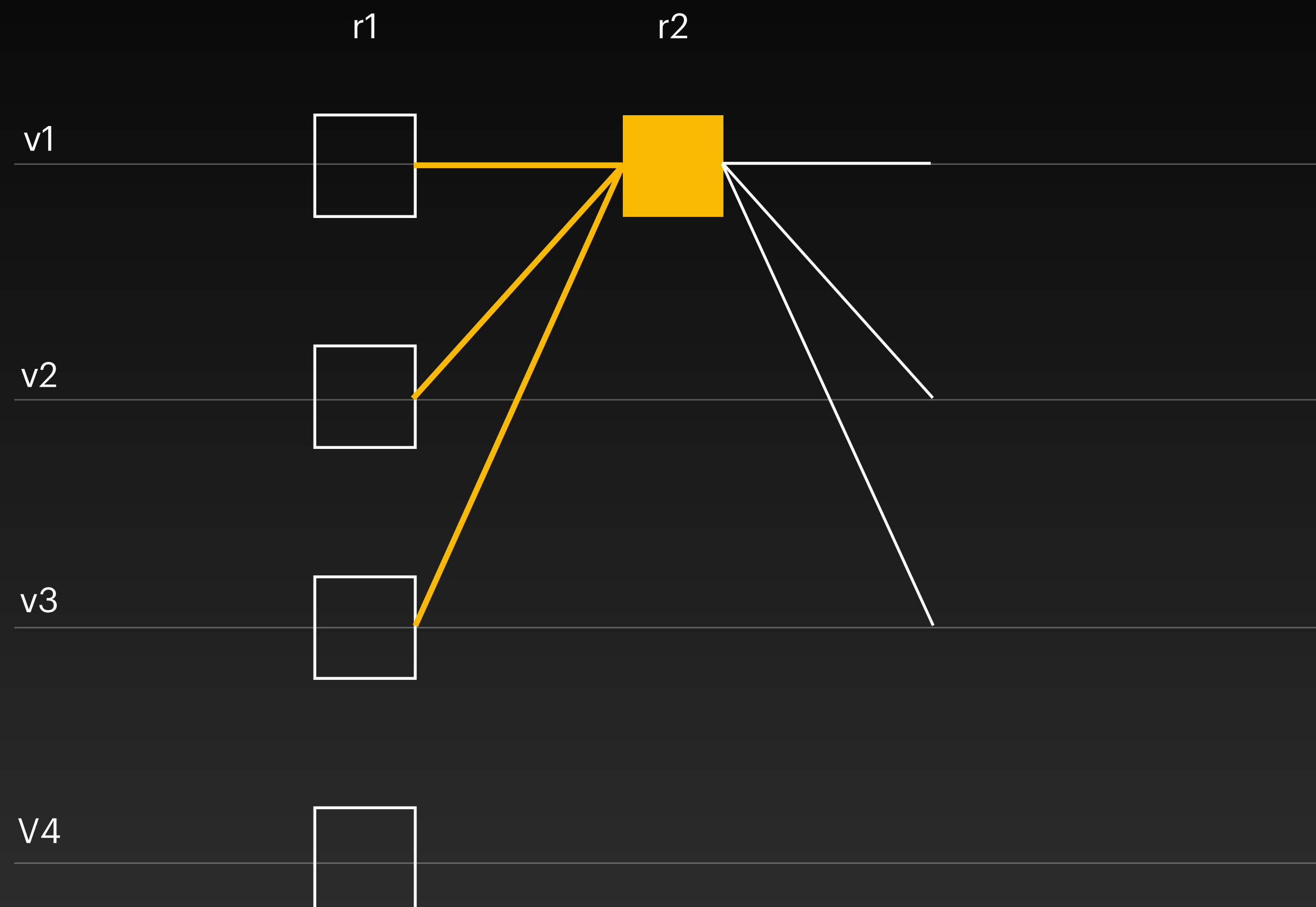






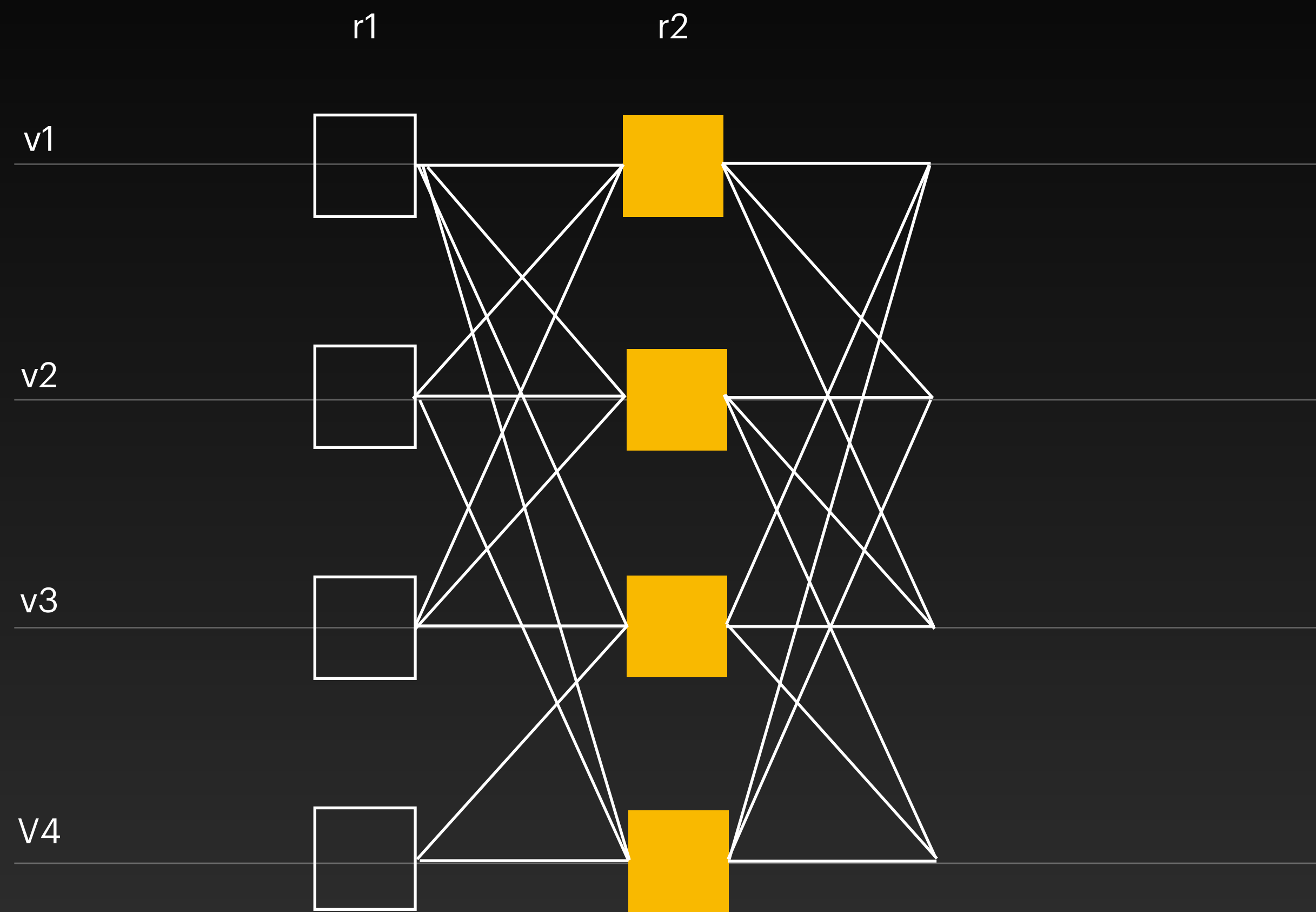






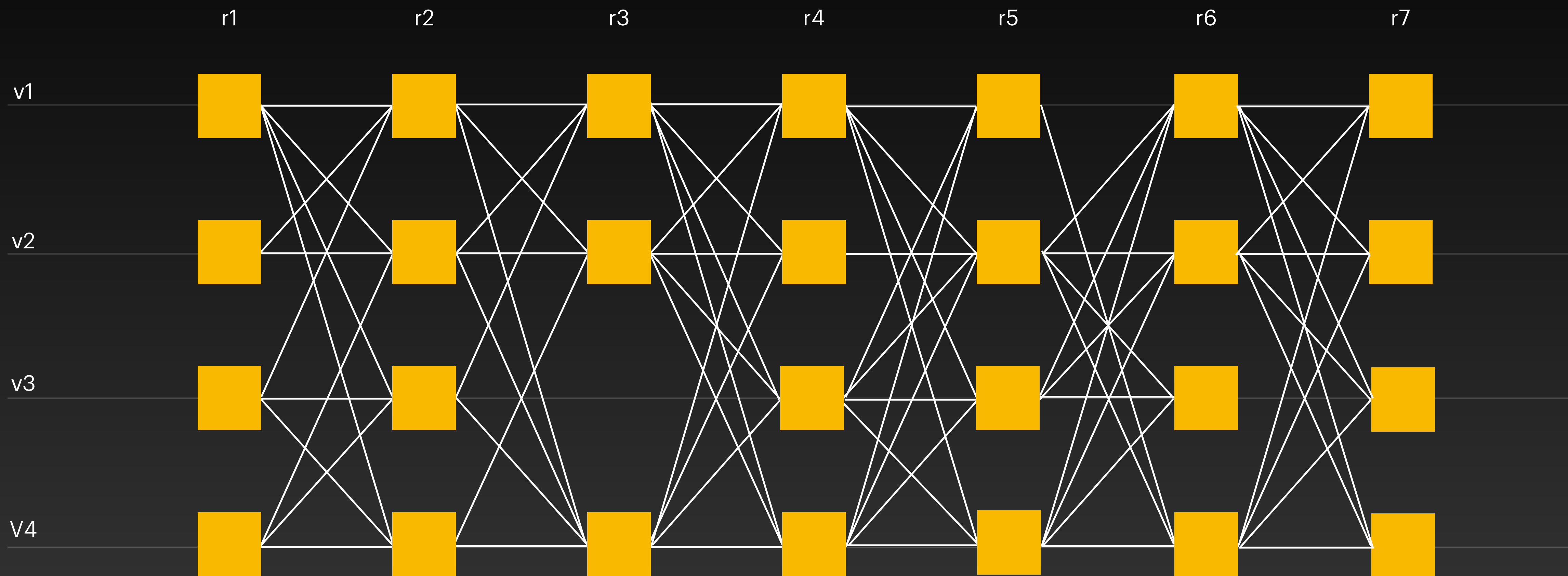
- Round number
- Author
- Payload (txs)
- **Parents**
- Signature

Back Pressure



- Round number
- Author
- Payload (txs)
- Parents
- Signature

Multi Proposer

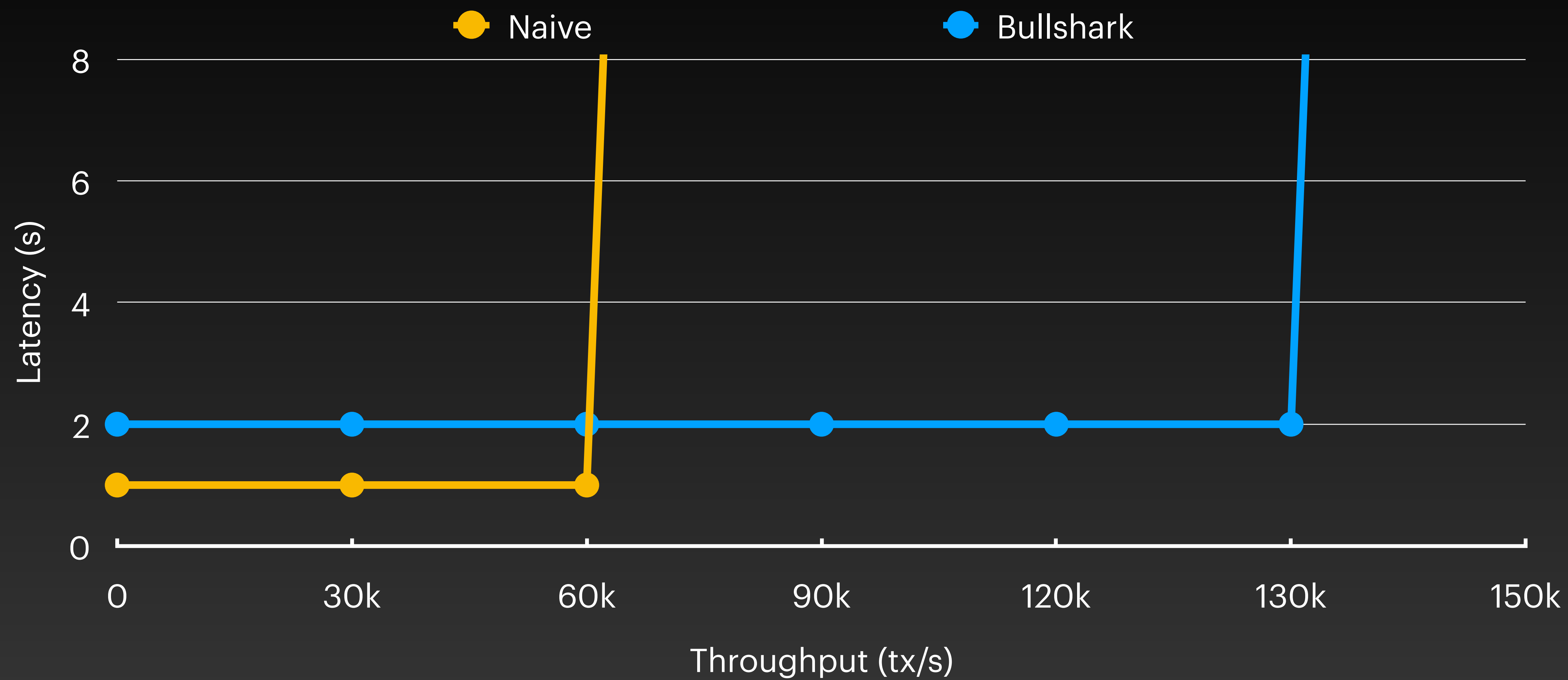


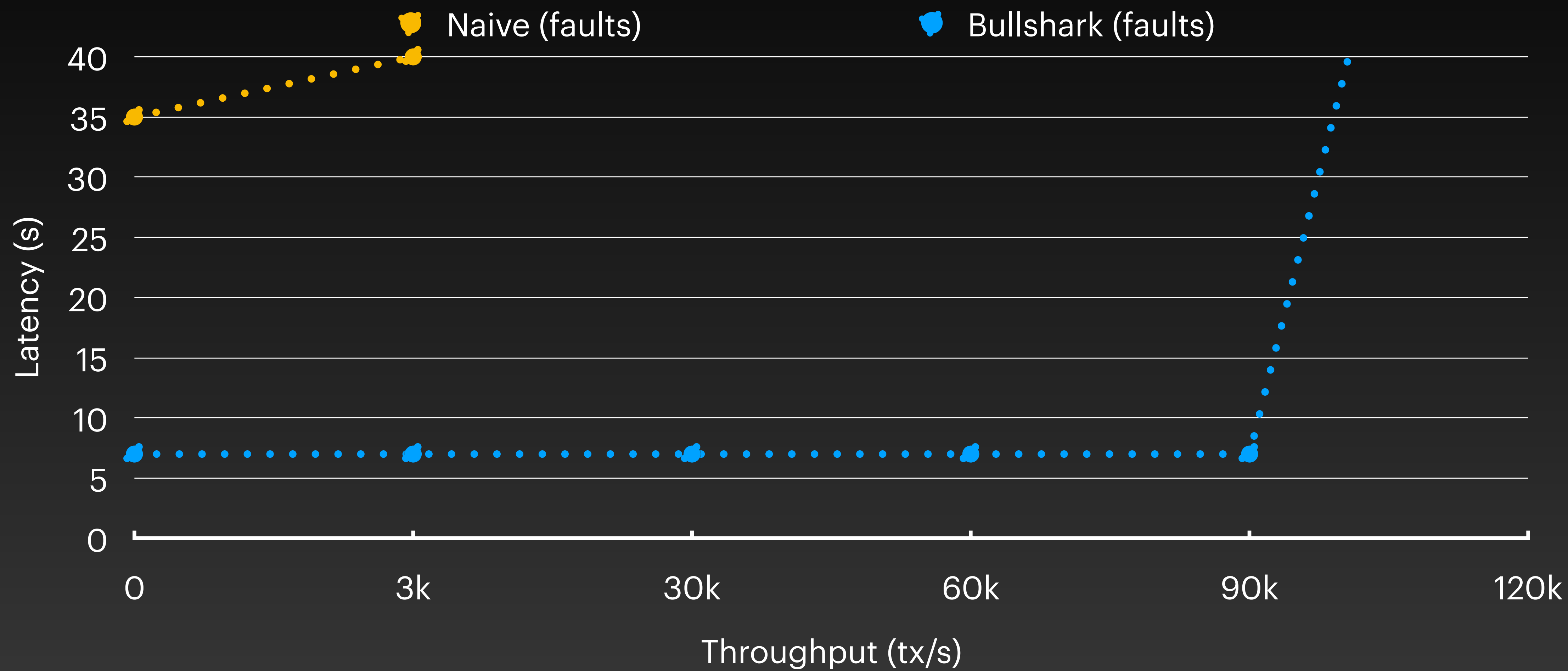
Back Pressure

Robustness

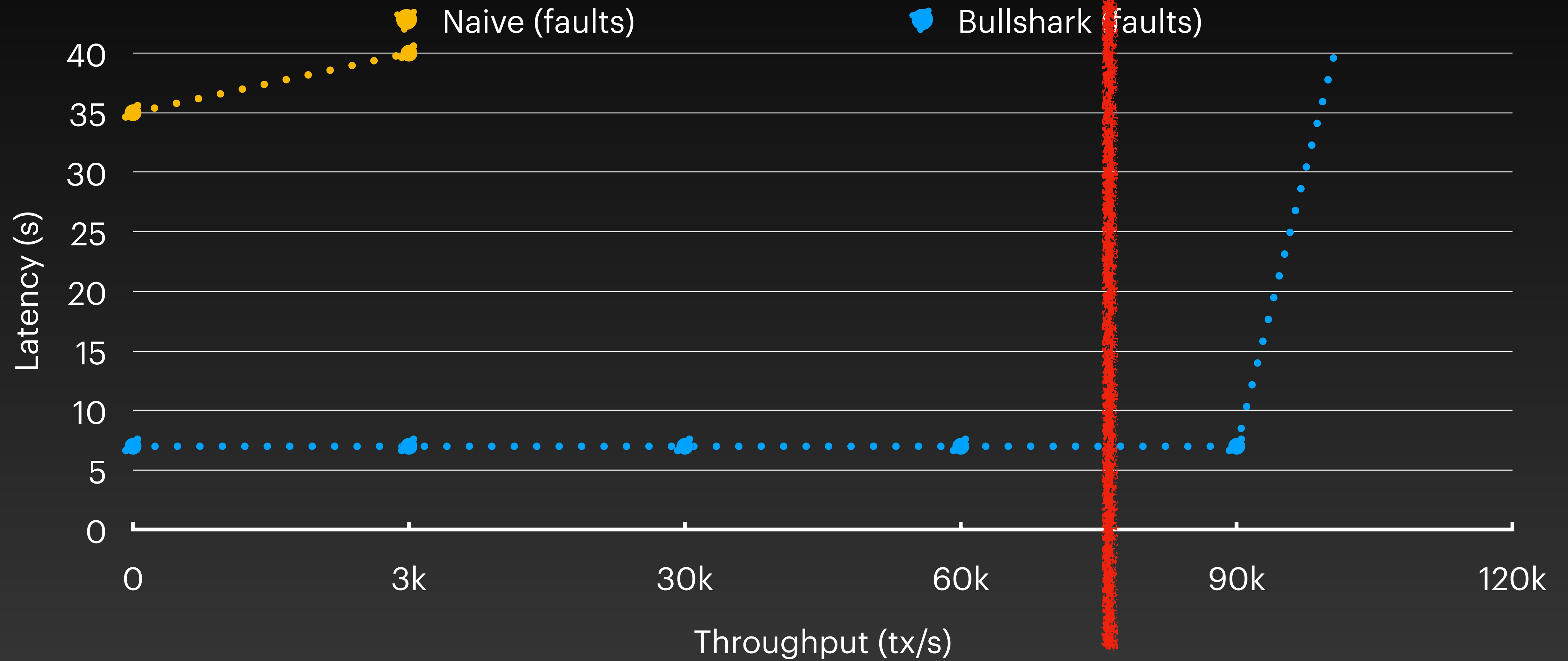
High Resource Utilisation

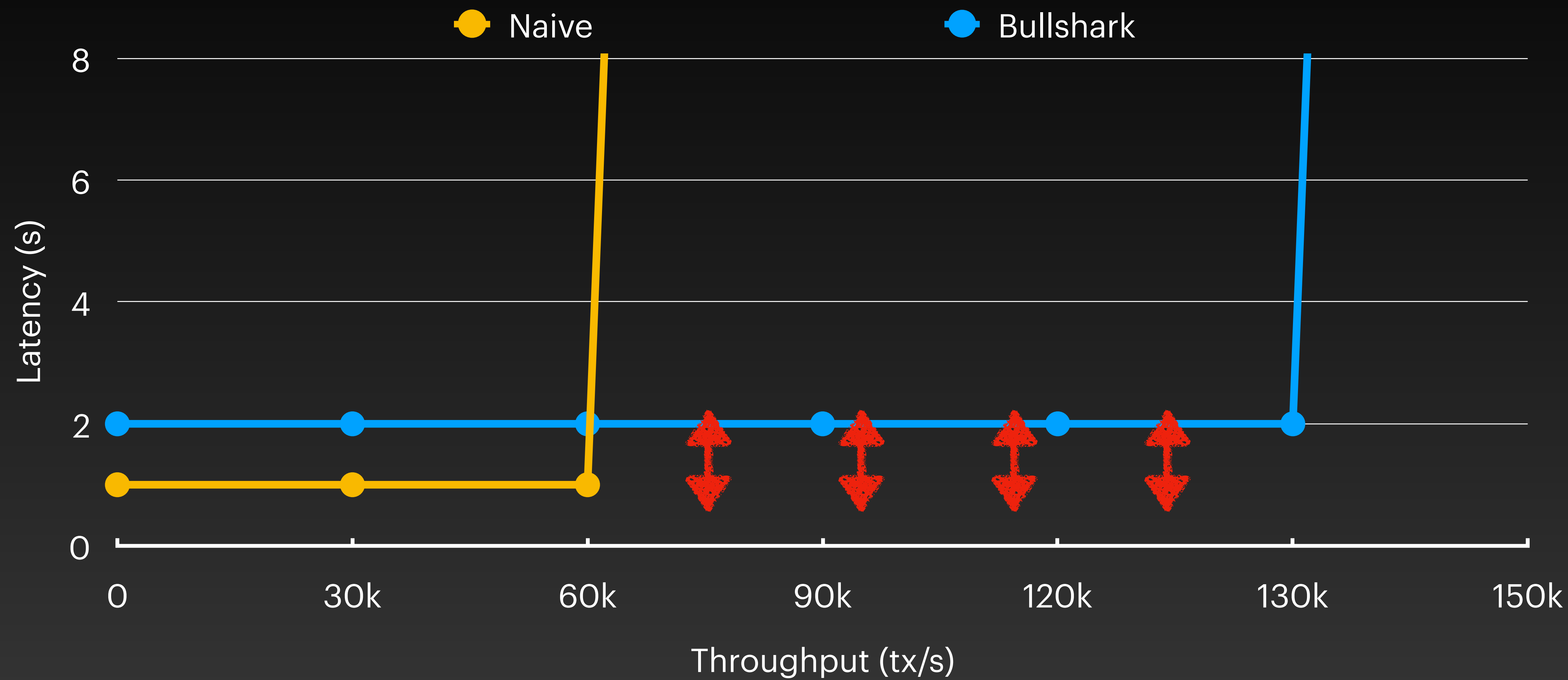
Throughput

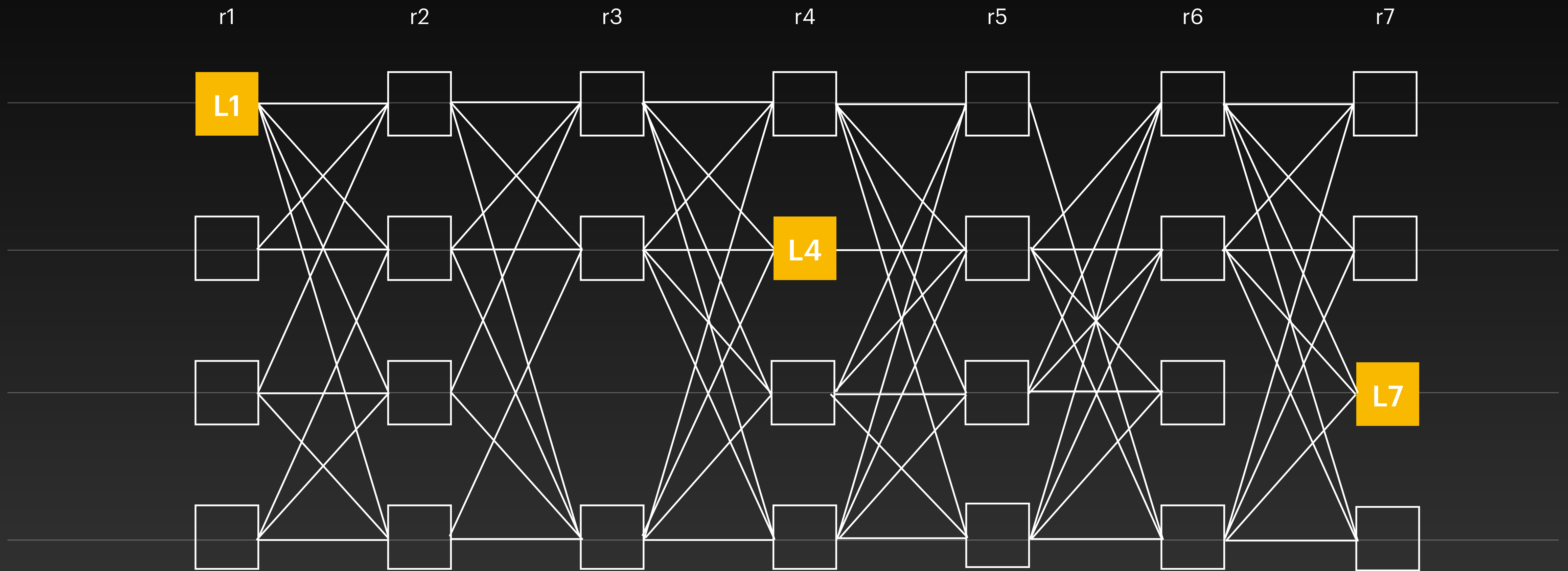




Visa + Master Card







Back Pressure

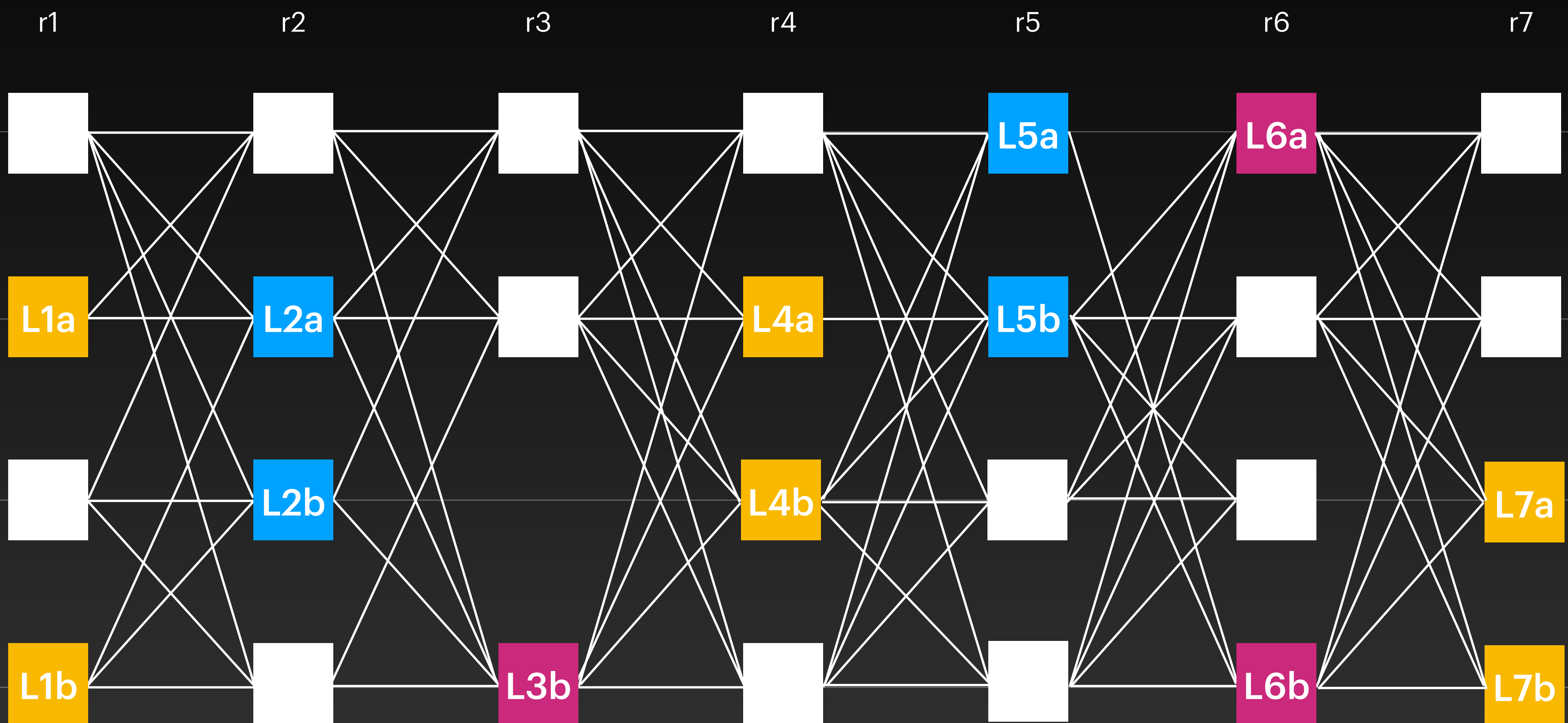
Robustness

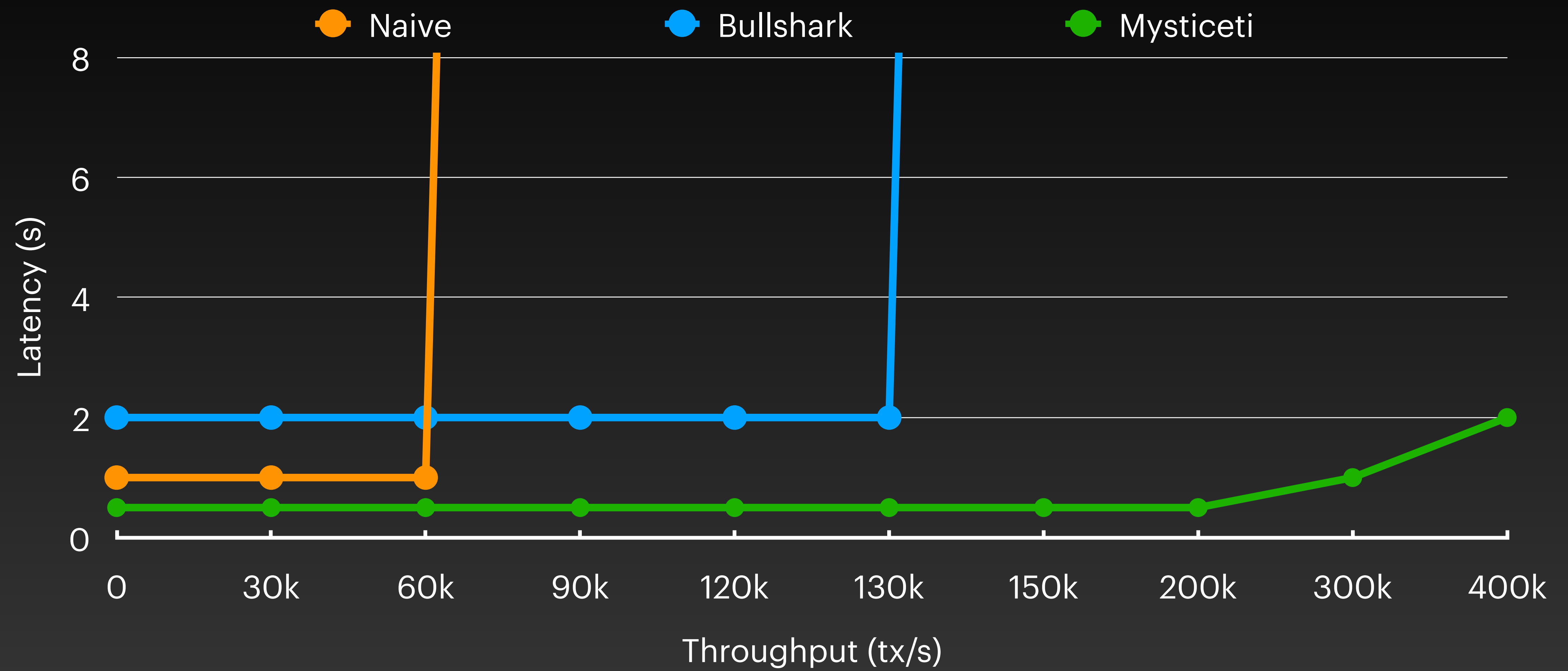
High Resource Utilisation

Throughput

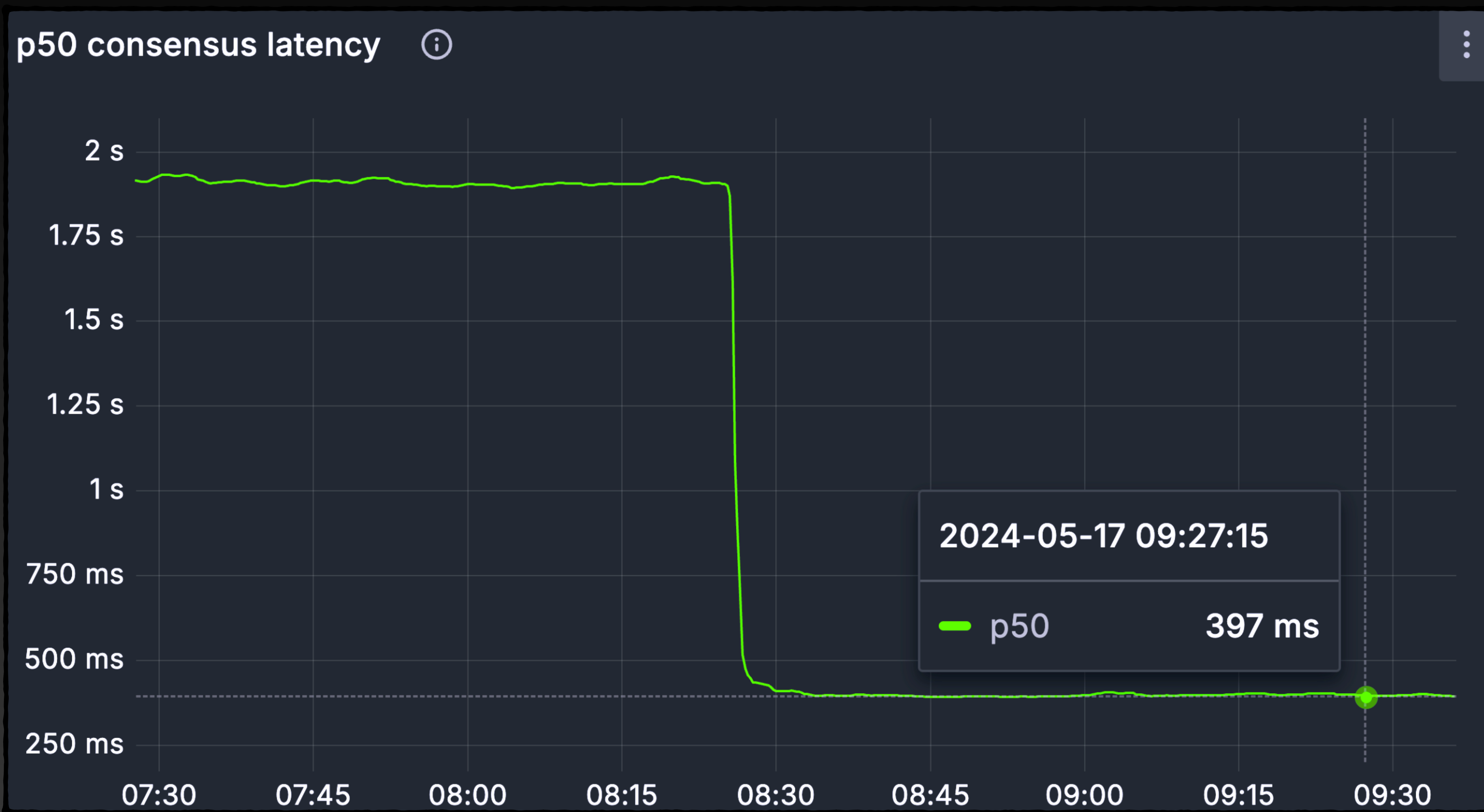
Frequent Commits

Latency



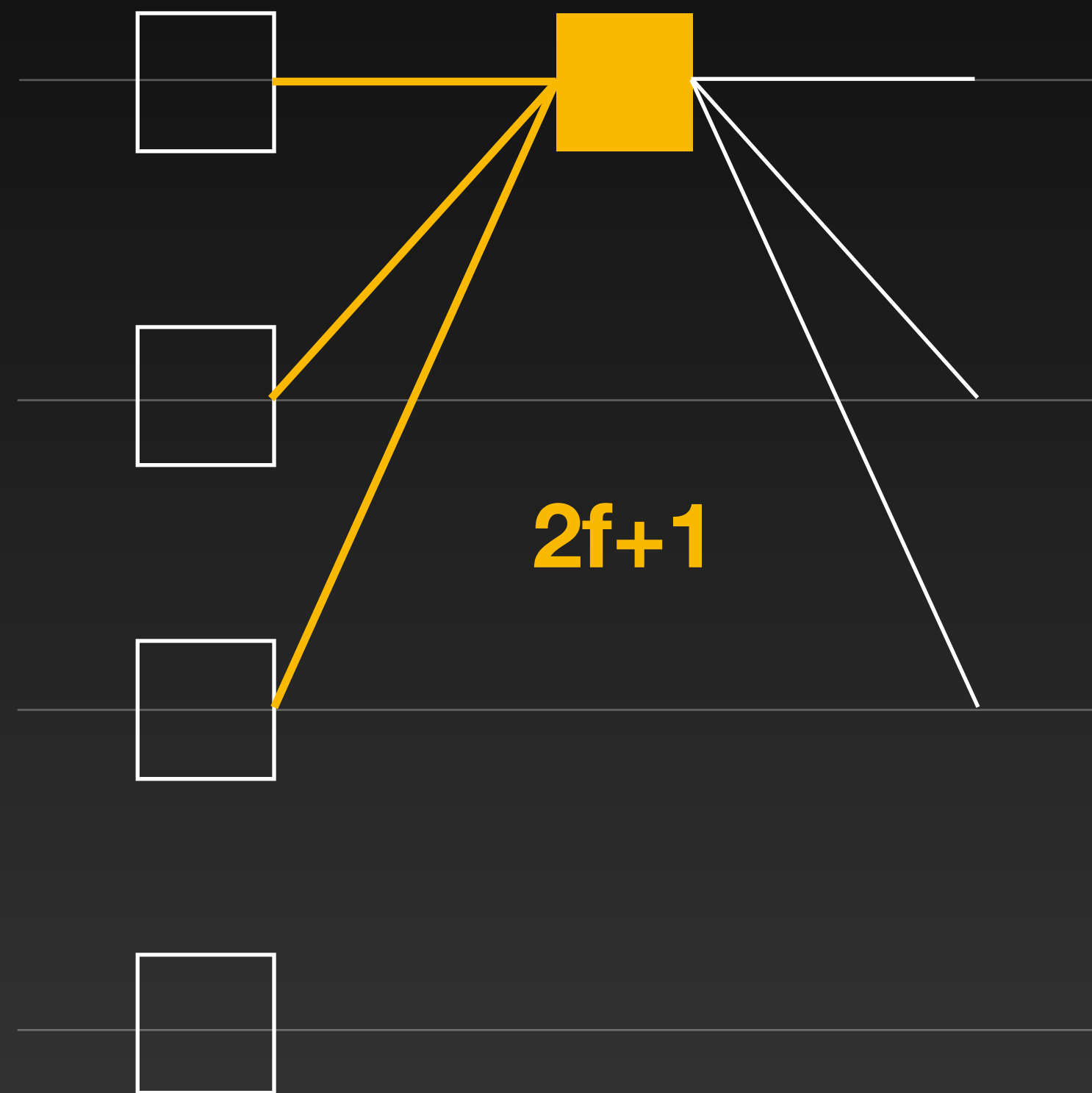


The Sui Mainnet

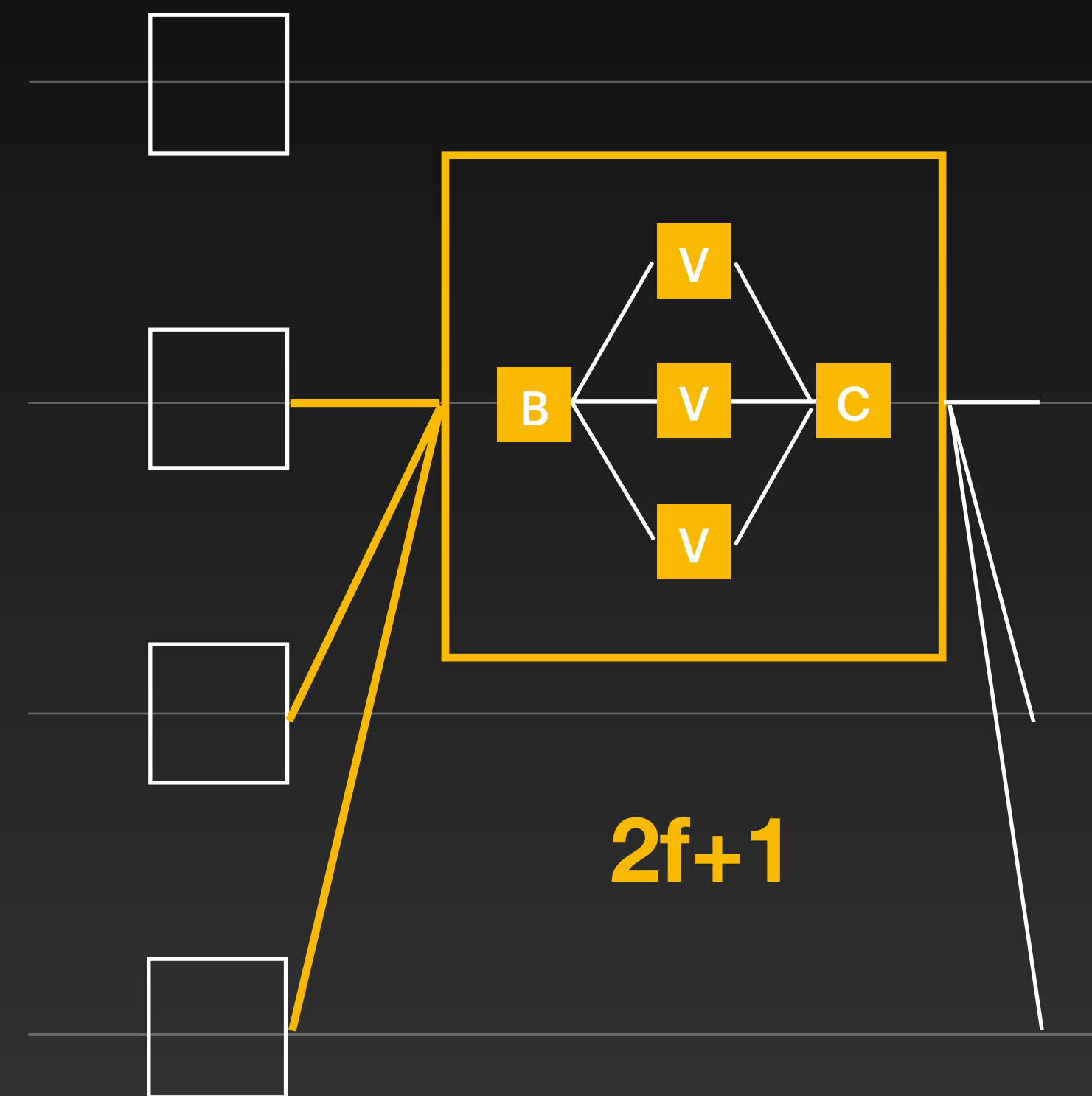


Structured DAGs

Uncertified

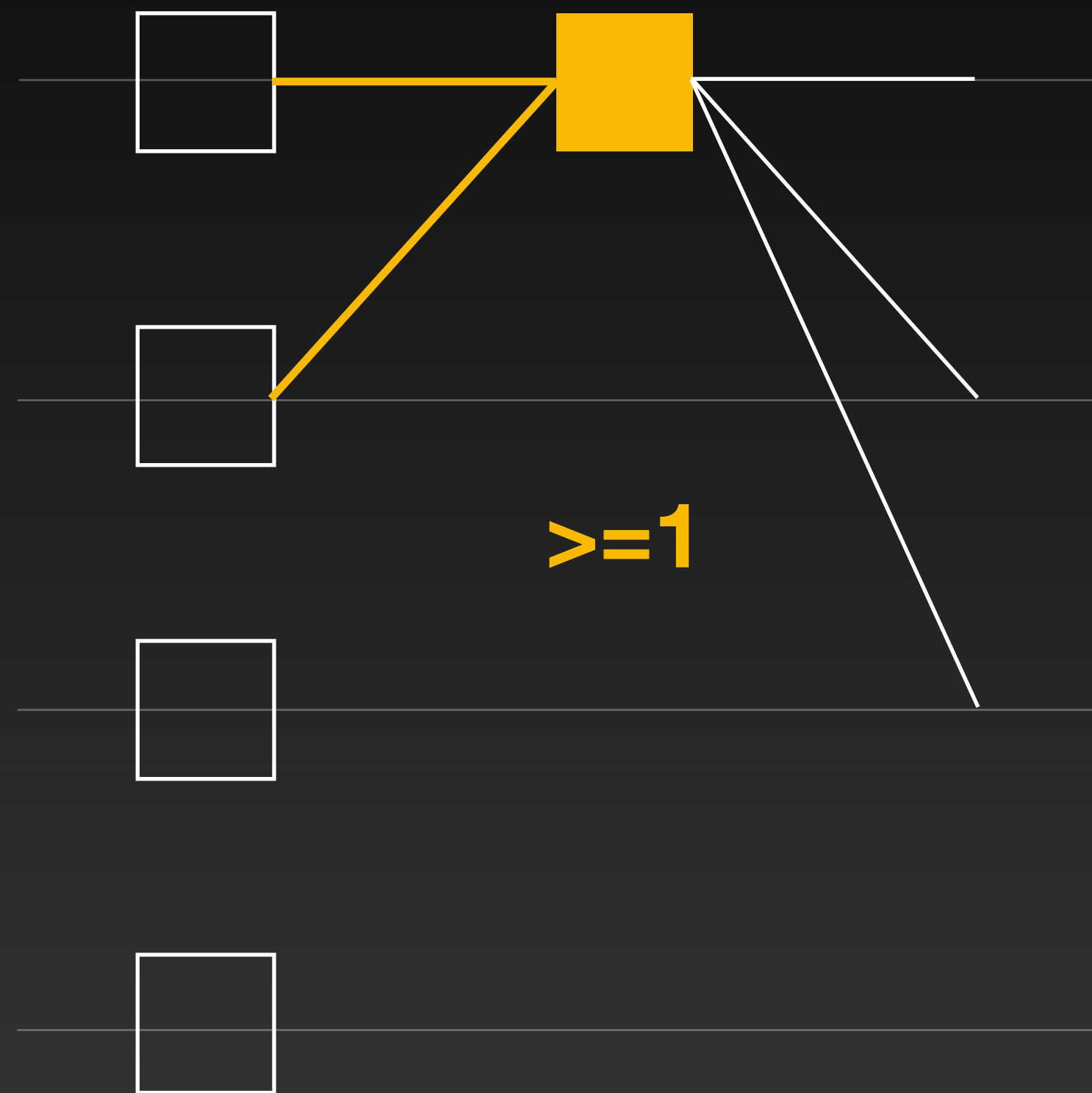


Certified

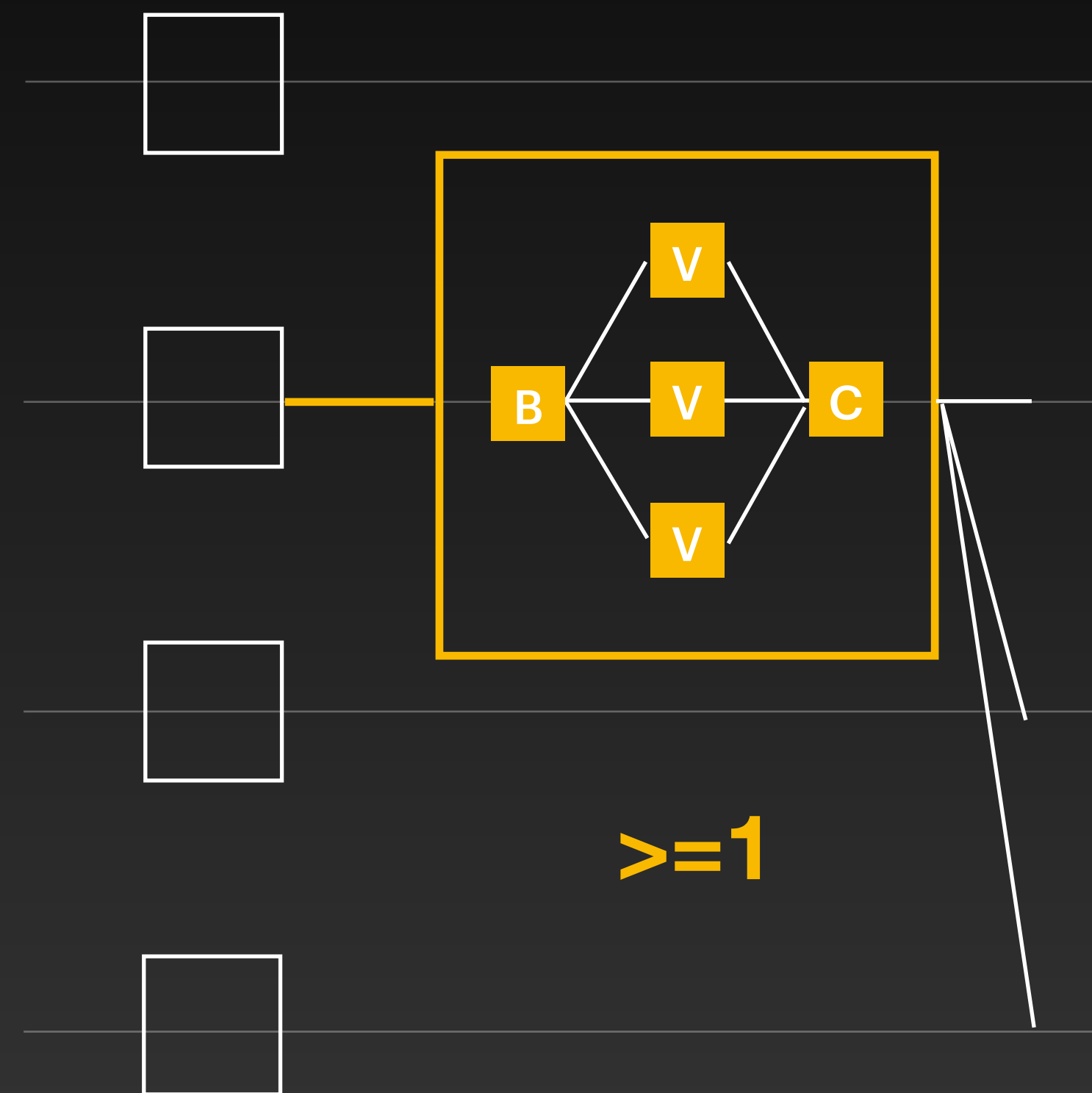


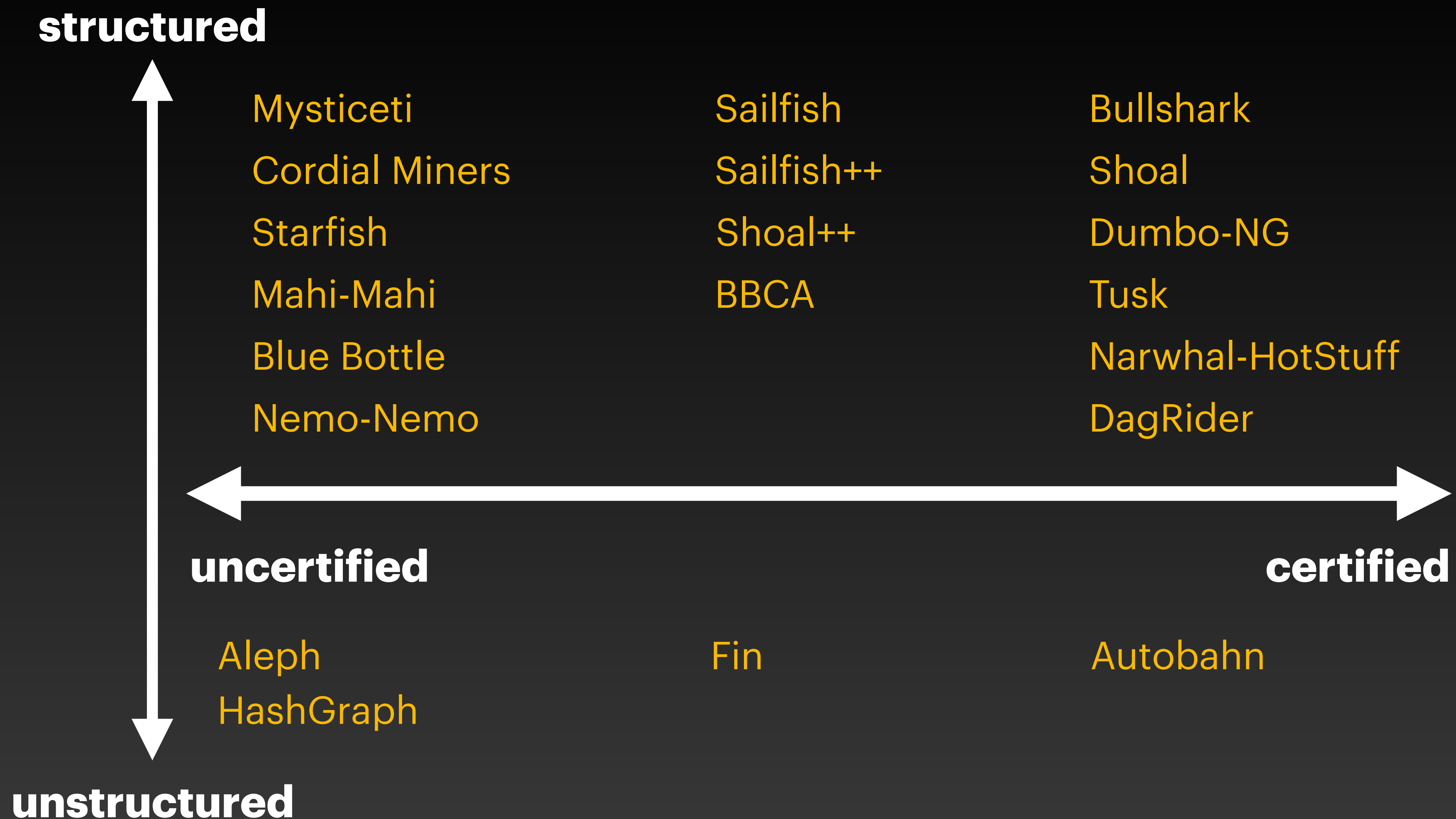
Unstructured DAGs

Uncertified



Certified





Certified

High Robustness • Easier Block Synchronisation

Uncertified

Low CPU • Graceful Crash Faults

Unstructured

Network Flexibility

Cordial Miners

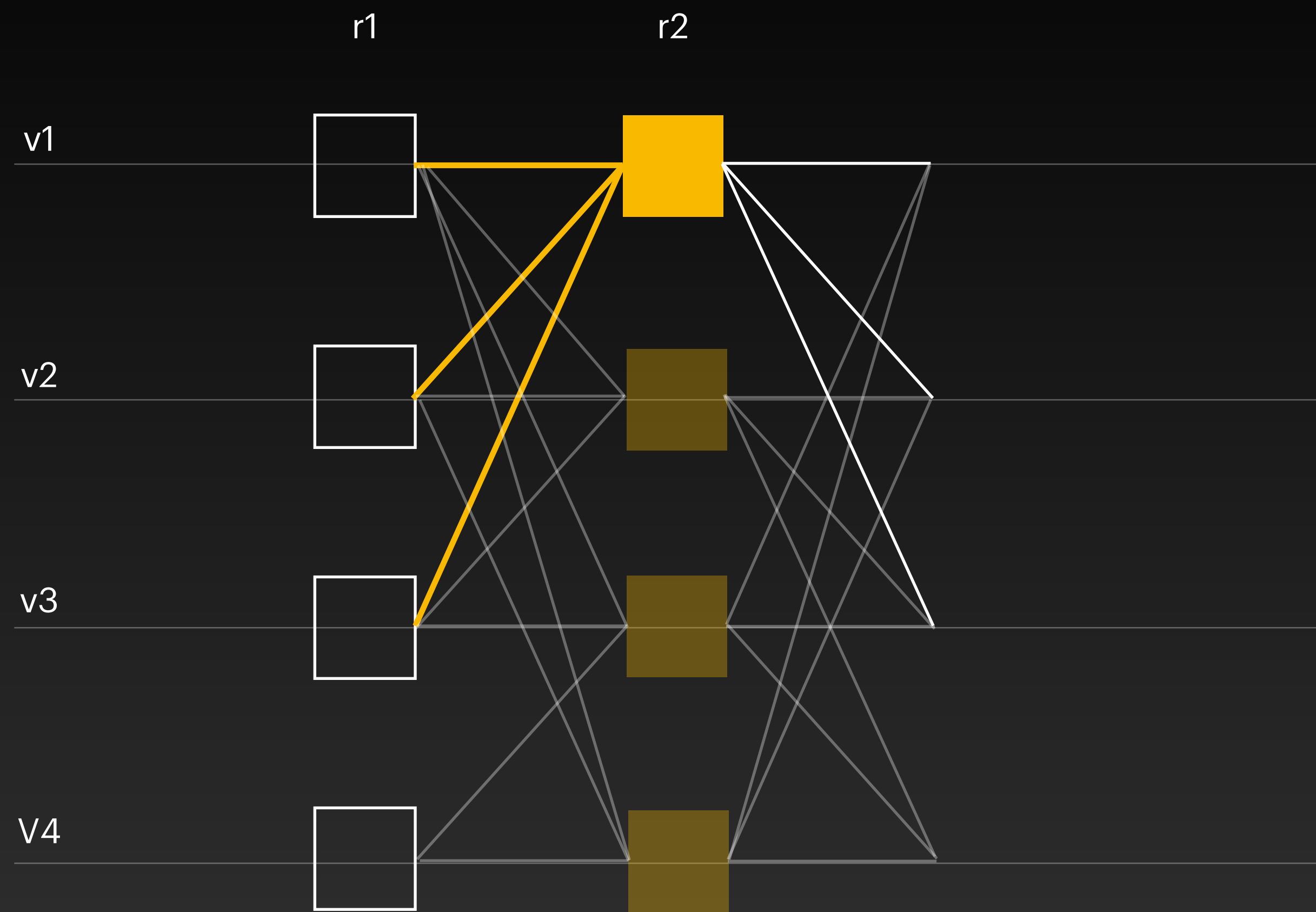
Bullshark

Shoal++

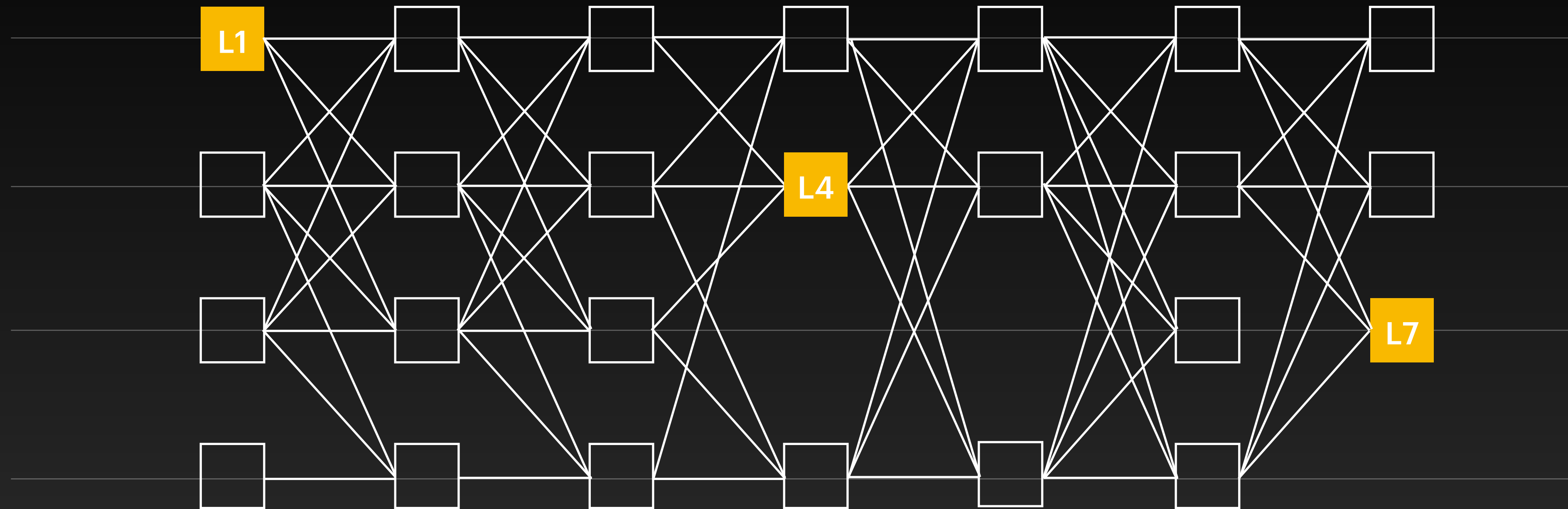
Mysticeti

...

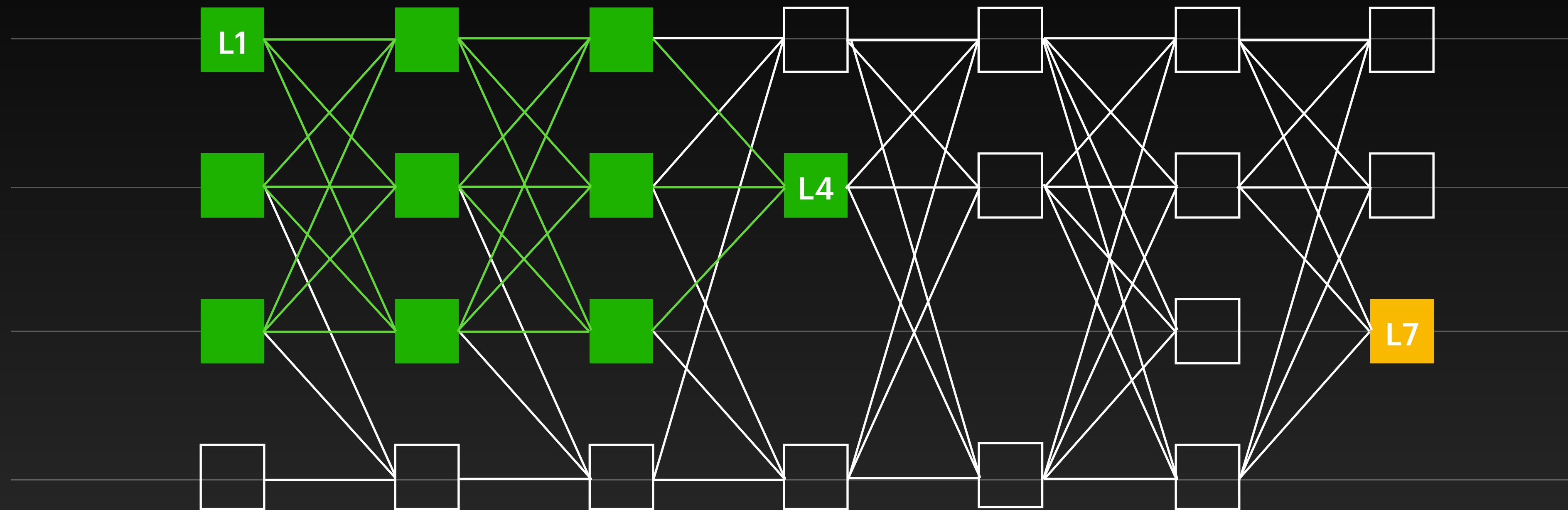
Structured • Uncertified



- Round number
- Author
- Payload (txs)
- **$2f+1$ Parents**
- Signature



- We focus on ordering leaders: L1 L4 L7



- We focus on ordering leaders: L1 L4 L7
- Linearising the sub-DAG is simple

wave 1

wave 2

wave 3

r1

r2

r3

r4

r5

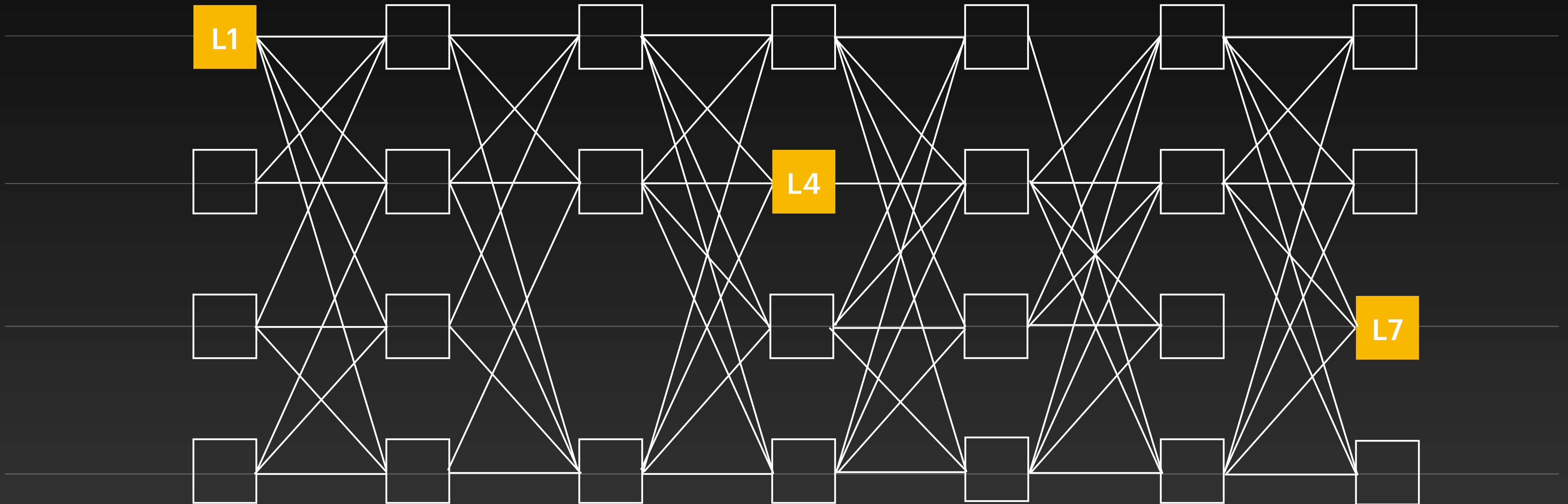
r6

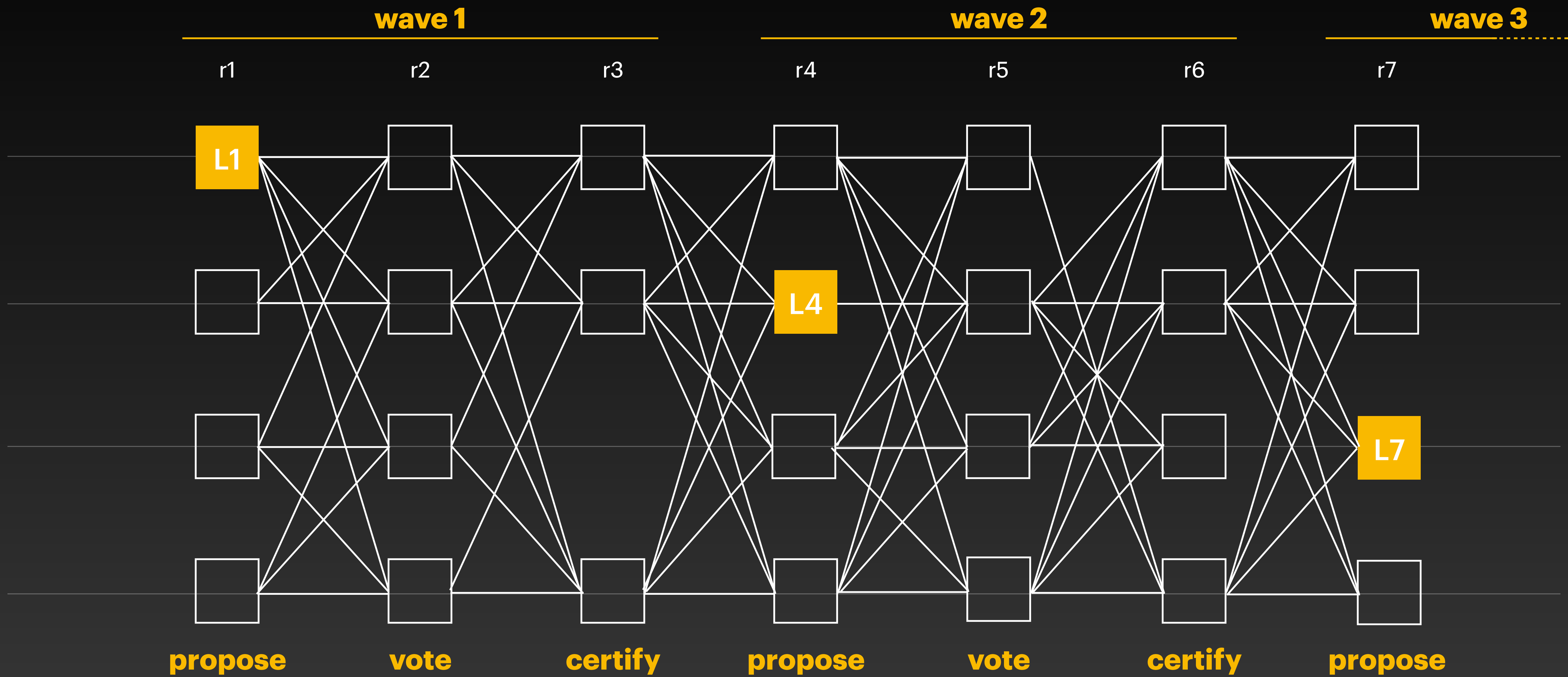
r7

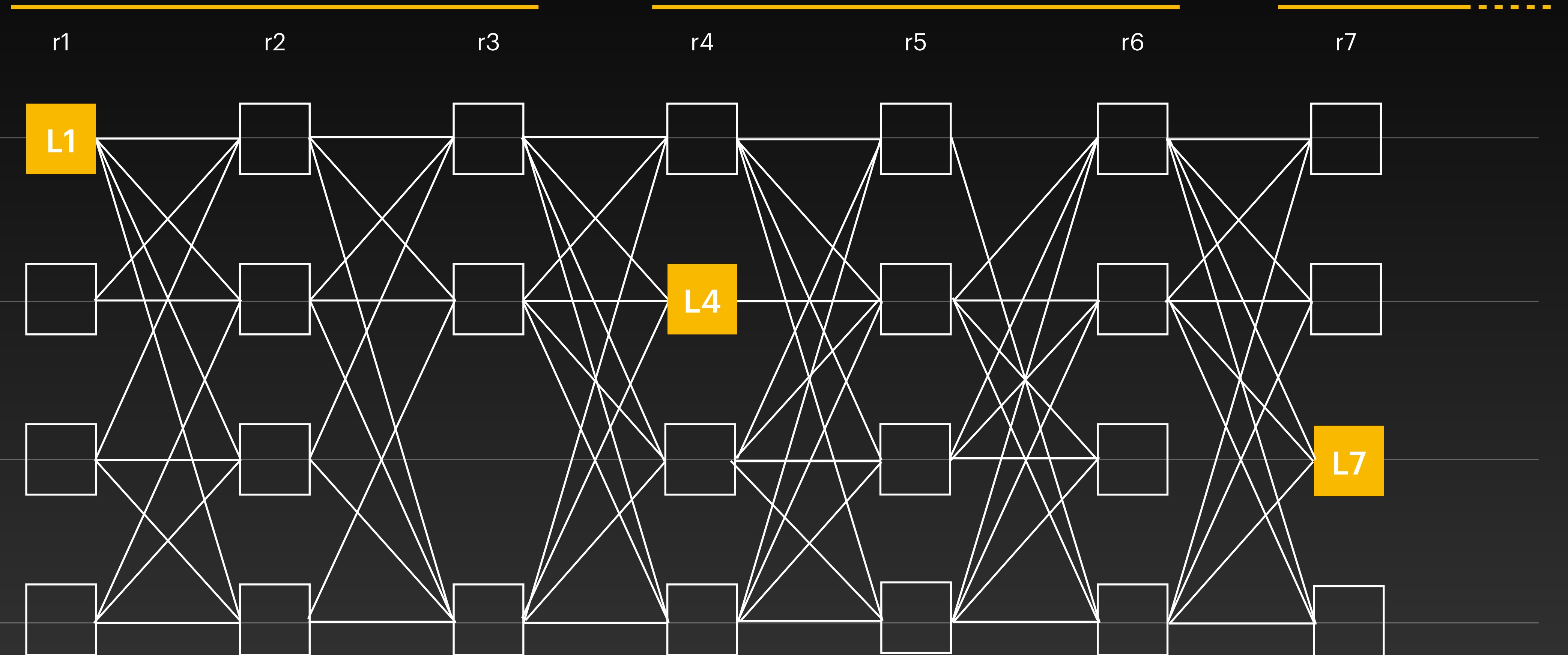
L1

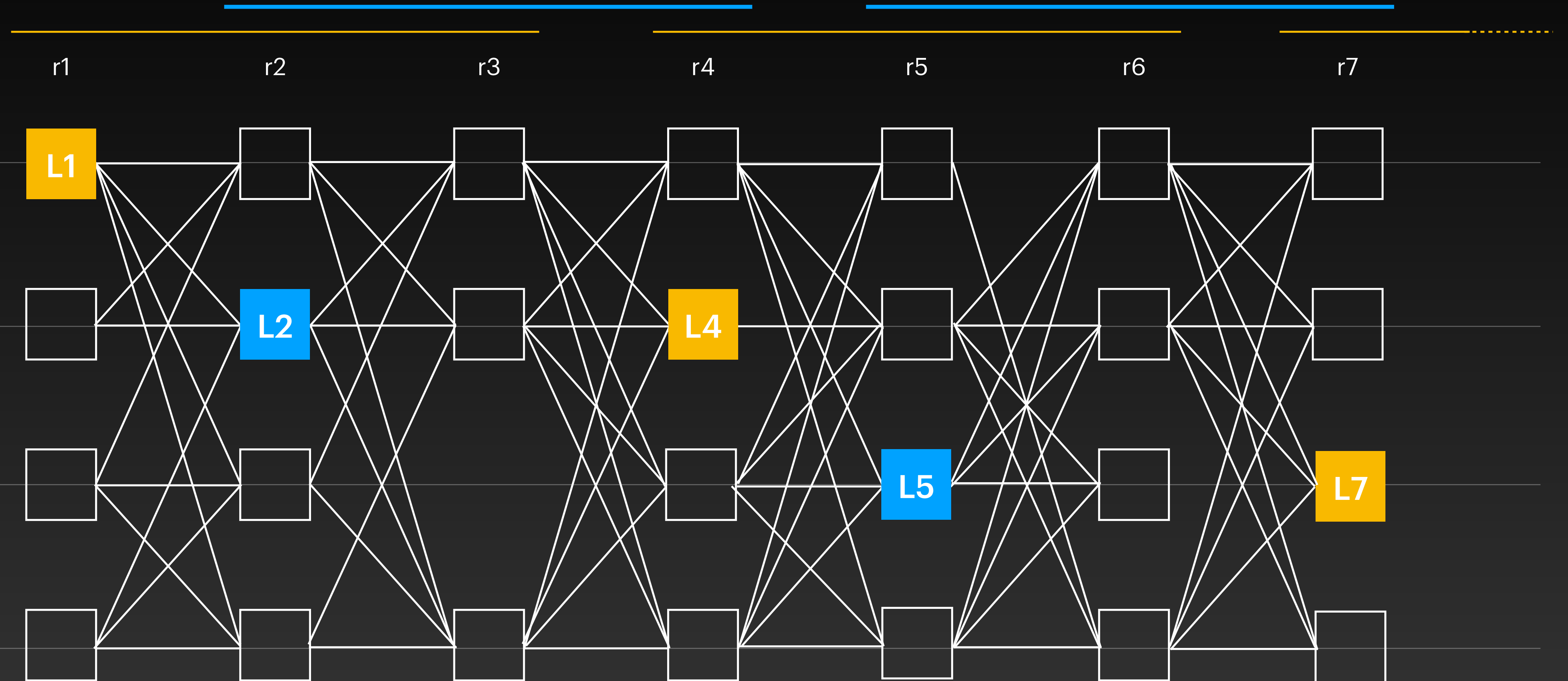
L4

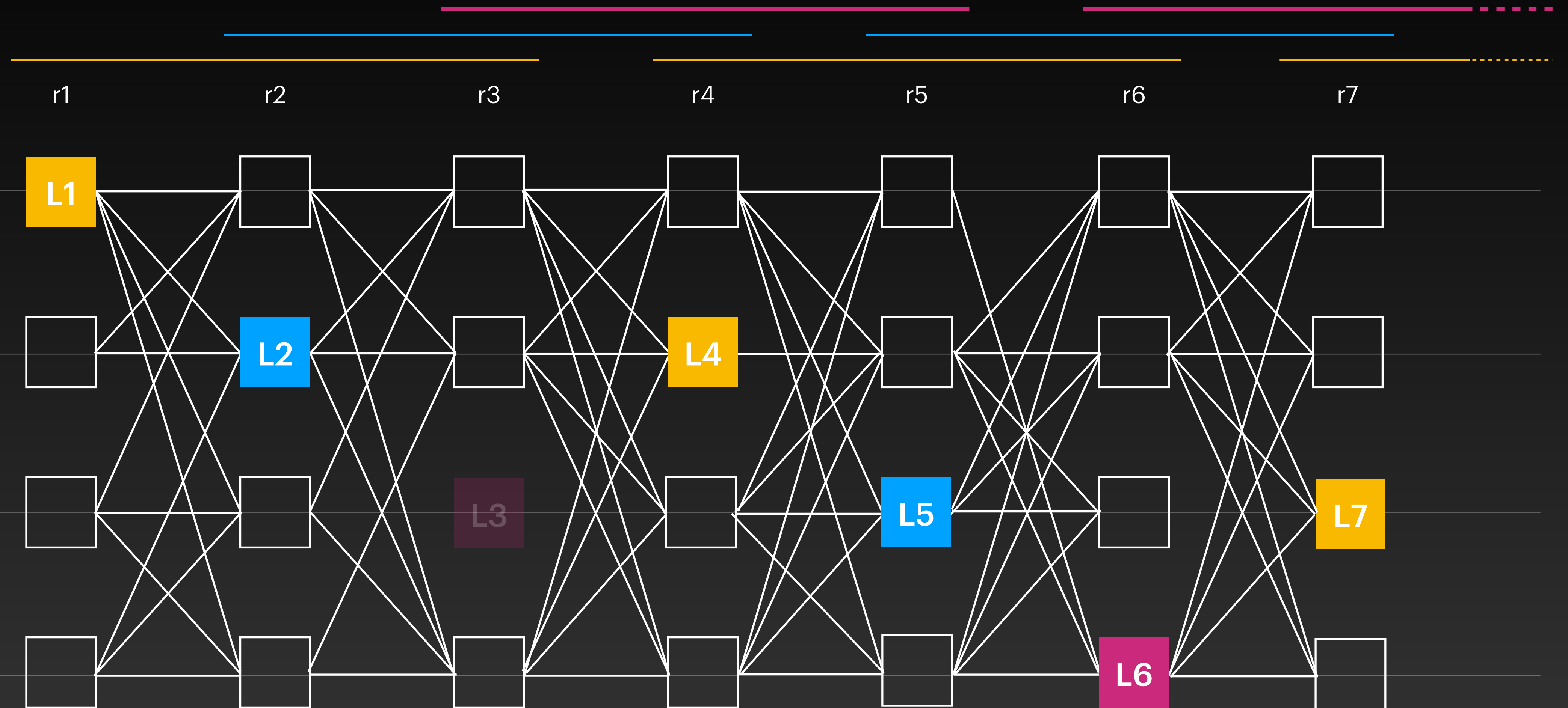
L7

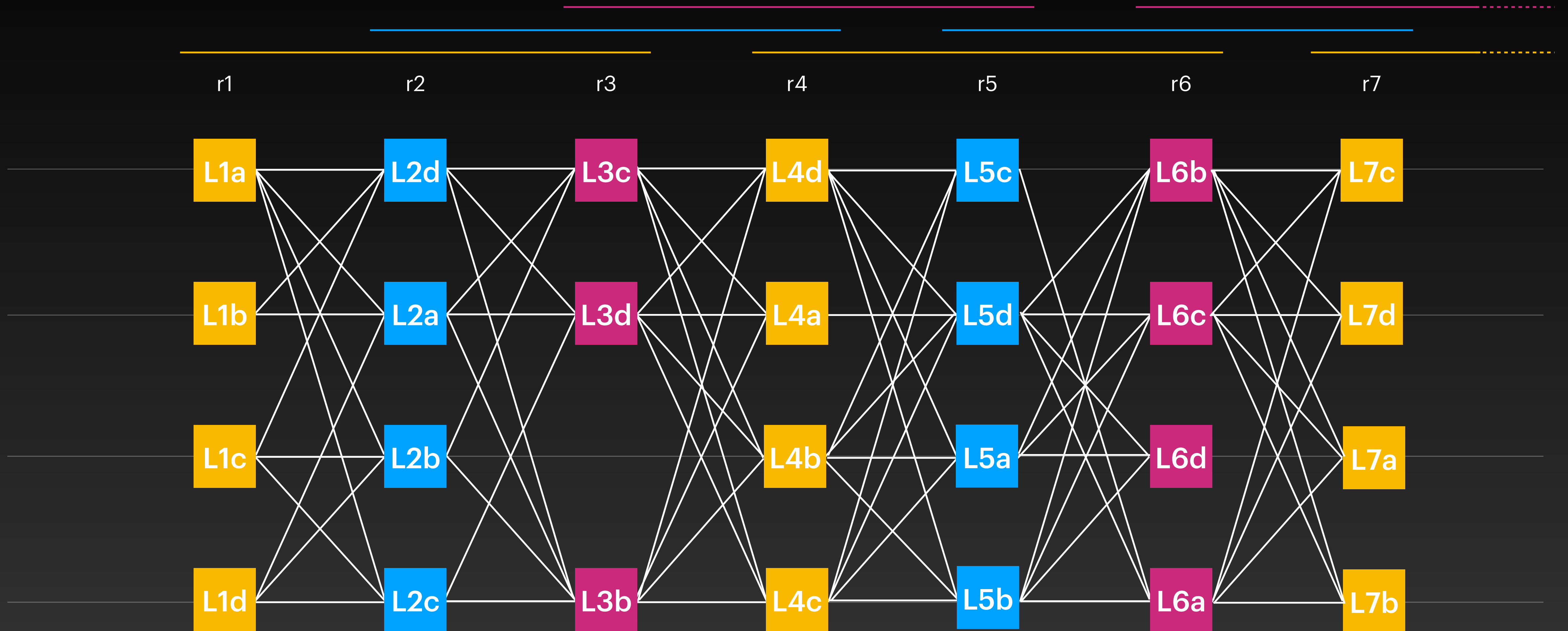


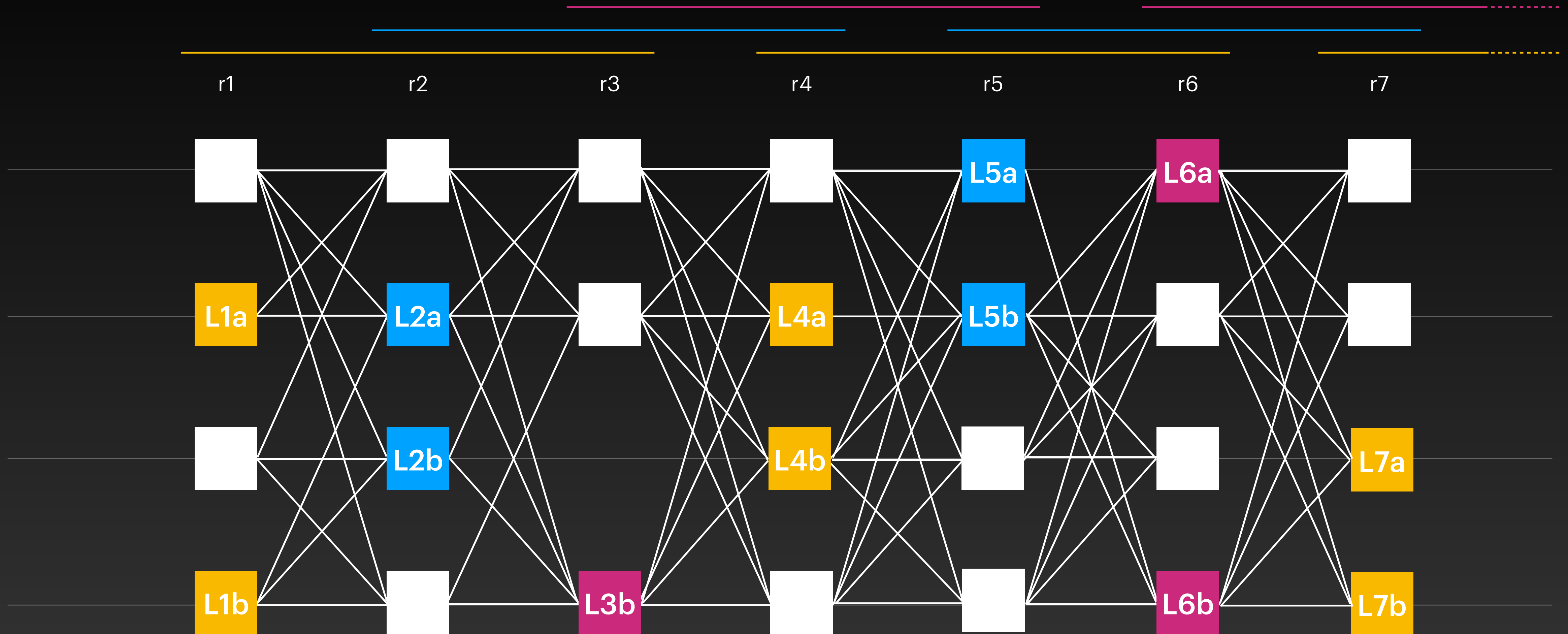


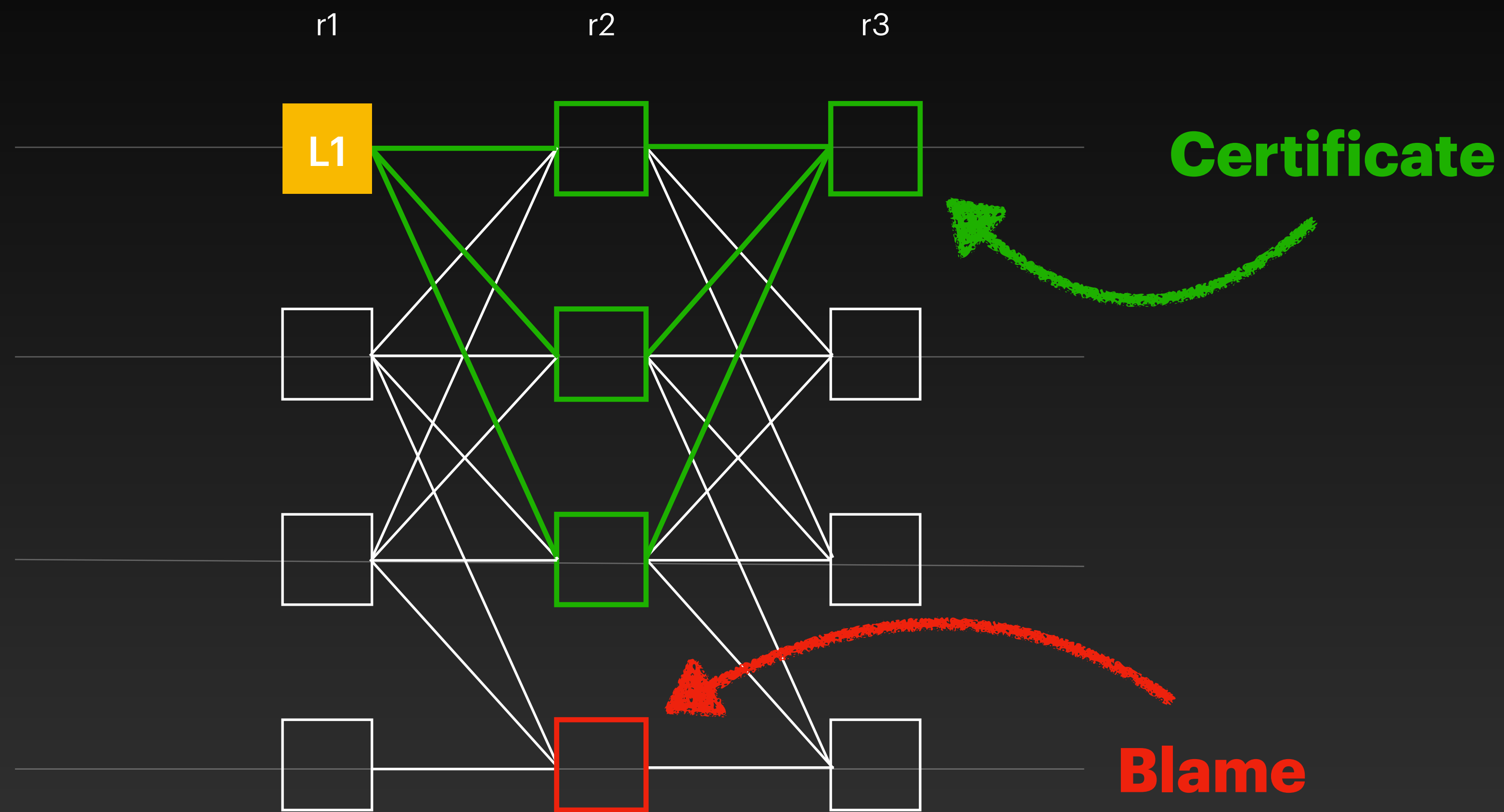












Direct Decision Rule

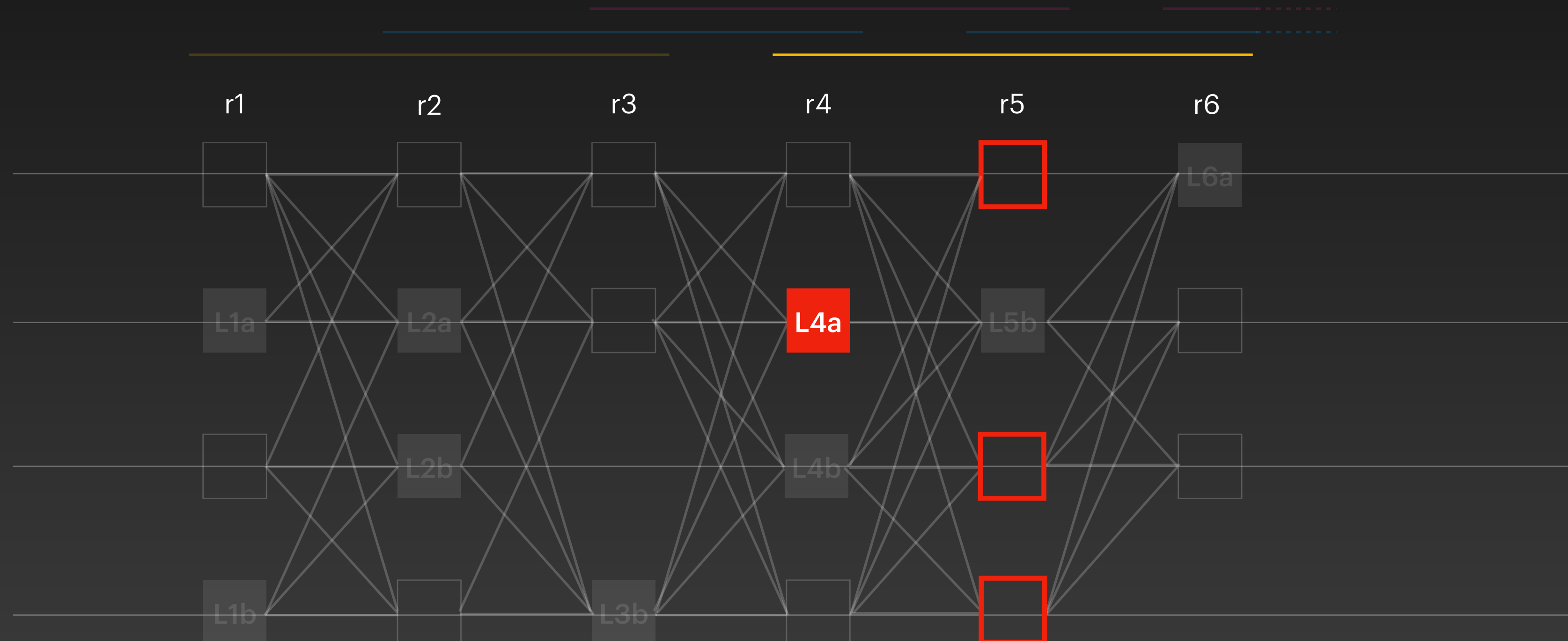
On each leader starting from highest round:

- **Skip** if $2f+1$ blames
- **Commit** if $2f+1$ certificates
- **Undecided** otherwise

Direct Decision Rule

On each leader starting from highest round:

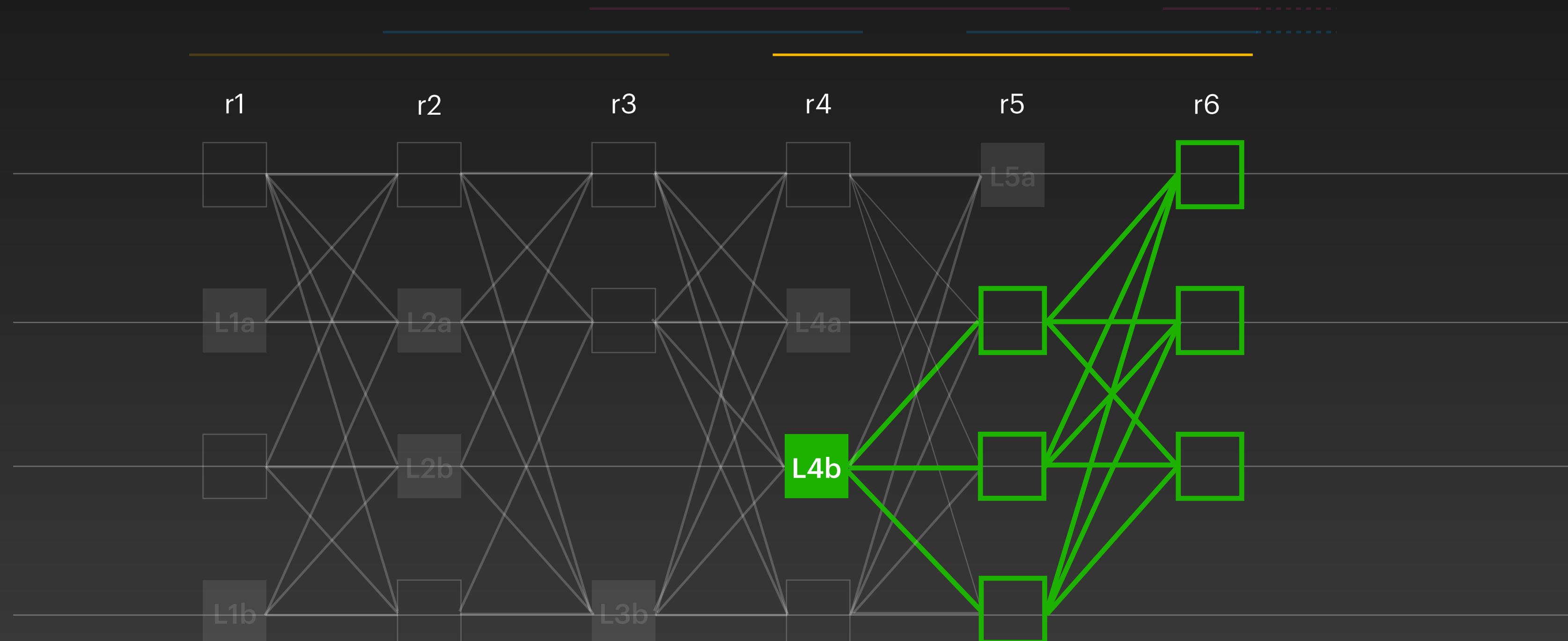
- **Skip** if $2f+1$ blames
- **Commit** if $2f+1$ certificates
- **Undecided** otherwise



Direct Decision Rule

On each leader starting from highest round:

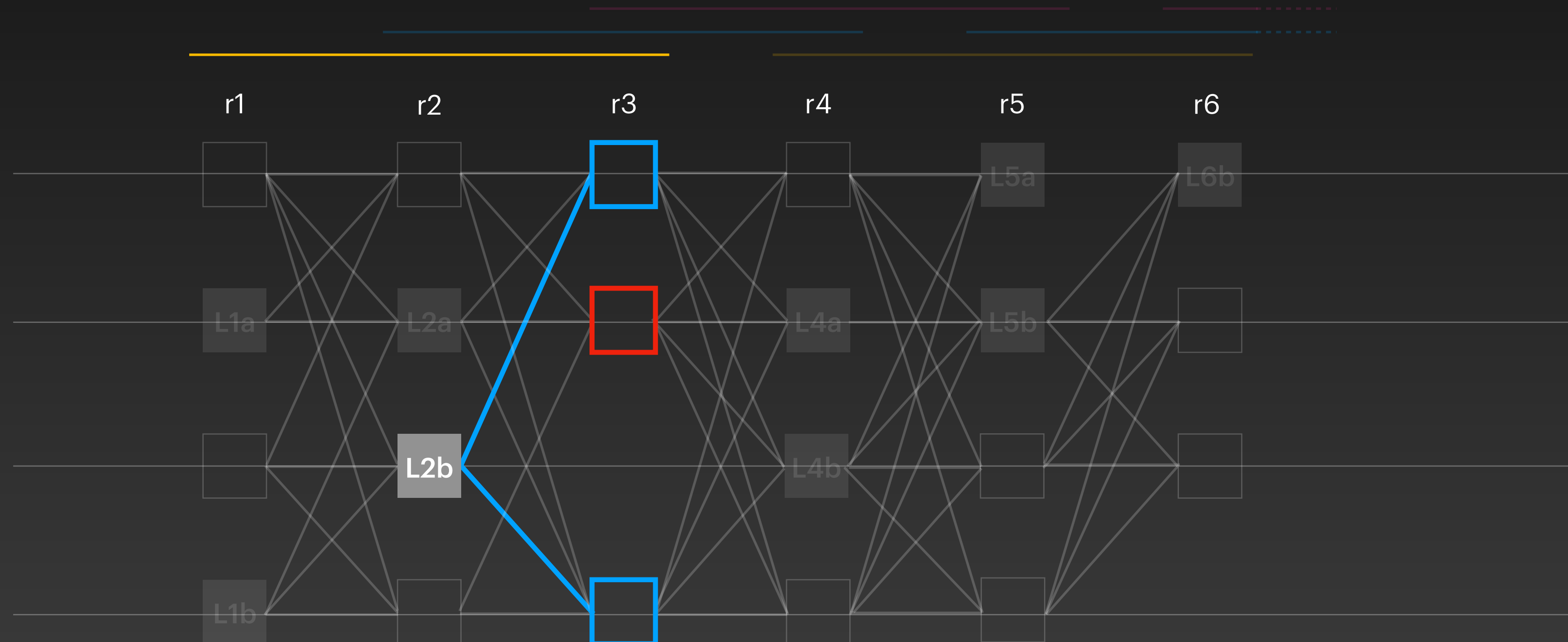
- **Skip** if $2f+1$ blames
- **Commit** if $2f+1$ certificates
- **Undecided** otherwise



Direct Decision Rule

On each leader starting from highest round:

- **Skip** if $2f+1$ blames
- **Commit** if $2f+1$ certificates
- **Undecided** otherwise

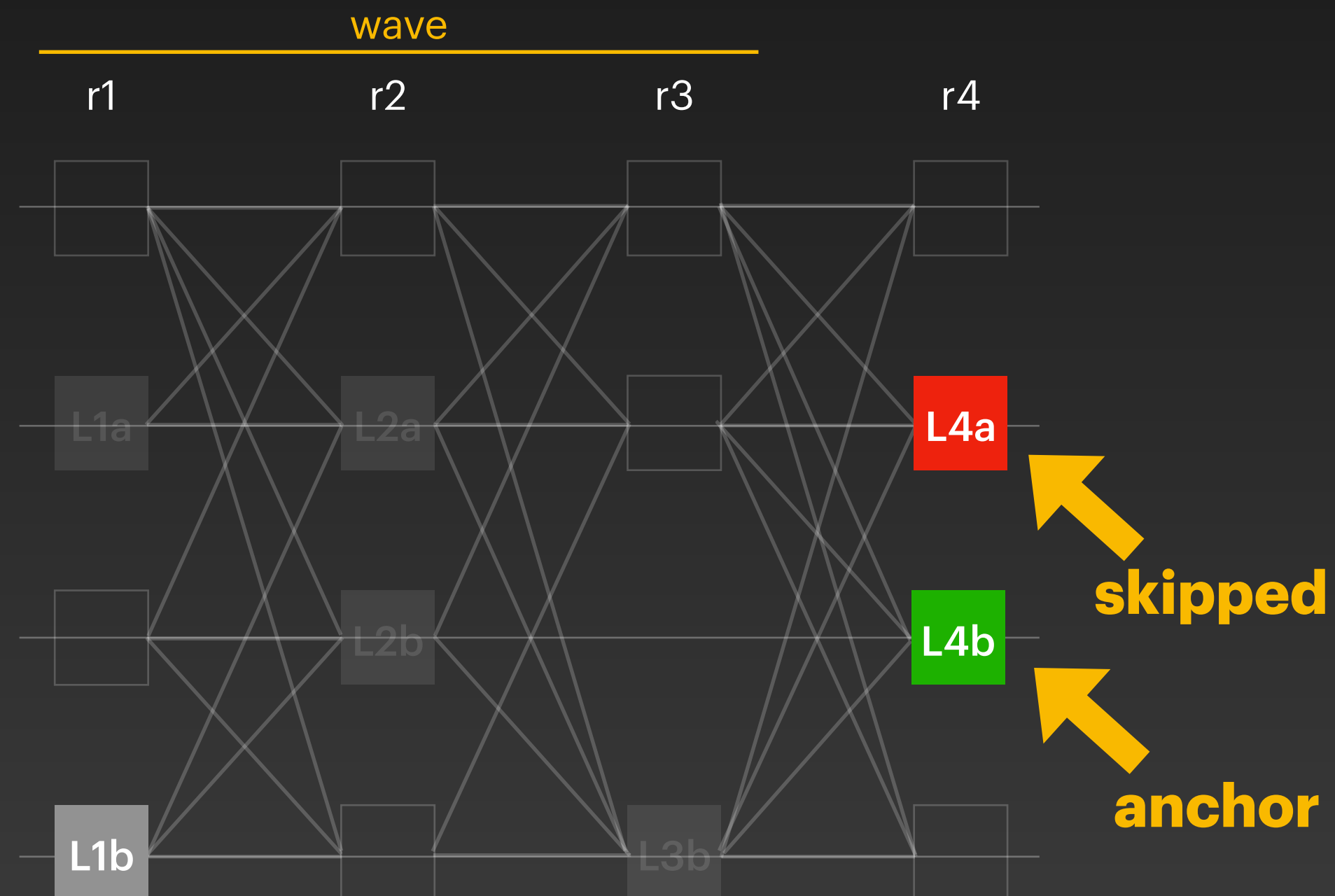


Indirect Decision Rule

Indirect Decision Rule

1. Find Anchor

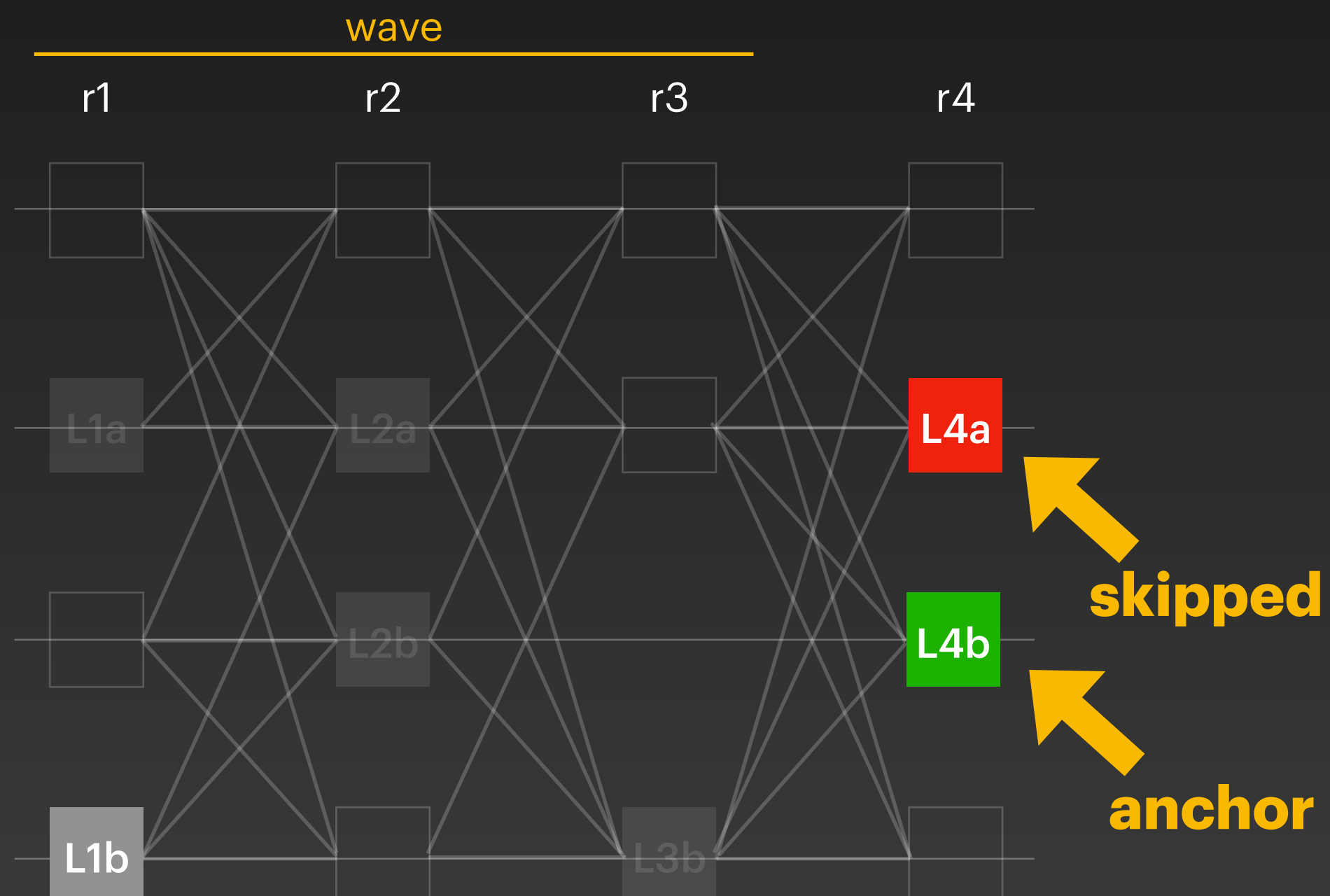
- First block with round $> r+2$ that is **Commit** or **Undecided**



Indirect Decision Rule

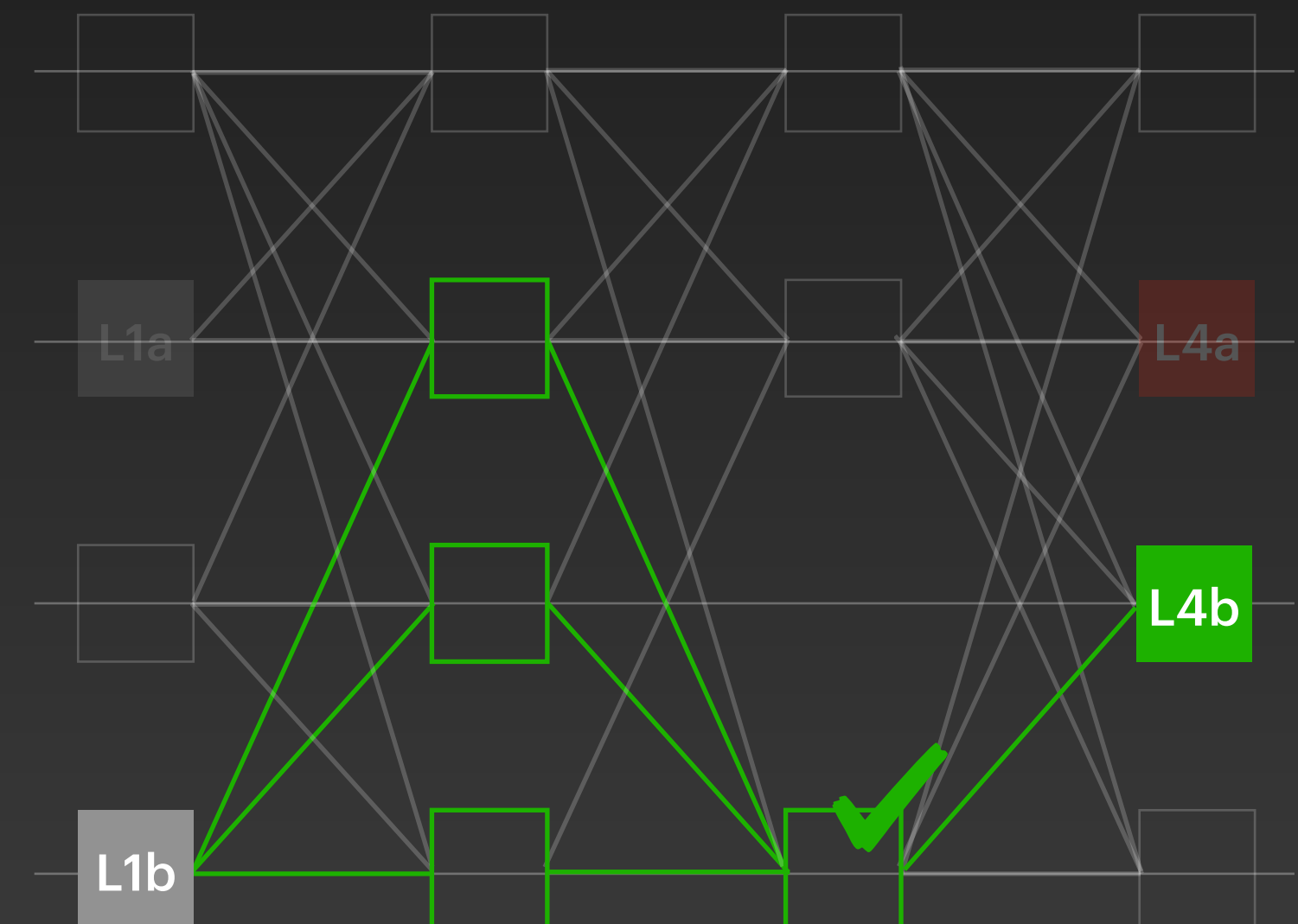
1. Find Anchor

- First block with round $> r+2$ that is **Commit** or **Undecided**

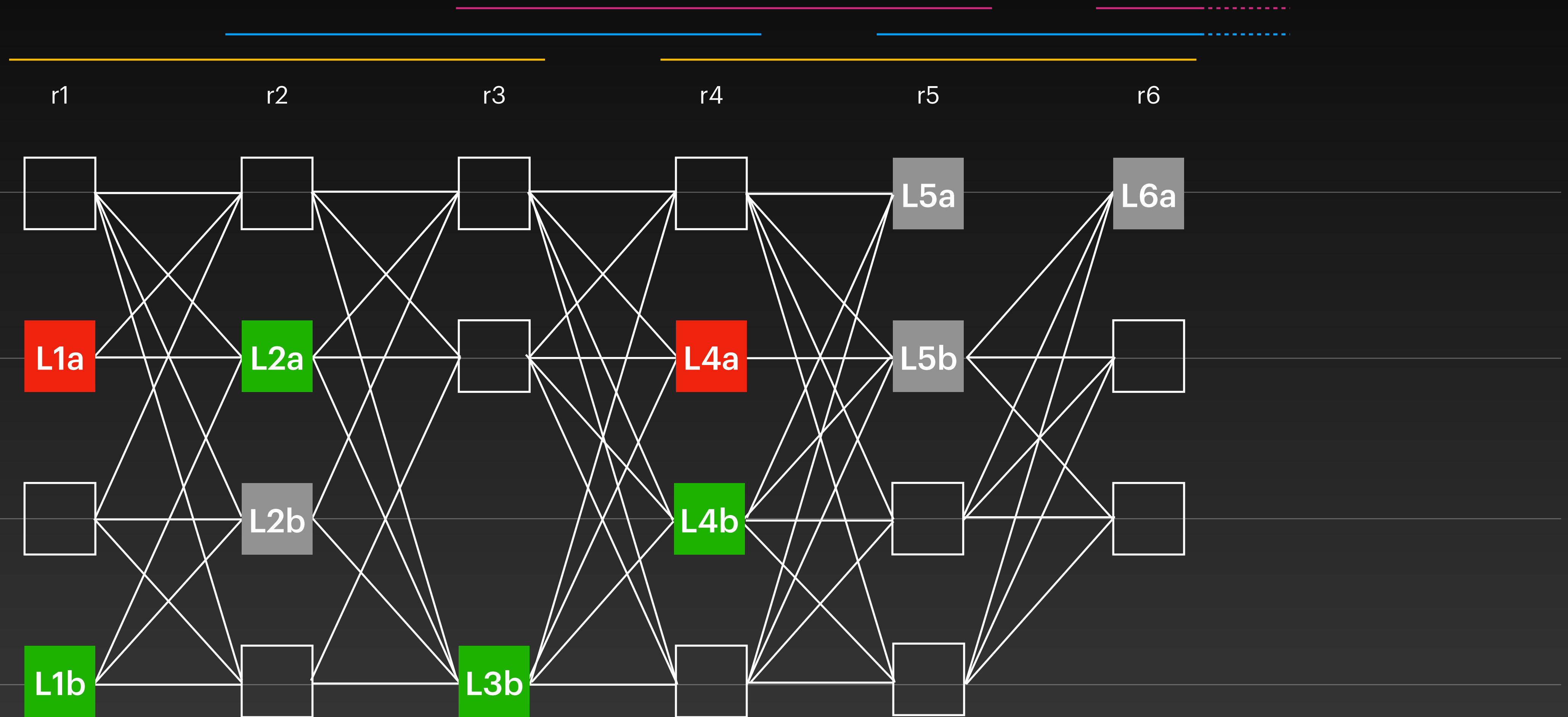


2. Certified link

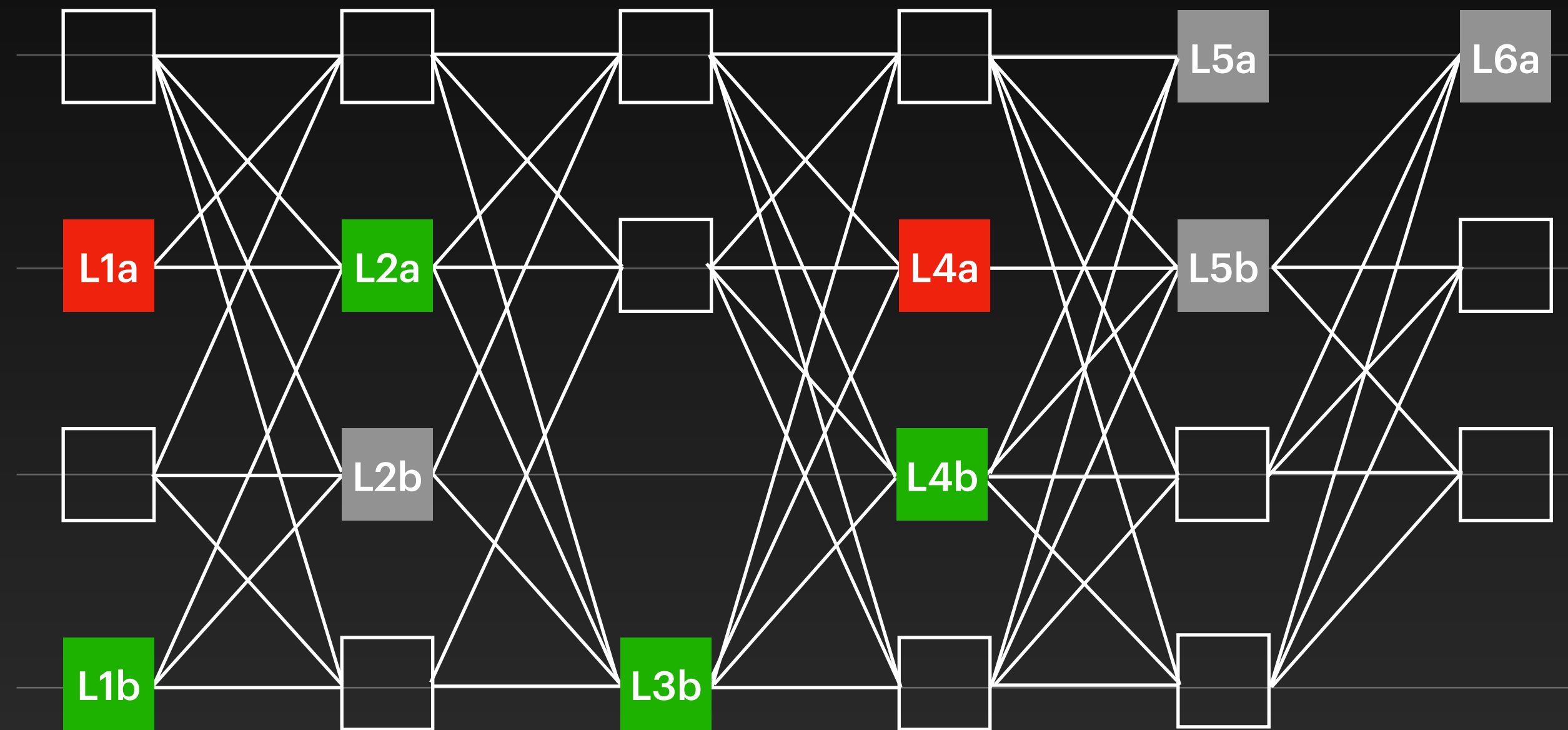
- Commit** if
 $B \leftrightarrow \text{certified link} \leftrightarrow A$
otherwise **Skip**



Current status



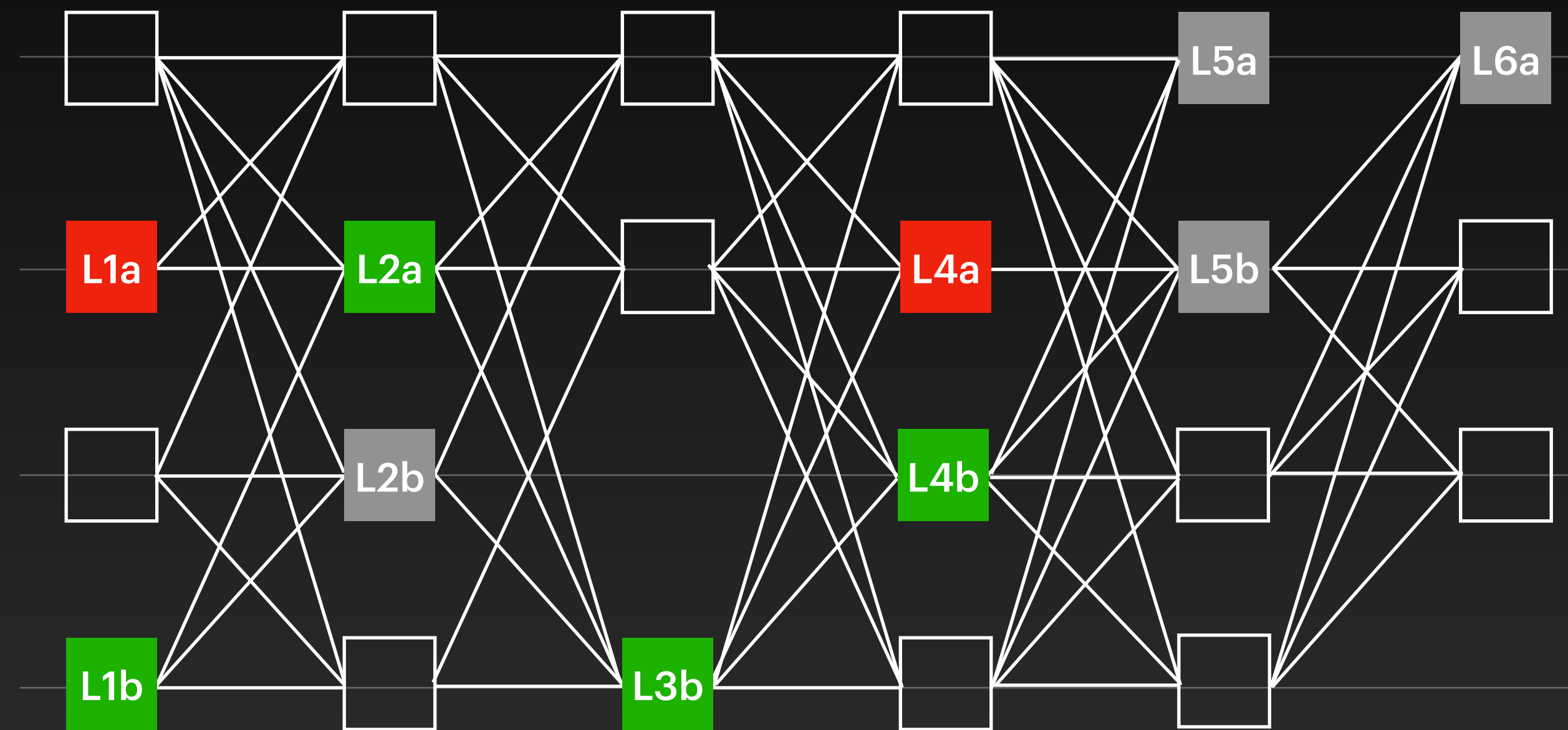
Take all leaders in order



leaders sequence:



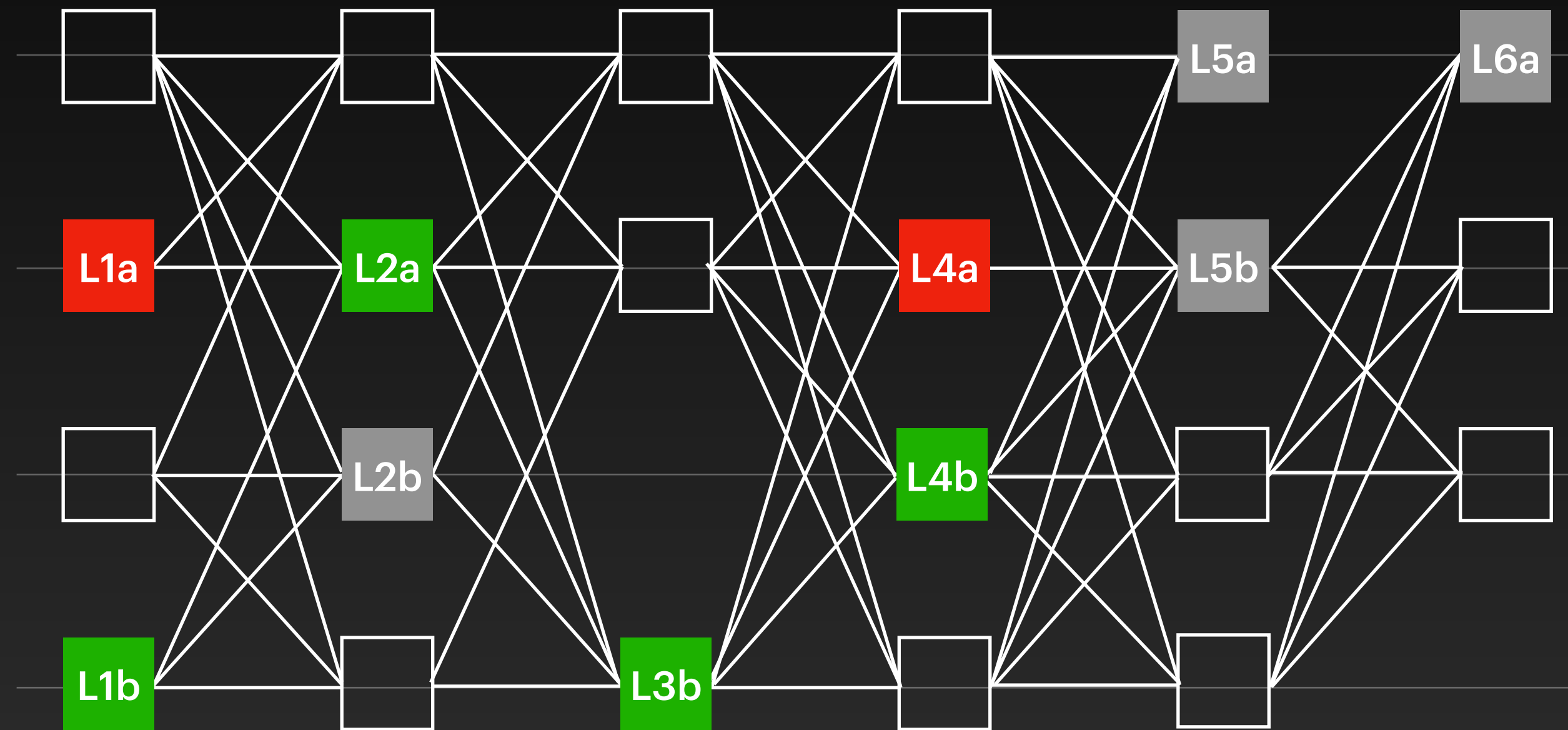
Stop at the first undecided leader



leaders sequence:



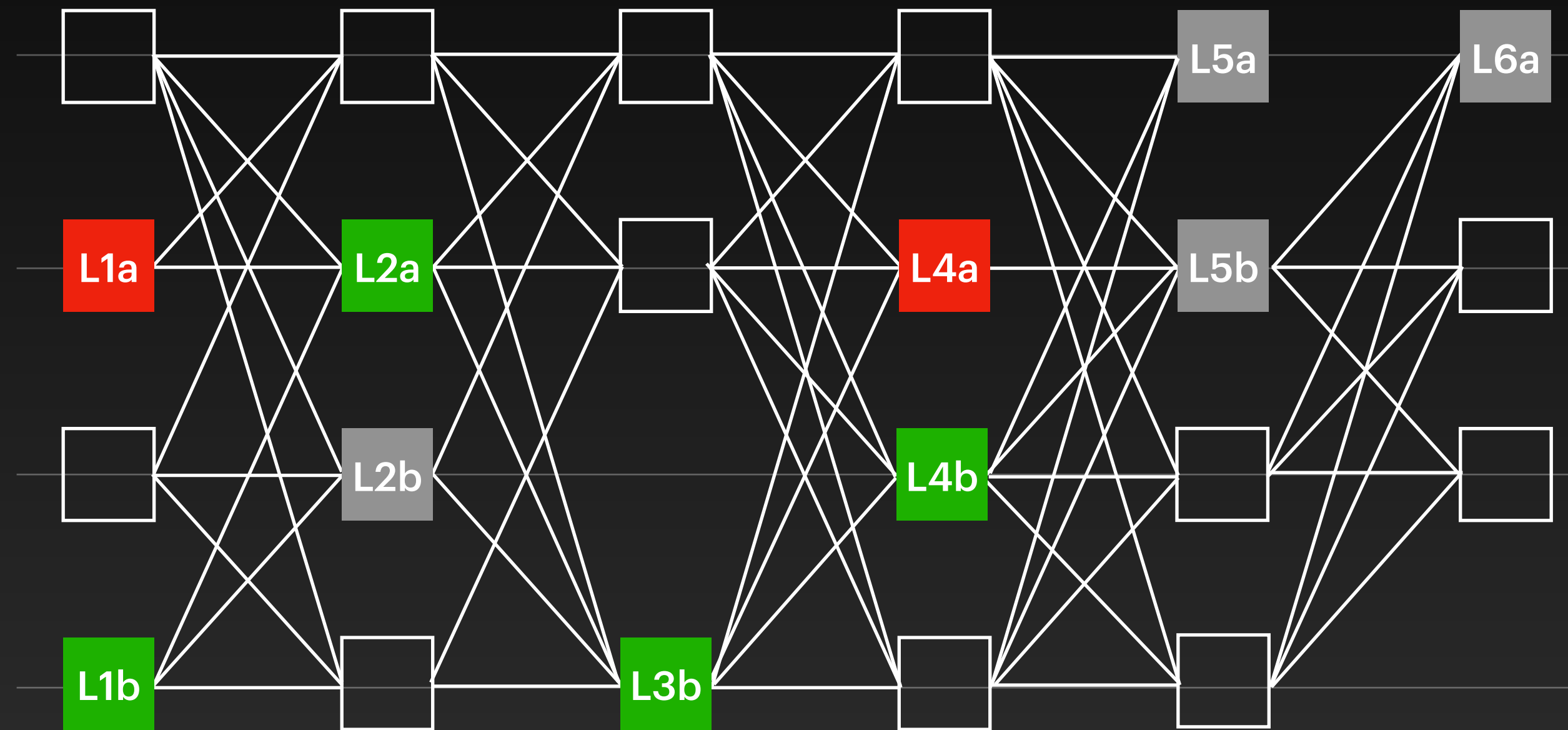
Remove skipped leaders



leaders sequence:



Final leader sequence

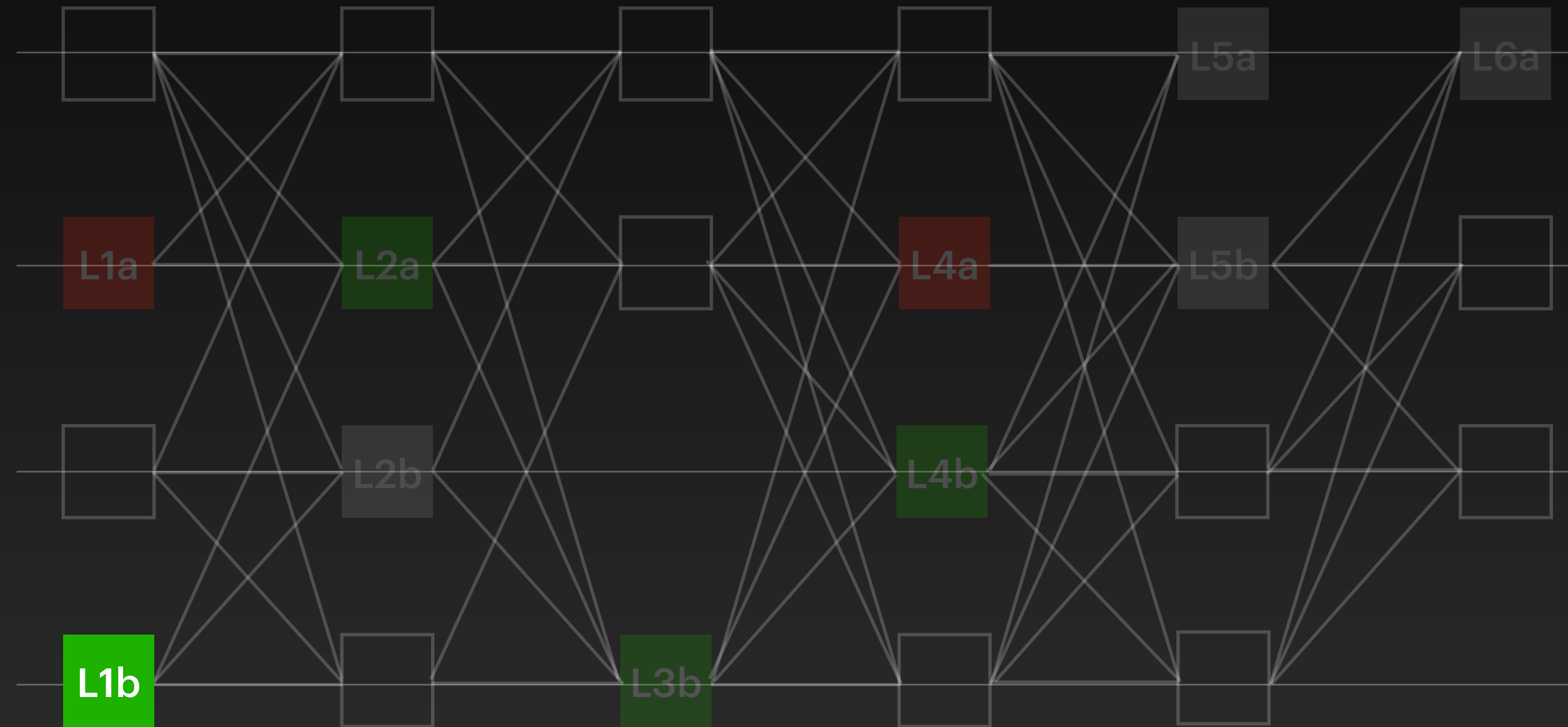


leaders sequence:

L1b

L2a

Commit sub-dag



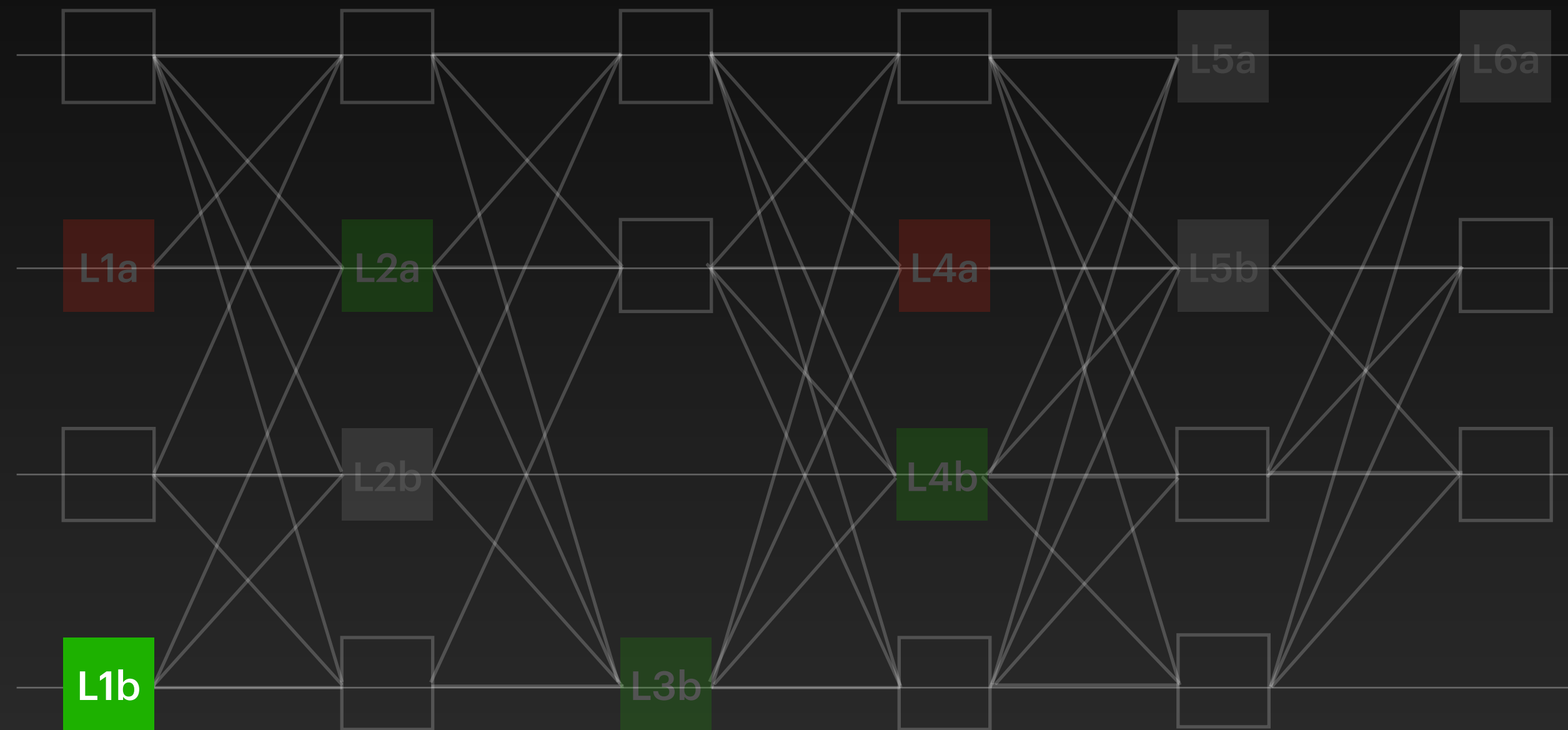
leaders sequence:

L1b

L2a

output sequence:

Commit sub-dag



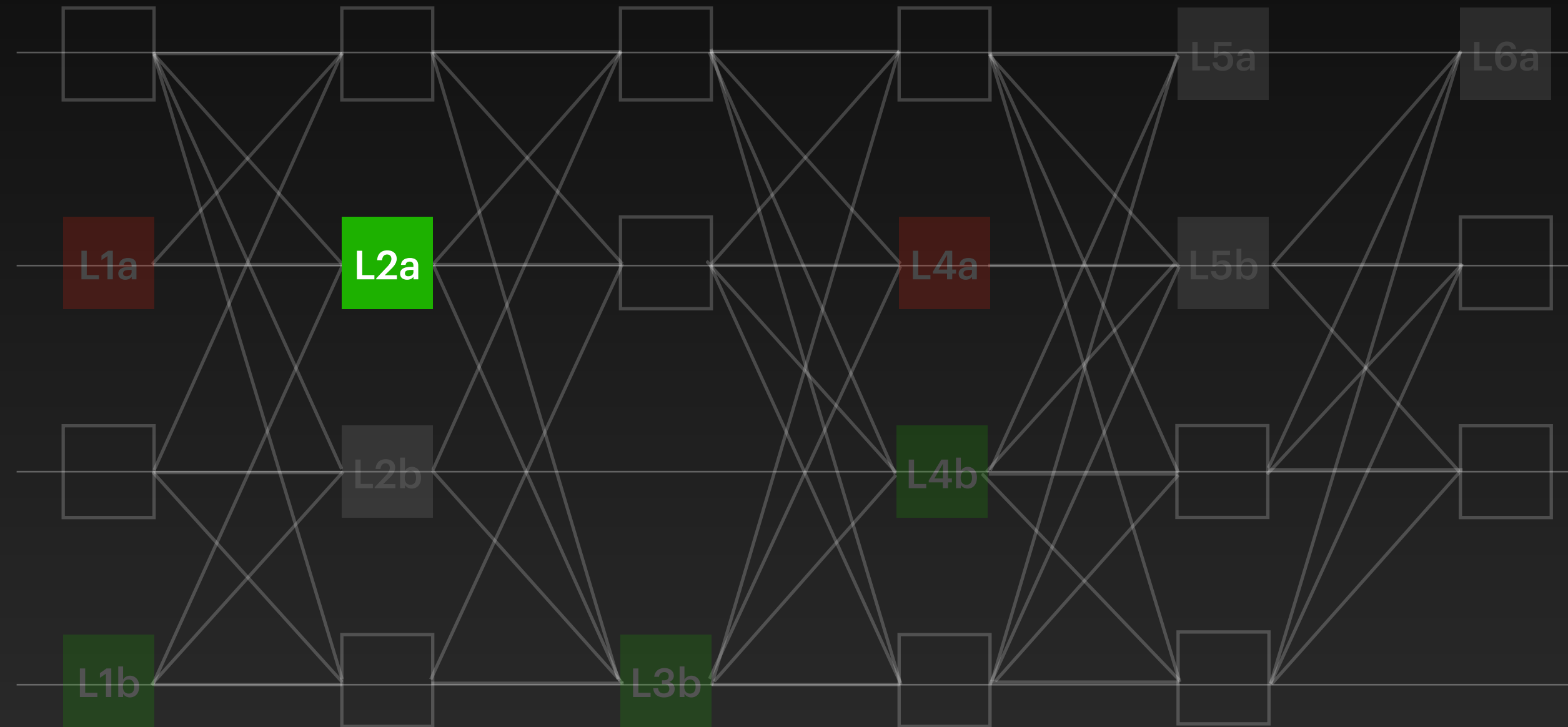
leaders sequence:

L2a

output sequence:

L1b

Commit sub-dag



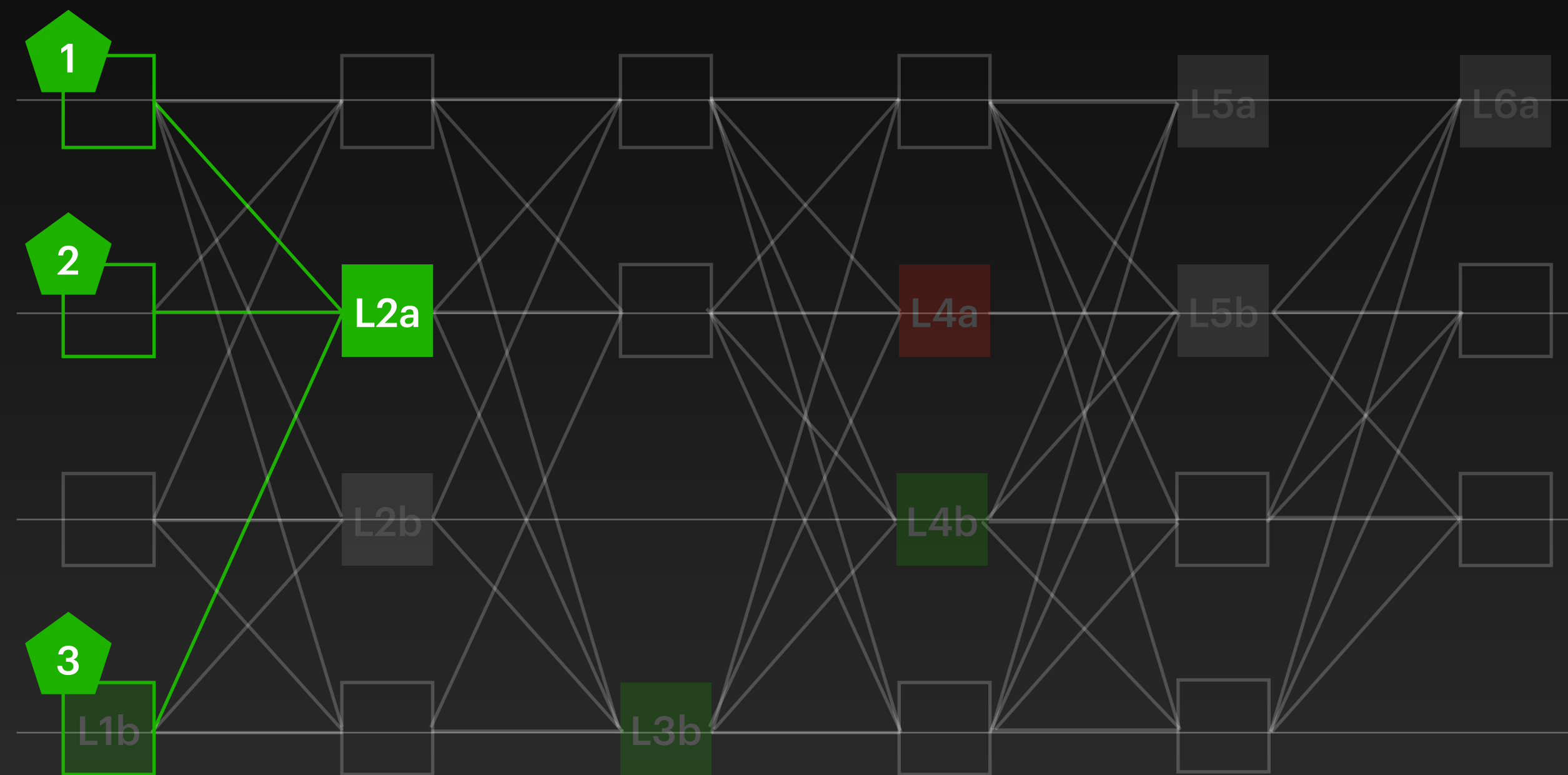
leaders sequence:

L2a

output sequence:

L1b

Commit sub-dag



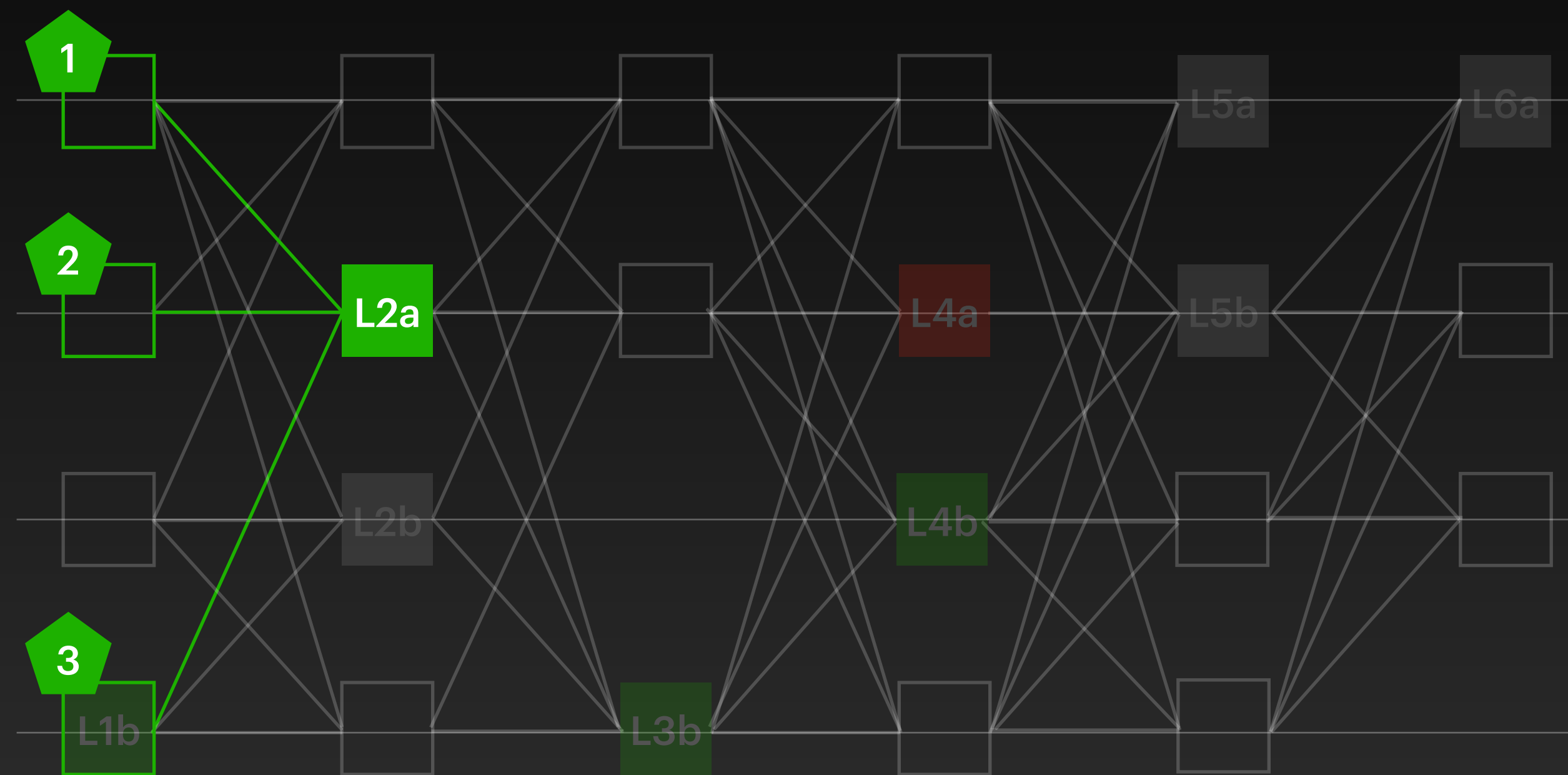
leaders sequence:

L2a

output sequence:

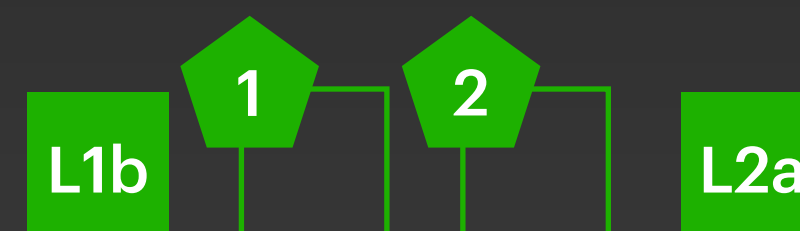
L1b

Commit sub-dag

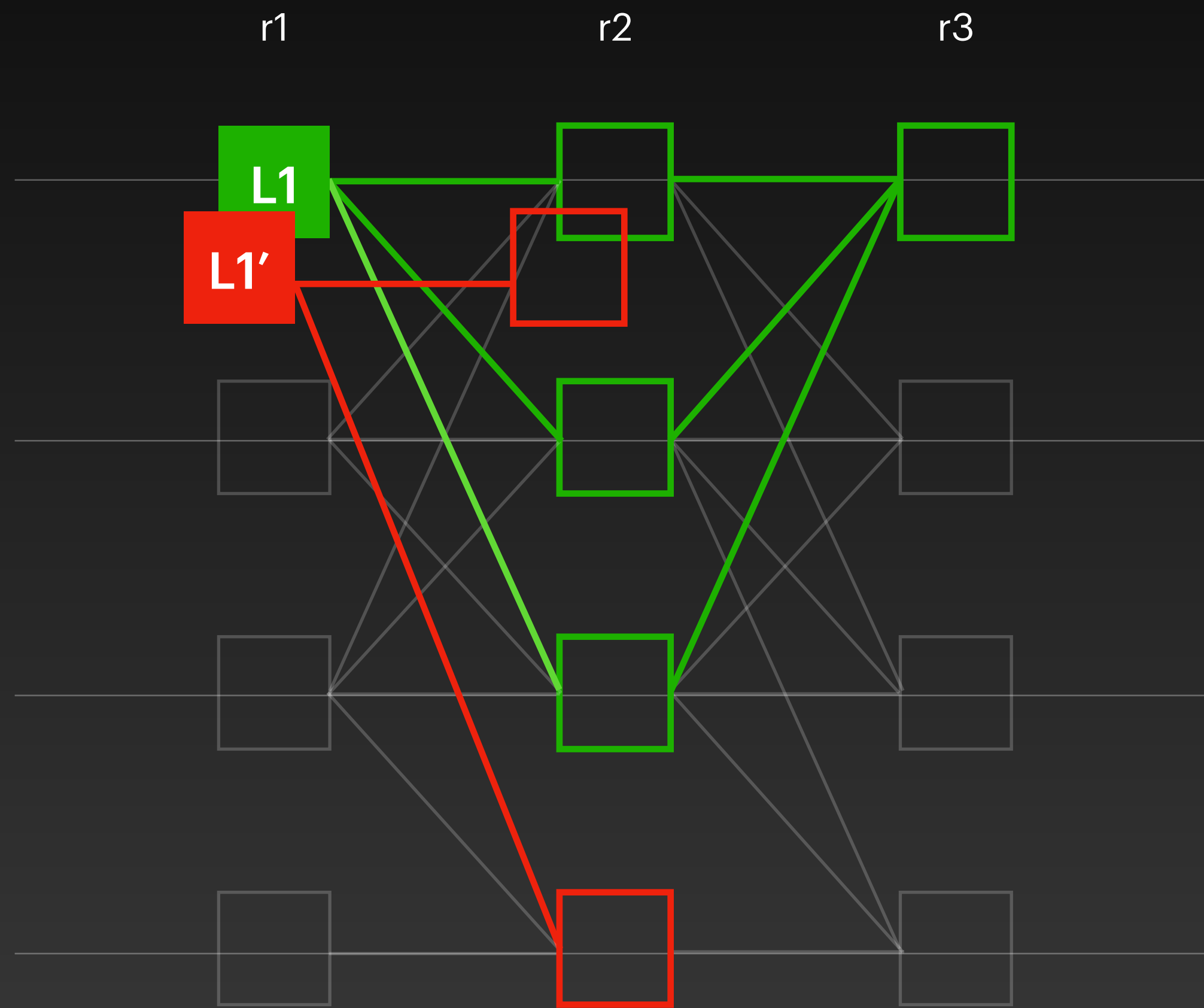


leaders sequence:

output sequence:

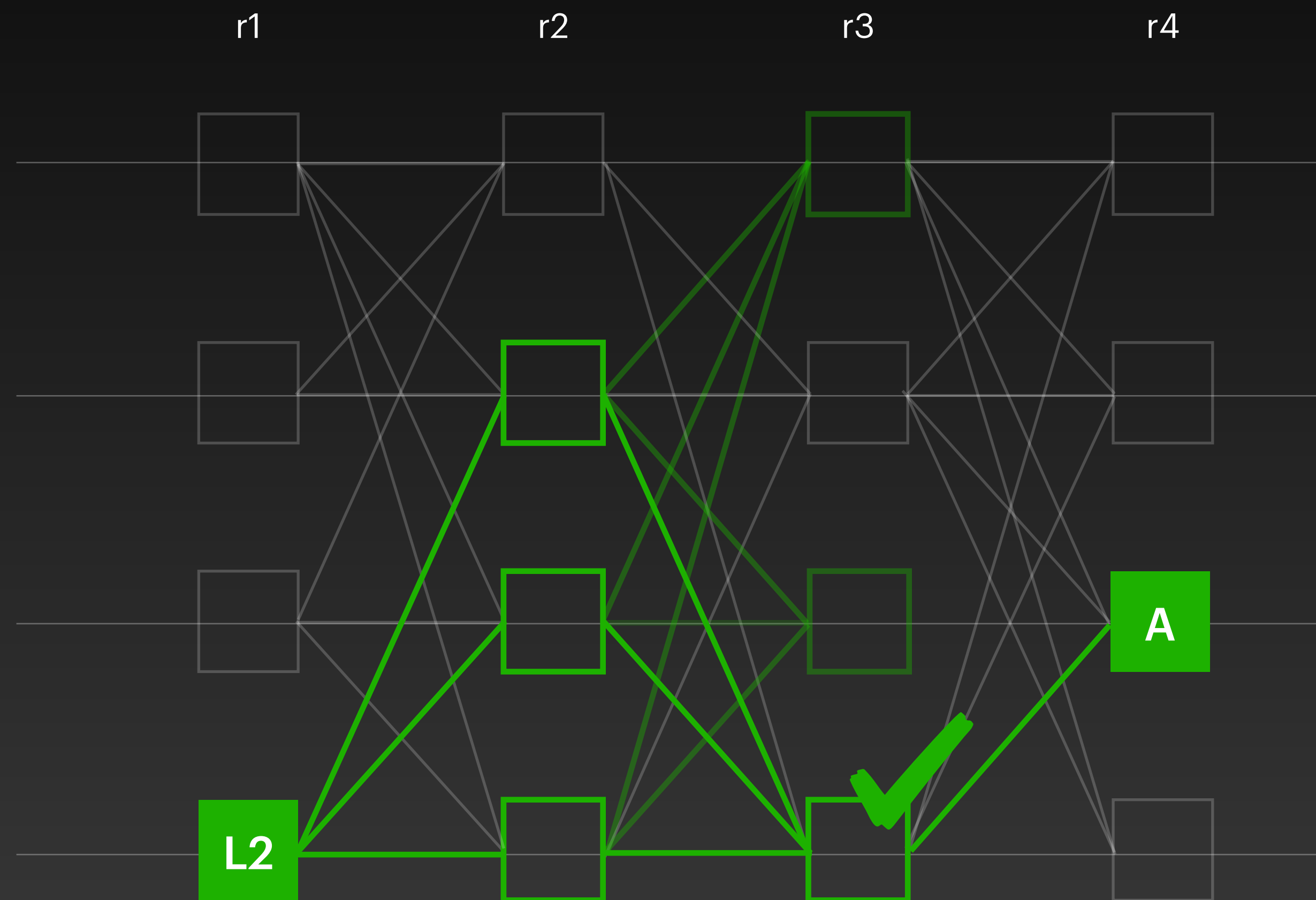


Safety



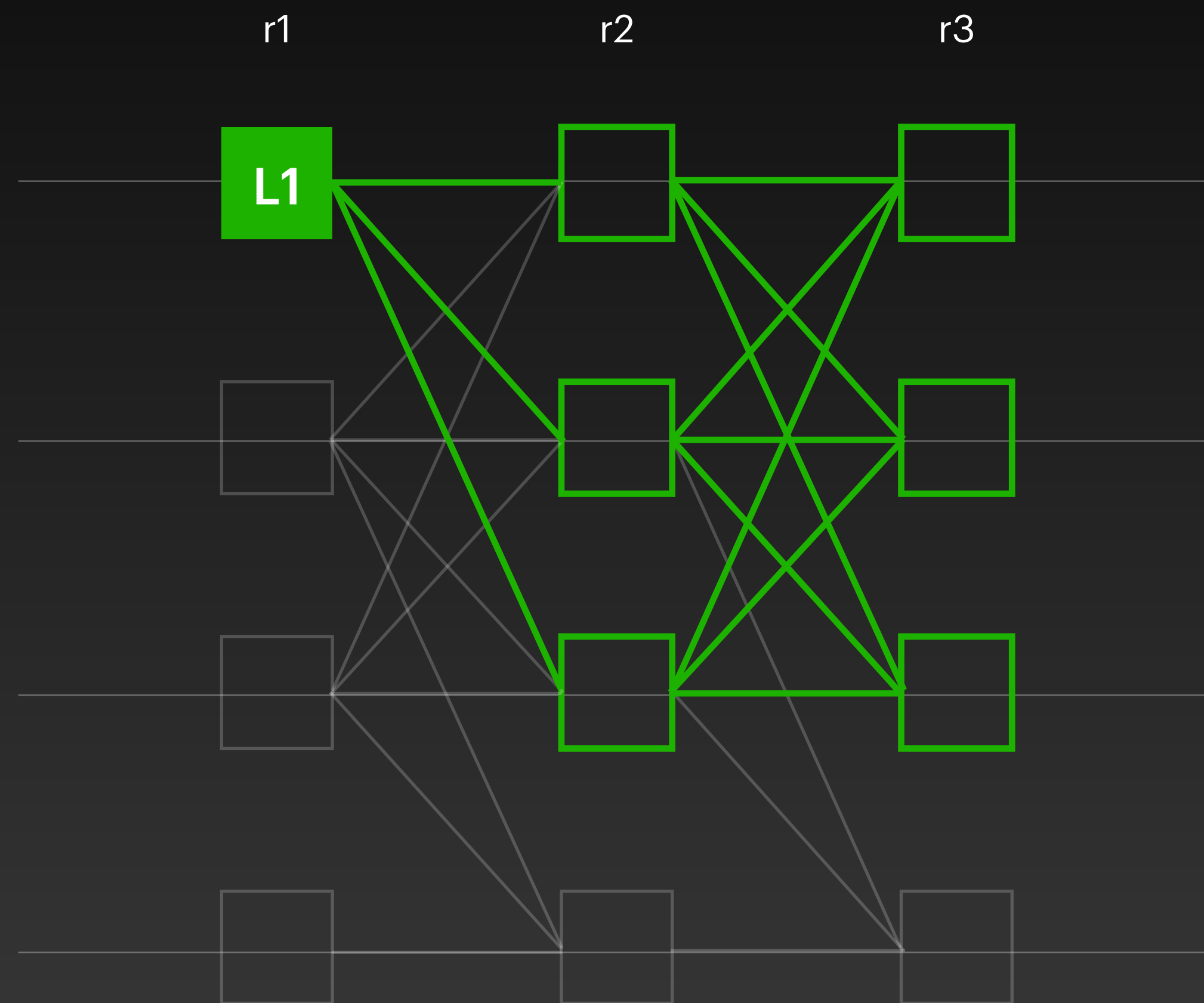
- At most **L1** or **L1'** can have a certificate pattern (quorum intersection)

Safety



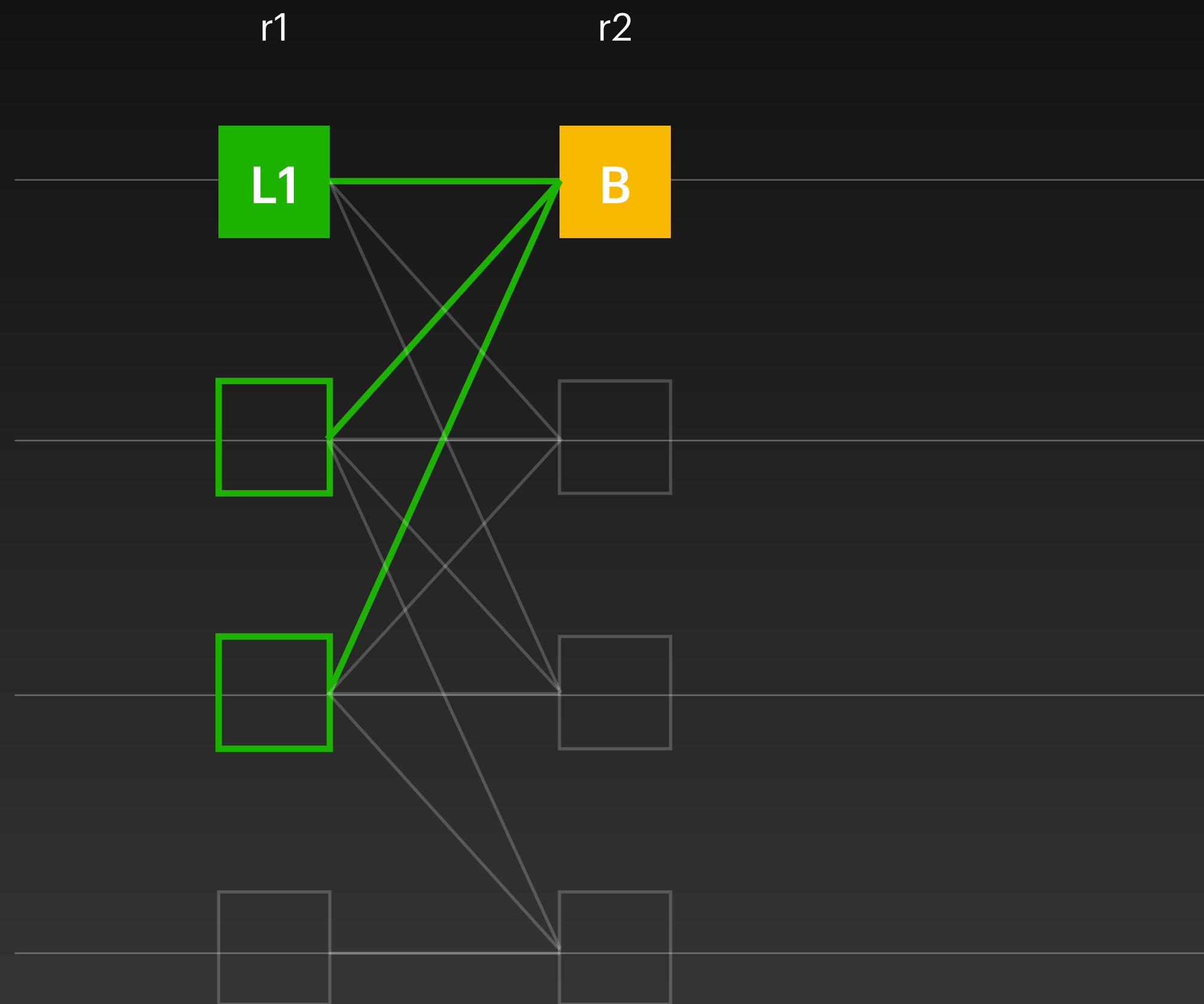
- At most **L1** or **L1'** can have a certificate pattern (quorum intersection)
- If **L2** has $2f+1$ certificate patterns, **A** always has a certified link to **L2**

Liveness



- After GST, the direct decision rule **commits** a block

Liveness



- After GST, the direct decision rule **commits** a block
- Leader timeout: **B** waits for $2f+1$ parents + 250 ms

**We are getting limited returns on
algorithmic improvements of BFT consensus**

controversial

High resource utilisation: throughput

Protocols commit often: latency

Engineering caught up with research

Change the assumptions (vs classic $3f+1$)

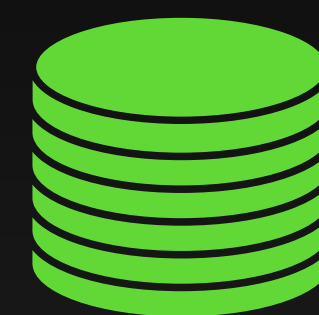
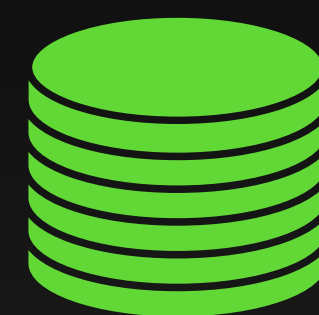
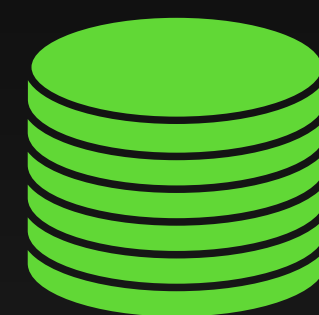
Specialised chains (vs one chain to rule them all)

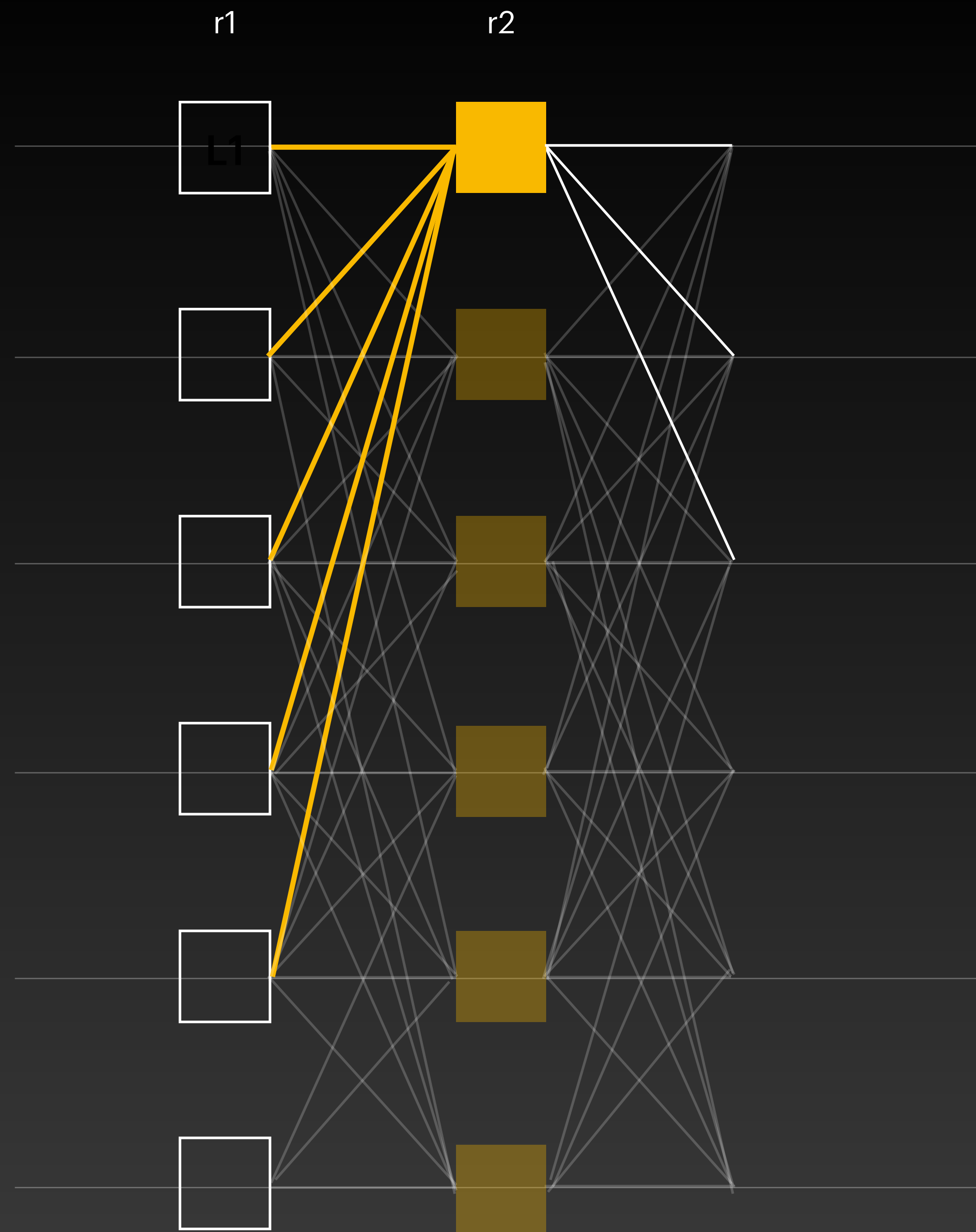
Other components are bottleneck (network, execution, etc)

$$5f+1$$

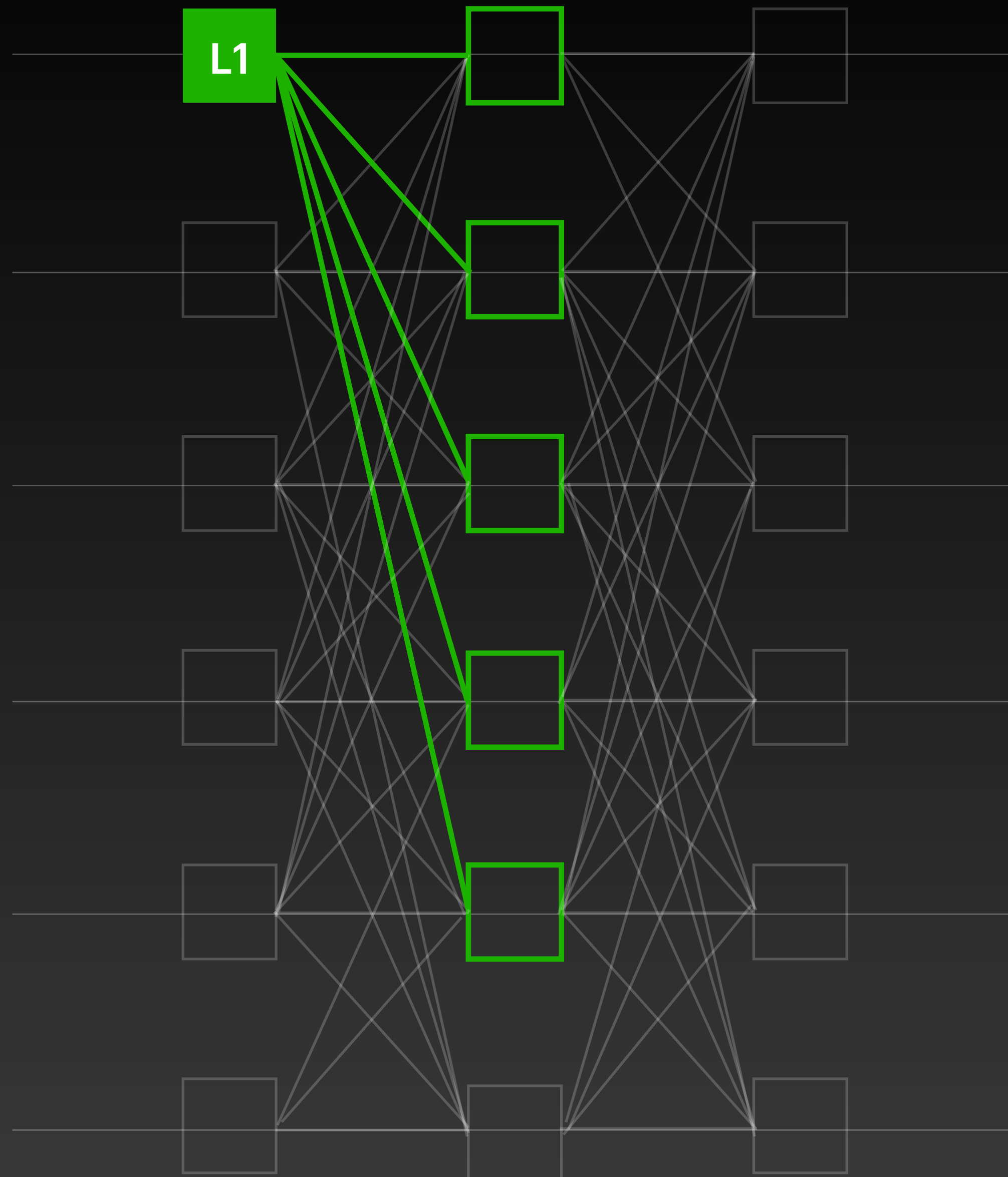


> 4/5

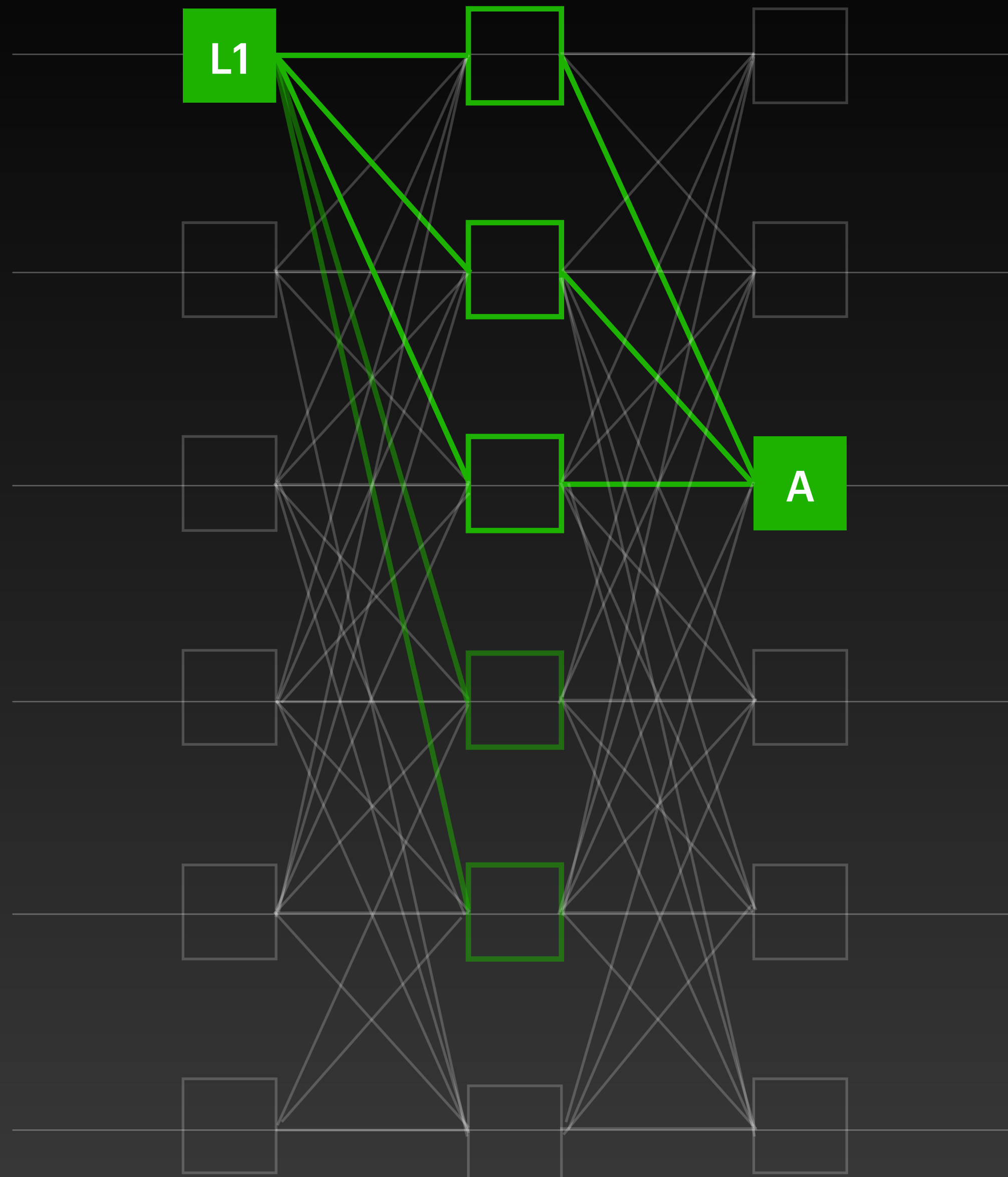




- Round number
- Author
- Payload (txs)
- **4f+1 Parents**
- Signature



- **Wavelength: 2**
- **Direct commit:** At least **$4f+1$** votes for **L1**



- **Wavelength: 2**
- **Direct commit:** At least $4f+1$ votes for **L1**
- **Indirect commit:** At least $2f+1$ links from **A** to **L1**

Tradeoffs

$3f+1$

More rounds to commit (3) • Smaller quorums ($2f+1$) • Balanced safety-liveness

$5f+1$

Fewer rounds to commit (2) • Larger quorums ($4f+1$) • Lower safety-liveness

Tradeoffs

$3f+1$

More rounds to commit (3) • Smaller quorums ($2f+1$) • Balanced safety-liveness

$5f+1$

Fewer rounds to commit (2) • **Larger quorums ($4f+1$)** • Lower safety-**liveness**

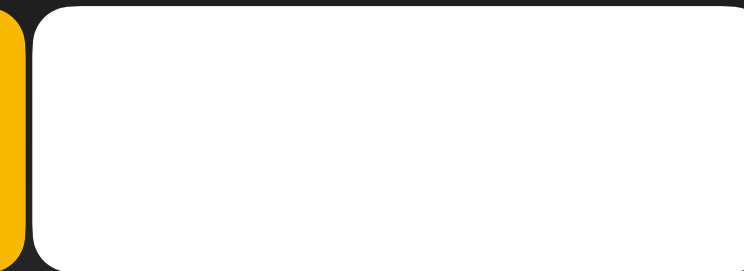
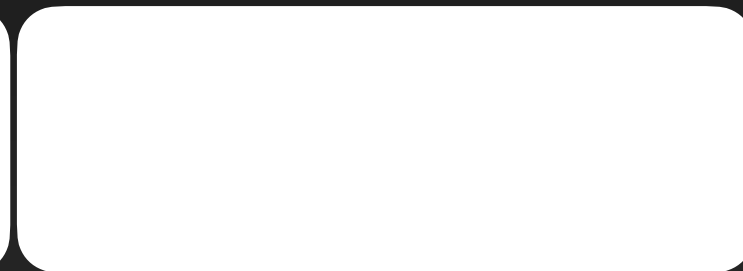
**Load-
induced**

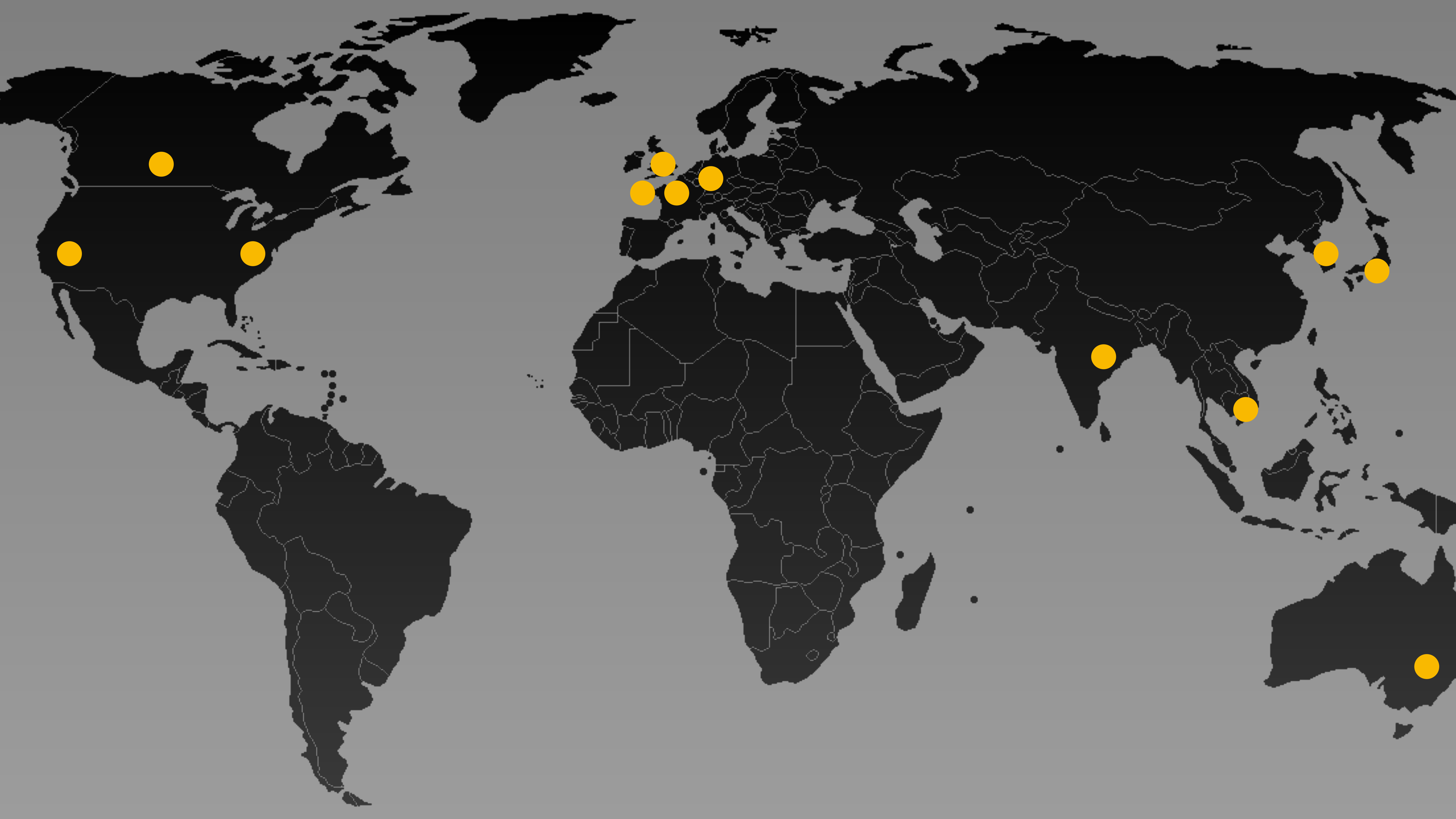
**Block-
Inclusion**

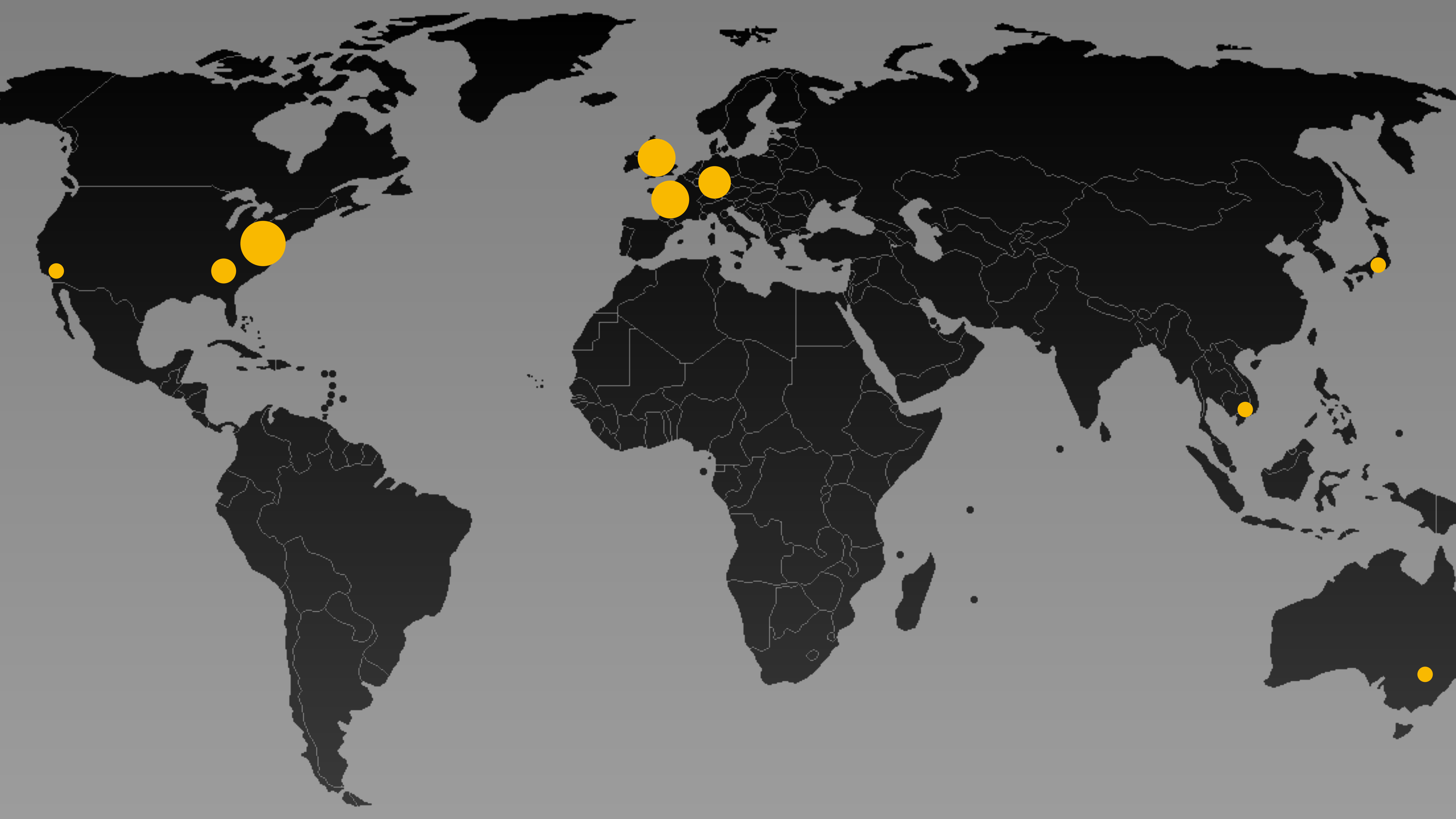
Consensus

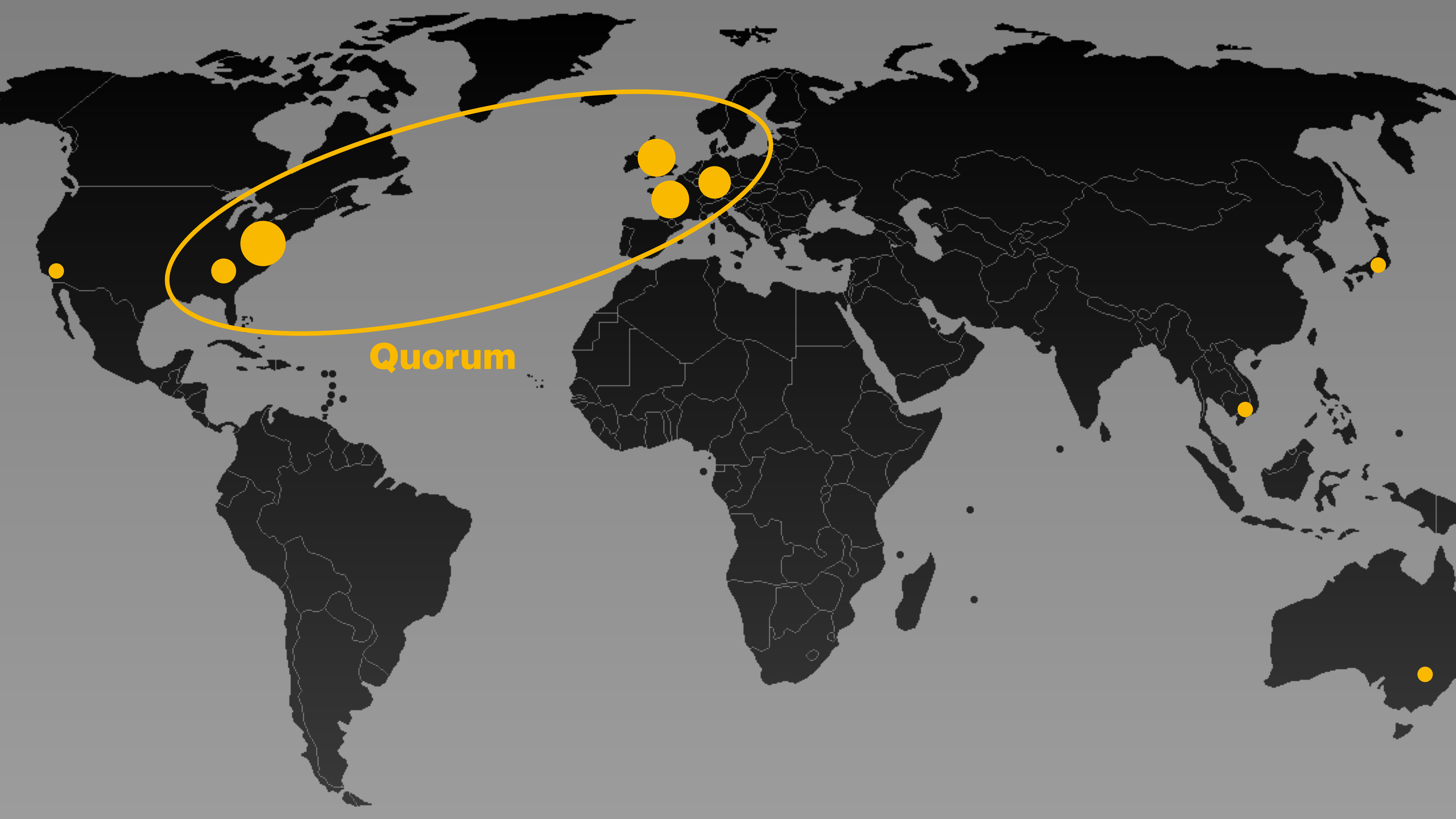
Execution

Checkpoint

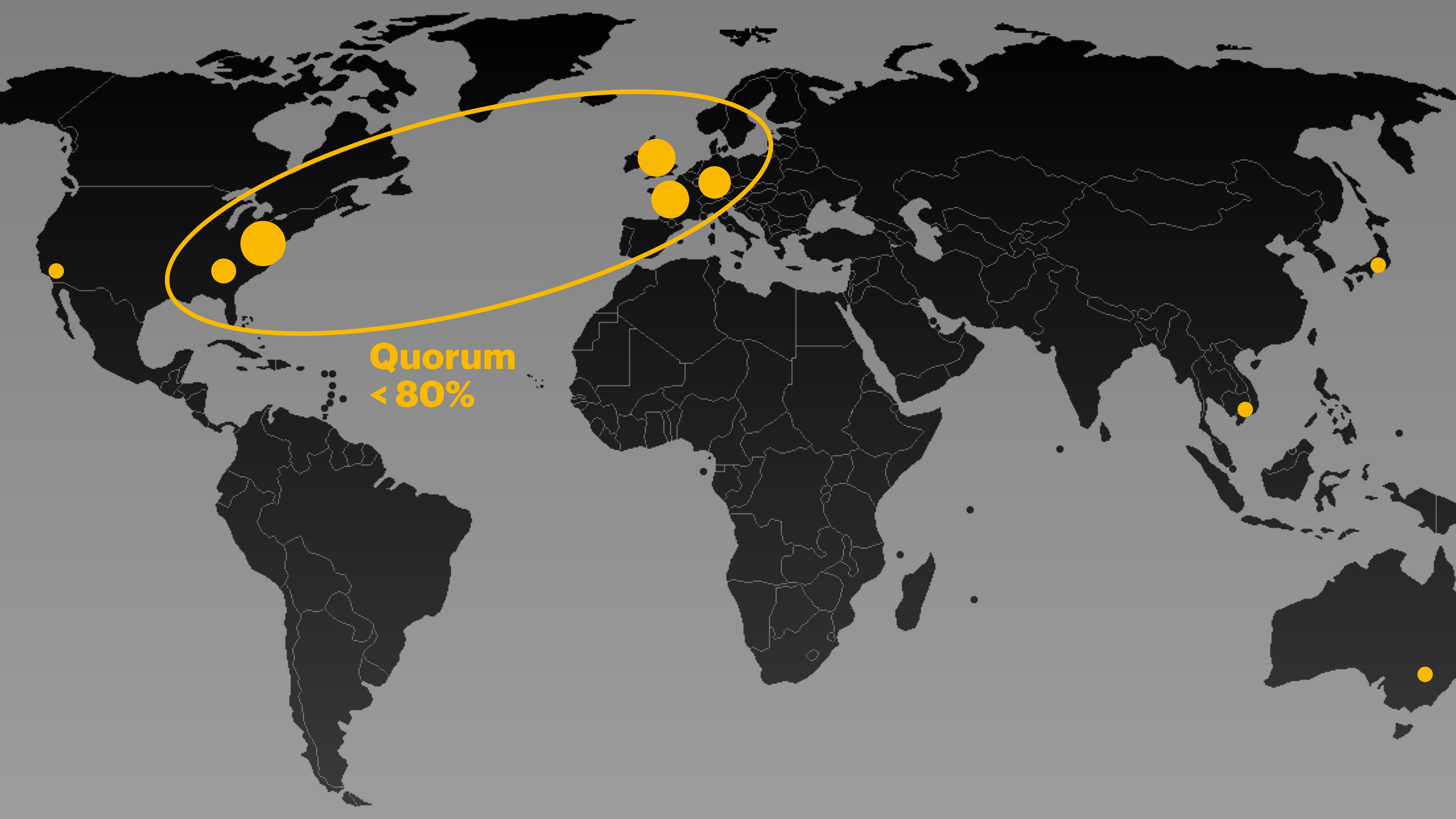








Quorum



Quorum
< 80%

$$5f+3c+1$$

crash faults

33

15

Configurable

Traded for
latency

$3f+1$

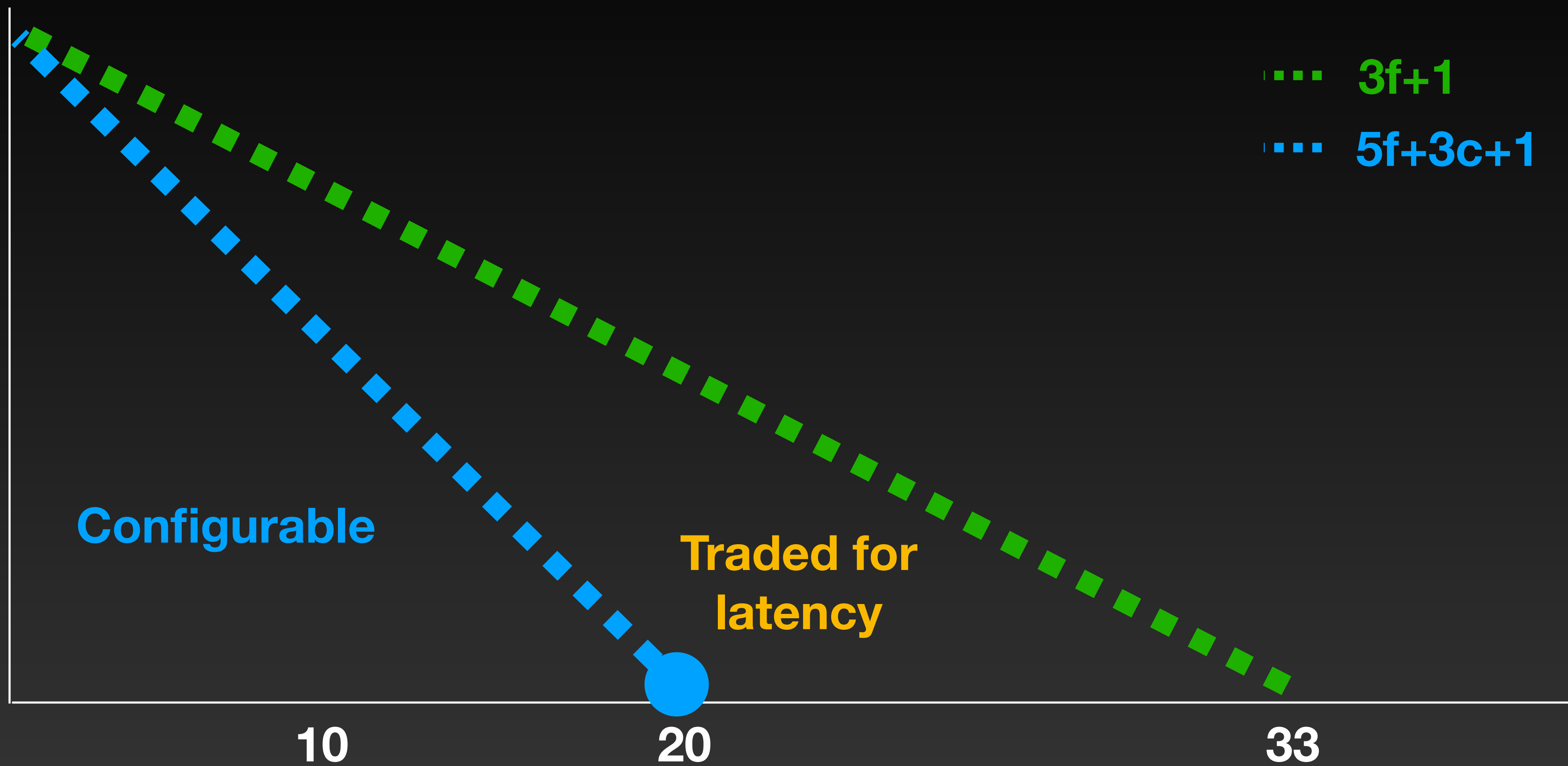
$5f+3c+1$

10

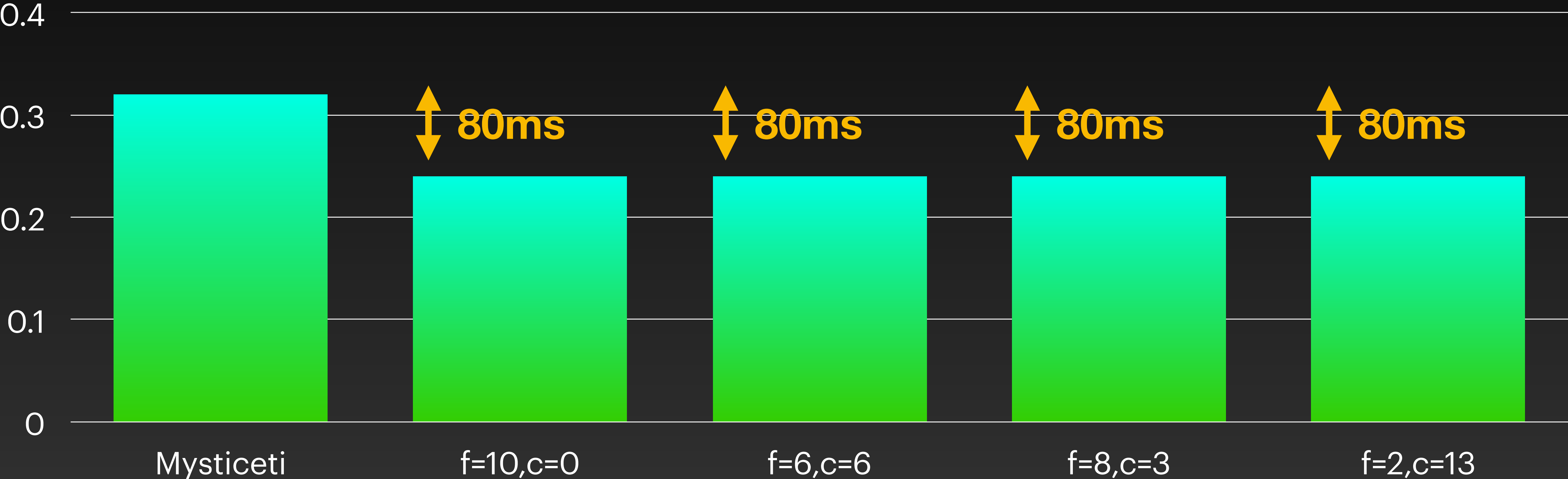
20

33

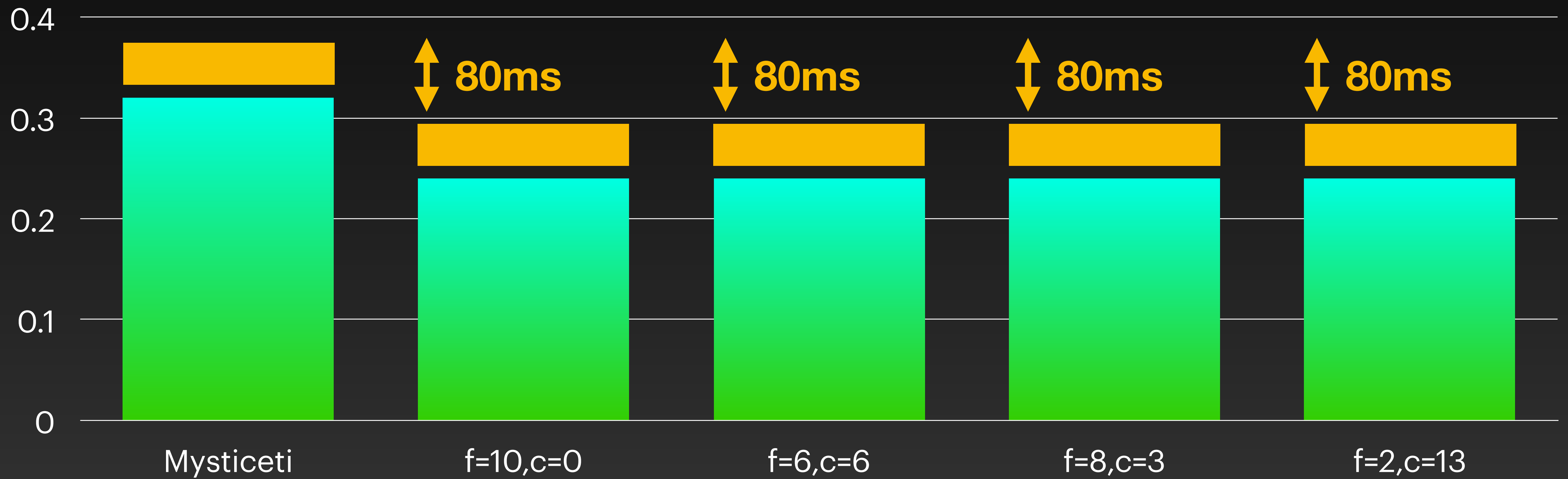
byzantine faults



EU-US: 10k tx/s



EU-US: 100k tx/s



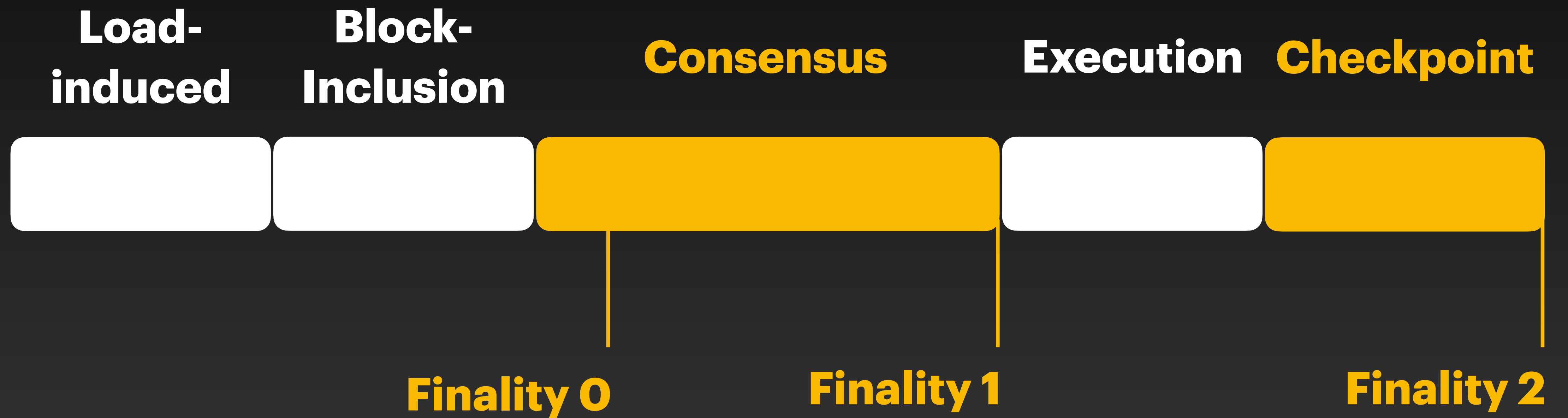
Mysticeti V3

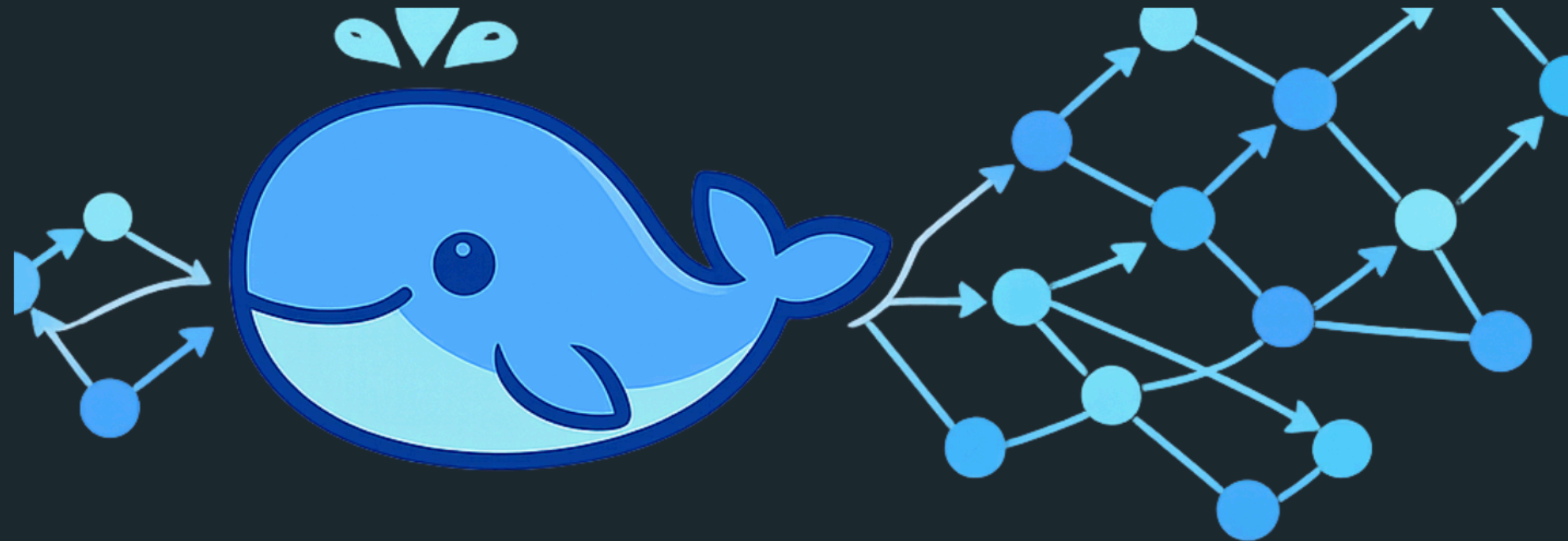
f=?,c=?

Mysticeti V3

f=12.5, c=12.5







Uncertified DAG Consensus

 build **passing**  rustc 1.92+ license **Apache**

This repository provides reference implementations of several uncertified DAG-based consensus protocols. The codebase is designed to be small, modular, and easy to benchmark and modify. It is not intended for production use, but relies on real networking, cryptography, and a persistent storage. It also ships with a discrete-event simulator for exercising the protocols under custom network and failure scenarios.

This repository currently supports [Mysticeti](#), [Mahi-Mahi](#), [Blue Bottle](#), [Cordial Miners](#) (both the partially synchronous and asynchronous variants), [Nemo-Nemo](#), [Orcaella](#), and a DAG variant of [Hydrangea](#).

Quick Start

<https://github.com/asonnino/mysticeti>

```

254 /// "Mysticeti: Reaching the Latency Limits with Uncertified DAGs" NDSS 2025.
255 /// <https://sonnino.com/papers/mysticeti.pdf>
256 pub fn mysticeti(total_stake: Stake, leader_count: NonZeroUsize) -> Self {
257     let quorum = 2 * total_stake / 3 + 1;
258     Self {
259         direct_commit_quorum: quorum,
260         direct_skip_quorum: quorum,
261         anchor_link_size: 1,
262         wave_length: 3,
263         leader_count,
264         pipeline: true,
265         leader_wait: true,
266         require_crypto: true,
267     }
268 }
269
270 /// Blue Bottle
271 ///
272 /// "BlueBottle: Fast and Robust Blockchains through Subsystem Specialization"
273 /// <https://sonnino.com/papers/bluebottle.pdf>
274 pub fn blue_bottle(total_stake: Stake, leader_count: NonZeroUsize) -> Self {
275     let strong_quorum = 4 * total_stake / 5 + 1;
276     let weak_quorum = 2 * total_stake / 5 + 1;
277     Self {
278         direct_commit_quorum: strong_quorum,
279         direct_skip_quorum: strong_quorum,
280         anchor_link_size: weak_quorum,
281         wave_length: 2,
282         leader_count,
283         pipeline: true,
284         leader_wait: true,
285         require_crypto: true,
286     }
287 }
288
289 /// Nemo-Nemo
290 ///
291 /// "Finding Nemo-Nemo: CFT DAG-based Consensus in the WAN"
292 /// <https://sonnino.com/papers/nemo-nemo.pdf>
293 pub fn nemo_nemo(total_stake: Stake, leader_count: NonZeroUsize) -> Self {
294     let quorum = total_stake / 2 + 1;
295     Self {
296         direct_commit_quorum: quorum,

```

Block Synchroniser

Theory

Reliable Broadcast • Proactive dissemination

Prototype

Optimistic dissemination • Random pull on a timer (if needed)

Production

Optimistic dissemination • Proactive re-sharing on timer • Bulk-sync of past commits

Transactions Duplication

DAG-based protocols

No deduplication until the first dissemination is complete

Multi-proposer protocols

No time to load and deserialise transactions for deduplication

Heavily pipelined protocols

No time to interpret previous leader's transactions for semantic deduplication

Why DAGs

Narwhal (sections 1-4) • <https://sonnino.com/papers/narwhal-and-tusk.pdf>

Robustness & Throughput

Bullshark (partially synchronous) • <https://sonnino.com/papers/bullshark-simple.pdf>

Latency

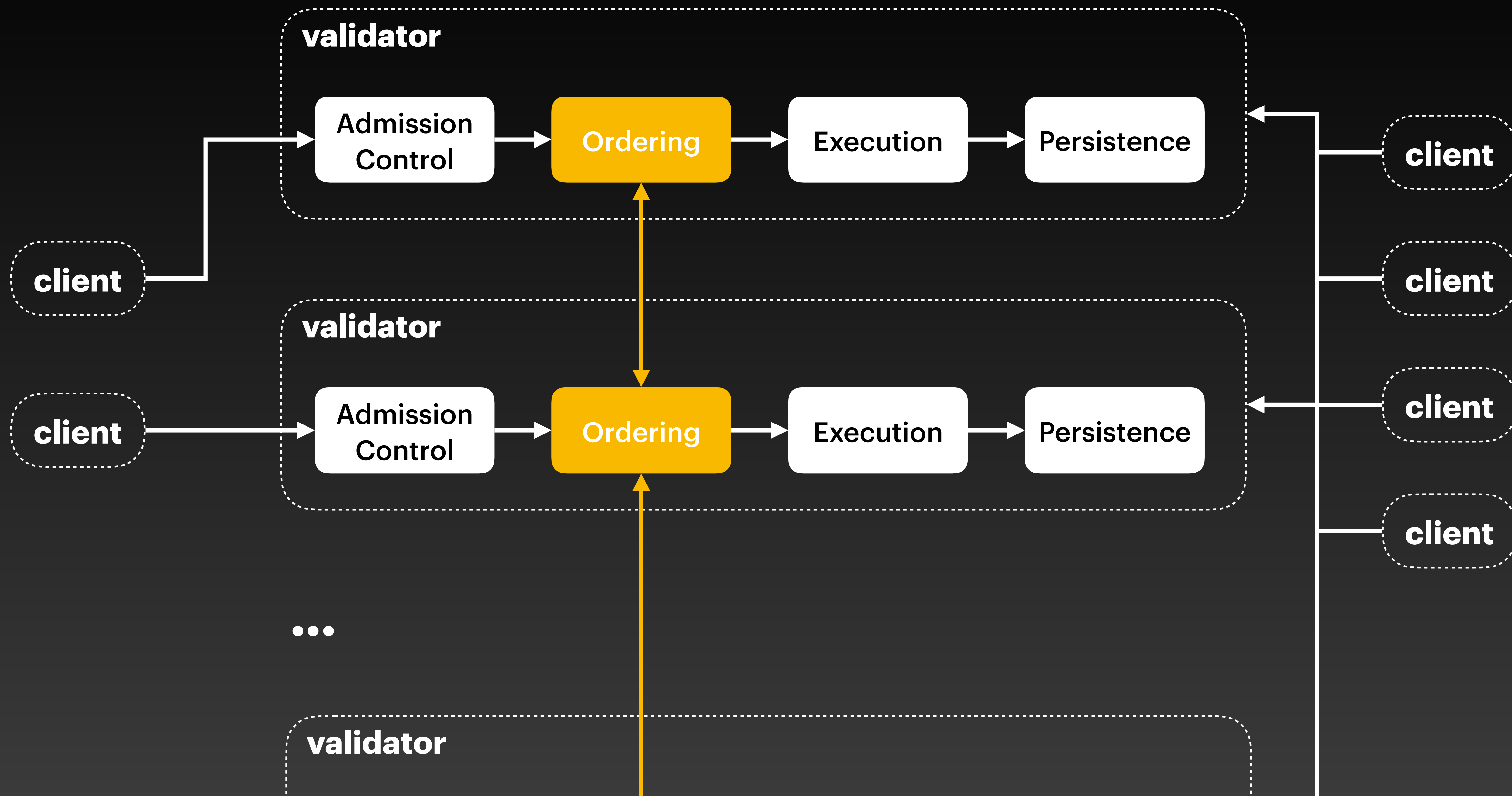
Shoal++ • <https://arxiv.org/pdf/2405.20488v1>

Uncertified DAG & Real World

Mysticeti (skip section 5) • <https://sonnino.com/papers/mysticeti.pdf>

Unstructured DAG

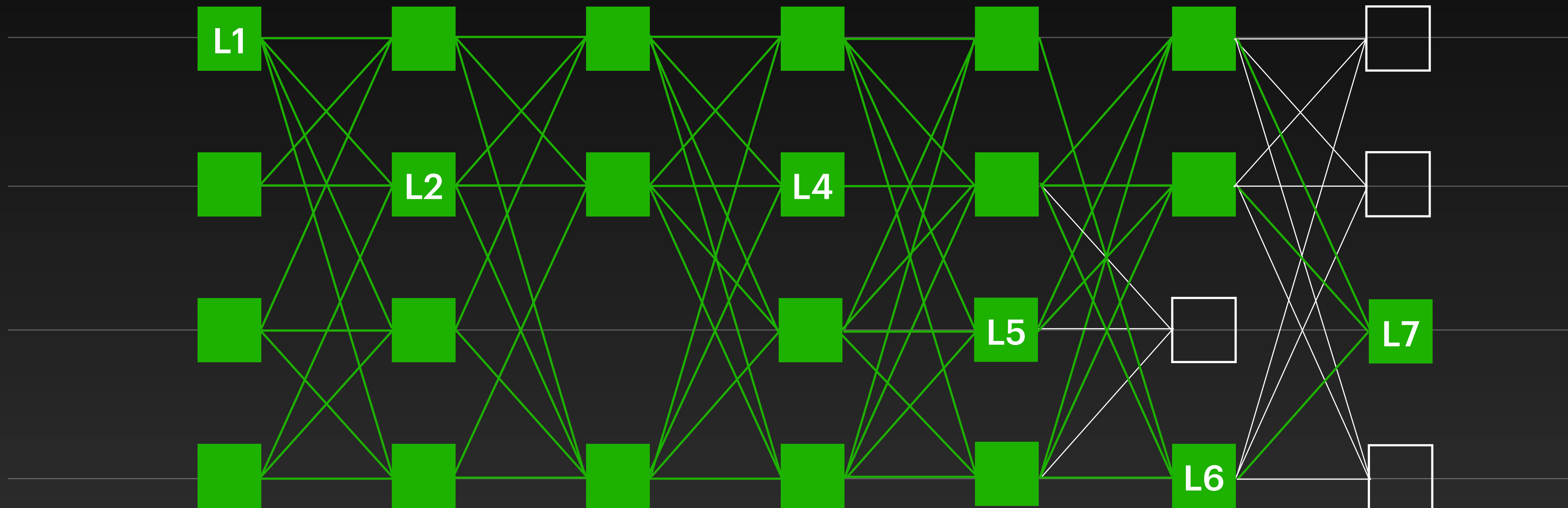
Autobahn • <https://arxiv.org/pdf/2401.10369>



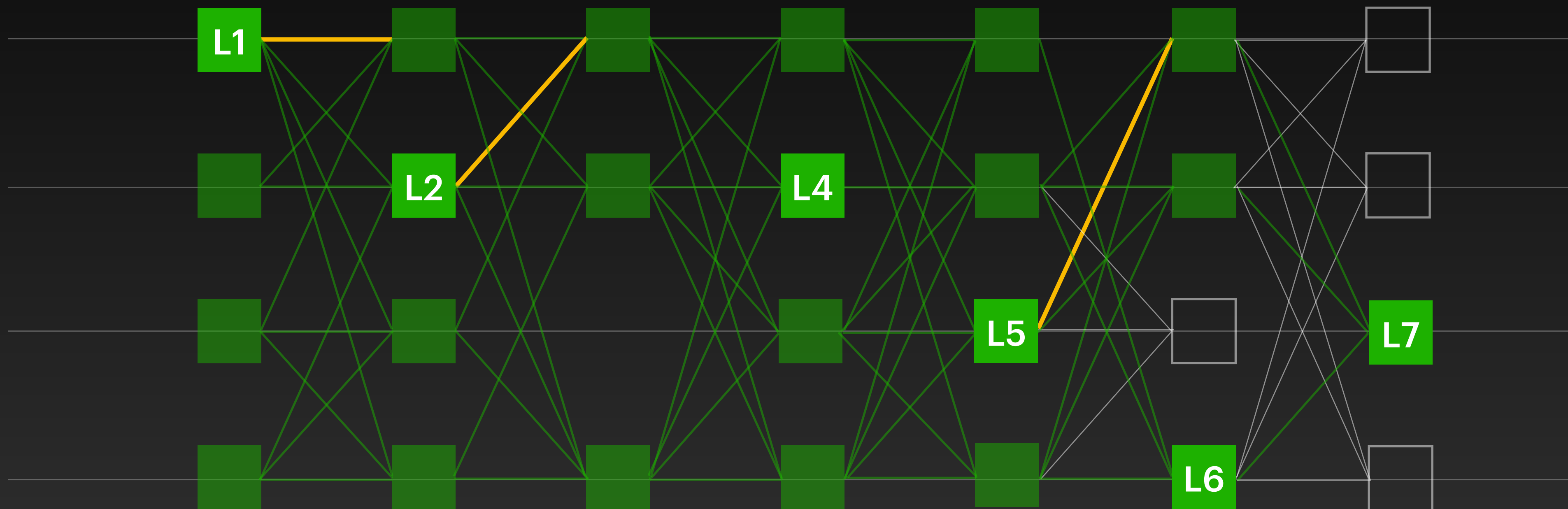
alberto@mystenlabs.com

EXTRA: **Leader Reputation**

Past Commits

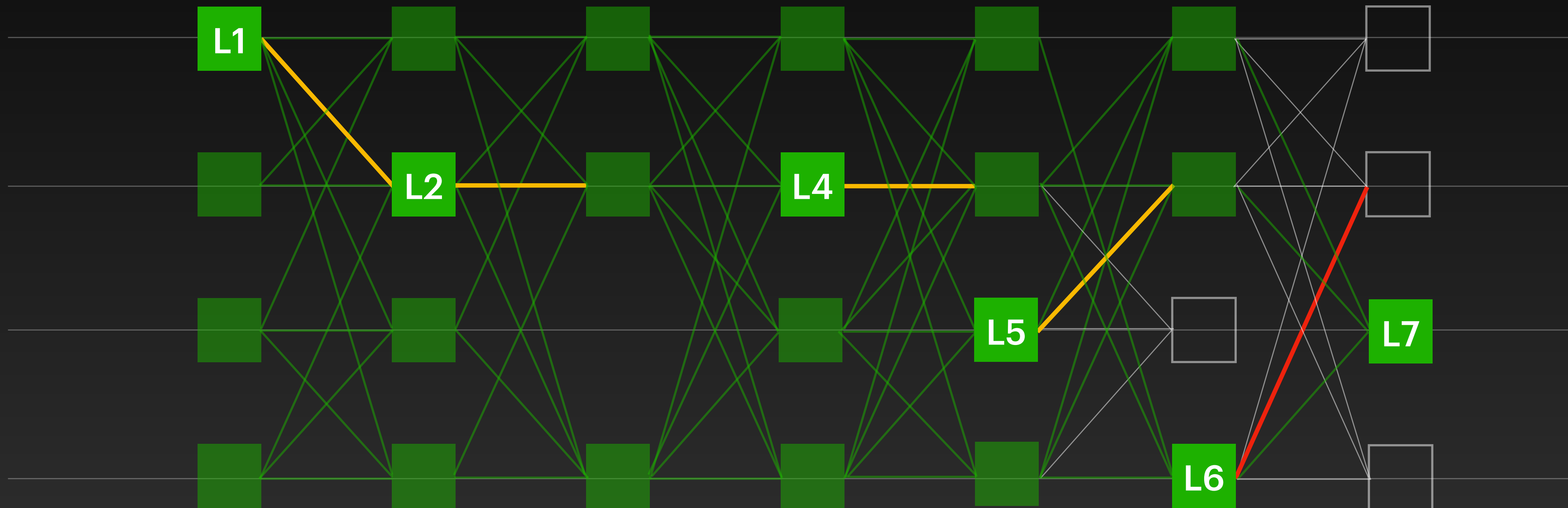


Compute Reputation Scores



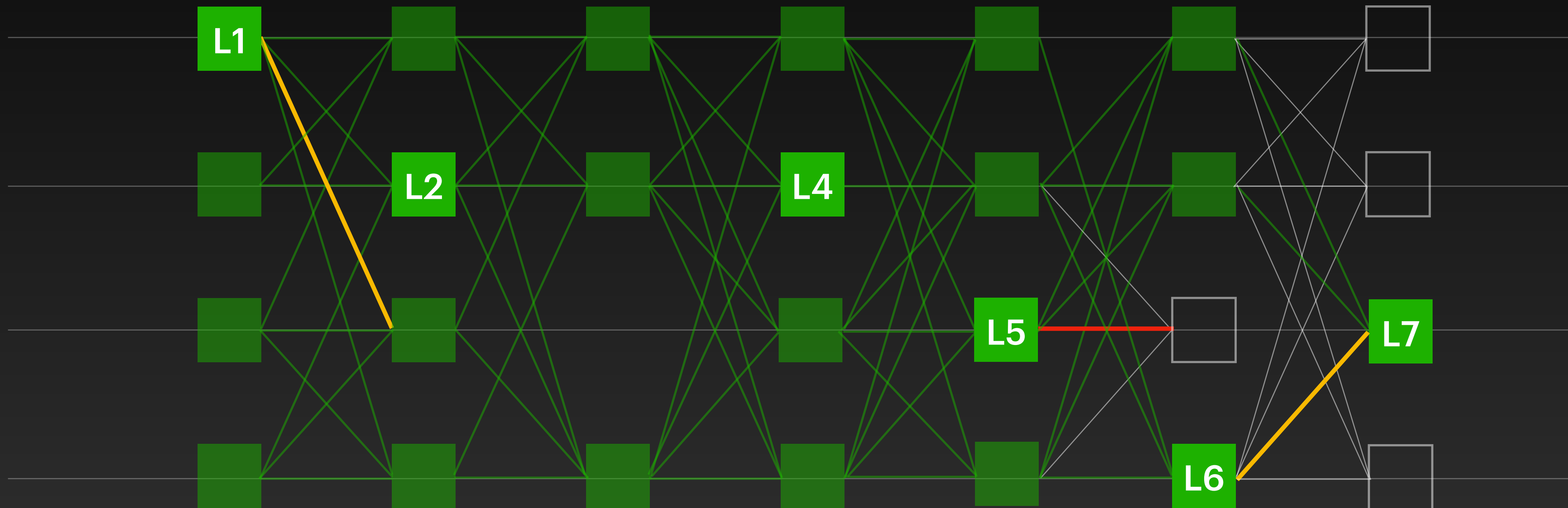
node 1: 3

Compute Reputation Scores



node 1: 3 node 2: 4

Compute Reputation Scores

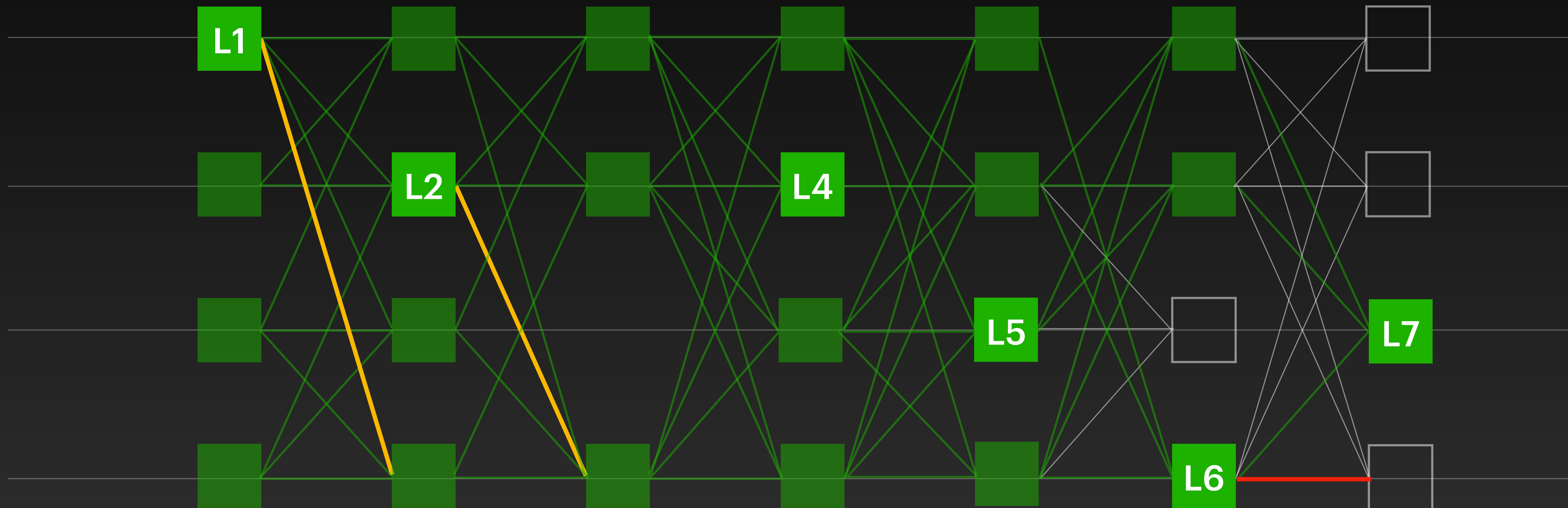


node 1: 3

node 2: 4

node 3: 2

Compute Reputation Scores



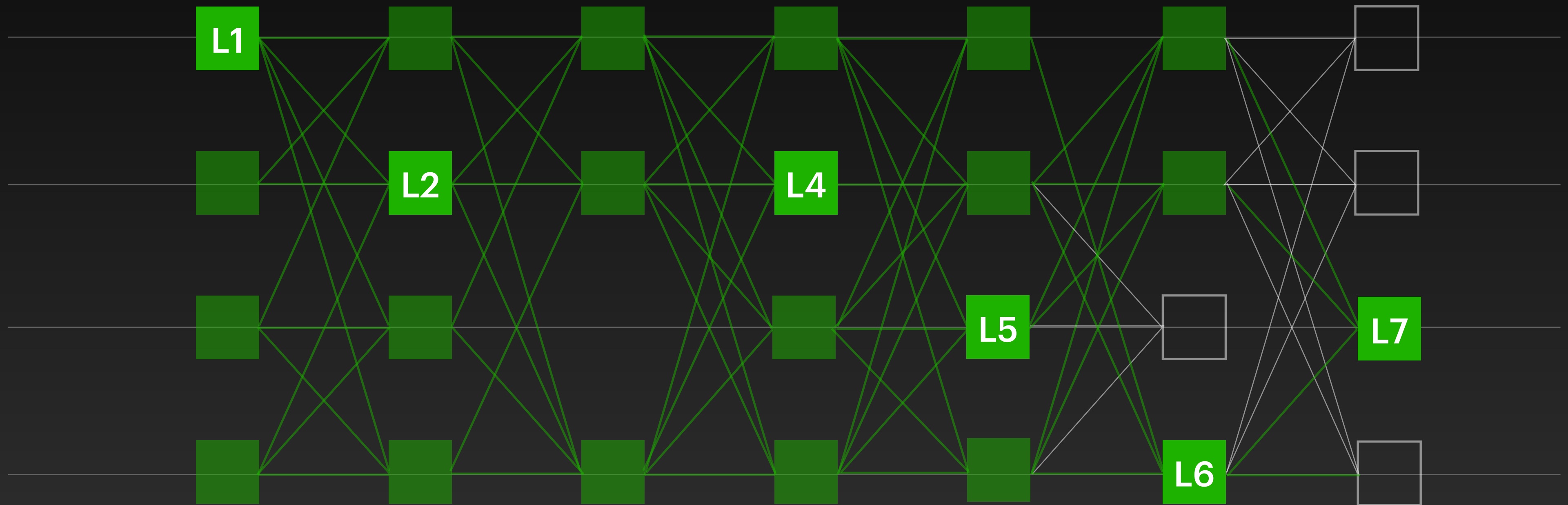
node 1: 3

node 2: 4

node 3: 2

node 4: 2

Future Leaders



node 1: 3

node 2: 4

node 3: 2

node 4: 2